सत्यमेव जयते

GOVERNMENT OF MAHARASHTRA

A REPORT ON FLOODS 2019 (KRISHNA SUB-BASIN)

by

EXPERTS STUDY COMMITTEE

(Analysis,Causes and Remedies)

Volume- I
Main Report

Volume- II
Annexures

Volume- III
Maps

May 2020

EXPERTS STUDY COMMITTEE: FLOOD 2019

Experts Study Committee : Report on Krishna Sub-basin

To,
The Principal Secretary (WR),
Water Resources Department,
Mantralay, Mumbai – 32

Subject : Submission of the study report by Expert Study Committee, for the Evaluation of Floods 2019 event, its Analysis and finding reasons thereof, and to suggest Flood mitigation measures for KRISHNA SUB-BASIN

Reference : शासन जलसंपदा निर्णय क्रमांक पूरनि-२०१९/प्र.क्र.२९१/२०१९/सिंव्य(म),
दिनांक २३/०८/२०१९

Respected Sir,

The experts study committee, formed by the GR under reference, to study the Flood event 2019, in Krishna and Bhima sub-basin has completed its study as per the ToRs given for **KRISHNA SUB-BASIN** and the same is being presented to the government herewith, in the form of 10 nos. of hard copies and a soft copy, as mandated, for further needful actions.

The same report for Bhima sub-basin is yet to be prepared and will be submitted separately, in due course.

Considering the vast data to be collected, compiled and studies to be done from various Expert Institutes regarding the flood event, and many other constraints like pandemic and all, the committee members has tried their best, to finalise the report on Krishna sub-basin in first phase.

The supporting documents viz. minutes of meetings, correspondence made by mail, whatsapp etc. and other related documents to be put on record have been compiled and are being submitted in a separate file for further reference.

The final draft of this report has been approved unanimously, during the meeting of the committee on 14.5.2020 on VC, considering all the suggestions of the members, which have also been kept on record.

Thanking to all concerned and with regards,

Er. Nandkumar Vadnere
Prin. Secretary, WRD (Rtd) and
Chairman, Experts Study Committee

Er. Rajendra Pawar
Secretary, CAD, WRD and
Member Secretary, Experts Study Committee

DA – 1) 10 hard copies & one soft copy of the Report.
2) Correspondence File

KRISHNA SUB-BASIN

BHIMA SUB-BASIN

Preface

Floods have been a recurrent phenomenon which brings devastation to settlements, misery to human lives and losses to infrastructure and public utilities. All of these were experienced in months of August & September 2019 in Krishna basin in Maharashtra & Karnataka state. Sangli, Kolhapur and few districts in Karnataka suffered heavily during this flood event.

Study Committee of 10 members appointed by Government of Maharashtra has brought out this detailed report on various TORs concerning analysis & reasons of occurrence of 2019 flood in Krishna basin, technical measures to be undertaken in future and also study on back water effect in Maharashtra due to Almatti & other structures in Karnataka.

Various experts from IMD/IITM, MRSAC, WRD, CWC, MWRRA, NGOs, individuals inducted in committee studied different TORs and through their valuable reports, put fourth their opinions/conclusions/recommendations. The Committee held discussions and visited flood prone areas in Maharashtra, Almatti / Hippargi projects in Karnataka, various renowned institutes/organisations and brought out this technical report in three volumes. Volume-I comprises of main report, Volume-II contains Annexures including study reports of experts and Volume-III contains maps.

Broadly, there are three factors which are responsible for flood- Meteorological which is natural where prolonged heavy rainfall during monsoon, cloudburst, typical cyclones play role, Topological viz. Lack of drainage from the area, saturated subsoil and Human_where uncontrolled encroachment on flood plains, drying and sedimentation of natural drainage channels due to human activities, damming & diversions of river, destruction of natural barriers to flooding take place.

In Krishna sub basin the Committee transpired that persistent & simultaneous occurrence of heavy precipitation in short durations both in dam & free catchment, typical topographic features, river meandering, large encroachments in flood plains, reduced flood carrying capacity of river channels, poor land drainage in urban areas, sedimentation are some important causes of flood. However inspite of all these constraints it is observed that the dam operators in Maharashtra had tried to keep the outflows from spillway on lower scale compared to inflows. Committee has proposed pragmatic & useful measures both structural & non-structural. Flood storages & diversions, straightening of meanders, restoration of natural drainage, raising river banks, control on development in flood plains, establishing flood lines, rehabilitation/removing encroachment in restricted/prohibited zones, providing spillway gates are some of the structural measures. Few of them are permanent and temporary. Integrated ROS, revising ROS and flood lines, Robust rainfall forecasting, Reinforcing RTDAS/RTDSS, vulnerability maps, Technology updation, Institutional Arrangements, policy measures especially for flood mitigation & climate *change, Adaptation of Maharashtra state climate change Adaptation Policy(MSAPCC 2017) are proposed as non structural measures.*

Committee with due consideration proposed useful recommendations based on detail studies done by various Institutes, Experts concerning Analysis and causes thereof. It request for immediate implementation of few important actions included for Preparedness, robust systems for forecasting, Flood resilient infrastructure, Operational Reforms and Disaster management, policy reforms/Legal provisions and other number of recommendations related to flood management.

Very importantly, Hydrodynamic analysis has been done using HEC-RAS software to study back water effect due to Almatti & other reservoirs from Karnataka on river Krishna. Specific chapter is included in report along with findings/conclusions. This study also showed effect of meandering, confluence of tributaries & inundation caused due to back water which has helped to better understand causes of submergence for longer periods in Sangli and Satara districts.

Flood management, Reservoir Management, Flood forecasting, Disaster Management, Dam Safety are all public interest activities. Thus all information in this regard must remain in public domain. Dams are tools to moderate flood to certain extent with adaptation of rule curves properly understood, along with rigid to semi rigid concept of ROS and switching over process from normal to emergency schedules backed by additional actionable efficient weather forecasting would help dam operators to manage floods in better way. Reservoir when not operated properly, may prove to be potential source of disaster if flood is taken as hazard. We also recommend revisiting designs of various structures constructed in river Krishna & its tributaries.

During last 2-3 decades the trend in flood damages have been growing exponentially. In the climate change scenario, one of the main threats would be floods at one side and droughts on other side simultaneously. So we will have to take it seriously in planning our goals.

Advanced Weather Forecasting System, NOW CAST, RTDAS/RTDSS, vulnerability mapping by Remote Sensing, well equipped disaster management units ready with evacuation plans, training/mock drills for volunteers, government staff & public is of prime importance. To acquire value added actionable information from IMD (more clear quantitative forecast) for efficient flood operation can be helpful. Proper co-ordination system amongst IMD/WRD/Disaster Management Units & various related government authorities is utmost essential during flood mitigation operations to look after the public safety and minimising the losses.

Encroachment, that is human interference in river plains is a serious concern causing flood disasters. Flood plains have been severely affected by urbanization particularly in Sangli & Kolhapur districts. State Government by an enactment may *establish Flood Zoning Authorities on pilot basis for Sangli & Kolhapur districts with necessary powers, functions & duties for regulation of land use & to overall control future encroachment/ unauthorised constructions etc.*

Krishna being an Interstate river, apart from present routine co-ordination process viz. exchange of flood data between Maharashtra & Karnataka state, it is prudent rather mandatory to establish advanced forecast modelling system like RTDSS on entire Krishna basin as a whole for effecting proper control on floods everywhere and minimising losses. In a state game of reservation and allocation of water, each state stand entitled to get their optimum share to safeguard the interest of that state. However, flood management in whole Krishna basin is a national game where all interstate players must play in harmony to achieve goal set for the nation.

It is necessary & my privilege to place on record that all honourable members & invitees contributed their best in preparing this report. Their deep sense of commitment is praiseworthy.

Submission of this exhaustive report got delayed owing to certain unavoidable reason and more so due to current pandemic/lockdown in our country. Nevertheless, the team worked from home & brought out this report. I hope that it would prove to be useful to various organisations, Institutes, Departments, Individuals, Dam operators and those all related to Flood Management.

Government of Maharashtra is requested to accept recommendations and may take quick steps to implement important measures at the earliest.

Clear her path, Let her flow...
Along with her, ensure we grow...
Respect the River!

Er NANDKUMAR VADNERE
(Retd. Principal Secretary WRD, GOM)
CHAIRMAN, EXPERT STUDY COMMITTEE

Acknowledgement

Preparation of such exhaustive technical report on various flood issues related to Krishna sub basin of Maharashtra is a stupendous task. It involved collection of large hydrological/ hydrometeorological data, geographical features, various reports, literature, references from many concerned organisations/institutions viz. IMD, IITM, CWC, MRSAC, WRD, MWRRA, NGO etc

Committee is grateful to Government of Maharashtra for entrusting the responsibility of preparation of this technical report to us.

At the outset, chairman want to place on record his deep appreciation & thanks to the following honourable members.

Er. Vinay Kulkarni, Technical member MWRRA painstakingly guided & carried out Hydrodynamic study using HEC-RAS software for back water of Krishna river along with Er. H.T Dhumal SE(WRD) without whom this study could not have been complete. They brought out excellent technical study report and drawn various conclusions related to main TOR and allied issues. Technical Member voluntarily offered his studious suggestions on other TORs too for achieving further refinements.

Er. Sanjay Ghanekar Secretary (P&M WRD) along with Dr. Subrato. N. Das, Director MRSAC Nagpur and Mr. Vivekanand Ghare, Senior Consultant MRSAC, Mumbai with assistance from various WRD officials & organisations formulated proposals on measures to be adopted. The study report by MRSAC on Remote sensing and GIS based Analysis of Flood 2019 was proved to be very much useful in proposing measures.

Prof. Ravi Sinha, IIT, Mumbai with the assistance from Mr. Rohan Shinde offered suggestions on mathematical modelling. Prof. Subimal Ghosh IIT, Mumbai examined & generally agreed with the conclusions drawn in back water study.

Er. Nitya Nand Rai, Director(Hydrology) South CWC , New Delhi also carried out back water study on river Krishna using MIKE-II software. He participated in all technical discussions and offered useful suggestions on preparing Rule Curves for depletion period also.

Er. Pradeep Purandare , Retd Prof WALM I Aurangabad along with WRD officers brought out a well studied report on Koyna ROS & put forth conclusions and recommendations on related issues in Flood Management and goverence.

Shri K.S Hosalikar, Deputy Director General of Meteorology (DDGM IMD) Mumbai and Mrs. Nayana Deshpande, Scientist, Indian Institution of Tropical Meteorology brought out an excellent report on Storm Analysis July-August 2019. In real sense, it appeared to be a triggered report of first cause of flood. On the request by Chairman, DDGM took initiative and arranged presentation by NCCR, Chennai & visit to

IMD office in Pune. He kept proper liaison with the Chairman. We appreciate him and his team. We are thankful l to Dr. Raman Murthy & Dr. Ranjan Kelkar Ex DGM for their valuable suggestions in the field of flood forecasting and need for joint working thereof . We are also thankful to scientists Ms.Neetha Sasidharan, Mr. K.N Mohan, Dr. Guhathkurta , Mr. O.P Sreejit , Dr. Kashyapi from IMD.

Er. R.D Mohite Chief Engineer(WRD) Pune alongwith WRD officials, compiled exhaustive information and brought out actual field report on Flood 2019 in Krishna sub basin. Great help came from Er. H.V Gunale SE(SIC) Sangli who contributed ample information/data as coordinating SE. Er. S.L Doifode SE(SIC) Satara, Er. R.M Sankpal SE & Er. Survey SE (KIC) Kolhapur, Er. S.D Chopde SE(PIC) Pune, Er. Pravin Kolhe SE(PIPC) Pune, Er. S.M Sangle SE(KDC) Pune Er. Prasad Narvekar SE(e-Governence) pune Er. Y.K. Bhadane(DSO) Nasik. We appreciate their overall contribution more so to SE PIPC who till submission assisted in technical matters and all allied formalities thereof.

World Bank Delegates Mr. Rajgopal Singh, Mr. Wsbrand Dejong and Mr. Hans Banggrab participated in committee meeting & deliberated on various issues. We are thankful to them.

Smt Shailaja Deshpande, Dr. Gurudas Nulkar NGO Jivit nadi & Ecological society Pune gave presentation and submitted their report concerning ecological and environmental aspects of flood. They gave useful suggestions and recommendations for protection of human life & environment . We are grateful to them.

We appreciate the whole hearted co-operation extended by Er. K.H Ansari, Executive Director MKVDC Pune and Invitee who extended all hospitalitity and made us available office premises of MKVDC & personally contributed in meetings held . .He shared field experiences and offered pragmatic solutions on mitigation of floods in Krishna valley.

Thanks are due to Er. Atul Kapole CE & S Jt. Secretary Mantralaya, Mumbai who offered useful suggestions on few policy matters in administration set up for flood mitigation. Er. V.G Rajput CE(SP WRD) studied various issues pertaining to EAP & SOP and suggested certain additions in present procedures. We appreciate his contribution. We are also thankful to Er. Ranga Reddy CE (CWC) Hyderabad for his participation in meetings and providing valuable technical inputs.

Aerial observations during peak flood was of great importence. We are grateful to Mr. Kishor raje Nimbalkar, Secretary (Rehabilitation) for such quick arrangements.

We are grateful to Dr. Deepak Mhaisekar, Divisional Commissionar pune , Mr Abhijit Chaudhari, Collector Sangli, Mr Daulat Desai, Collector Kolhapur and Mr. Mallinath Kalshetti, Mayor Kolhapur for their useful suggestions on vulnerability of flood in towns , encroachments, timely forecast, flood mapping and liasioning in disaster management.

Er. R.P Kulkarni CE (KBJNL) Almatti & his team welcomed us at Almatti project and arranged visit & discussions. We appreciate him and his team who made all logistic arrangements & participated in presentations and discussions.

Dr. (Mrs.) V.V Bhosekar Director CWPRS along with Shri R.S Jagtap Jt. Director, Shri R.G Patil Scientist, Dr. Neena Isaac Scientist, Shri A.A Purohit & other officials welcomed committee in CWPRS, held discussions & arranged visits to various hydraulic models. Useful guidelines on river modelling was discussed. Director readily agreed to take up some necessary hydraulic modelling projects in Krishna basin. Committee is grateful to CWPRS team..

Chairman is personally thankful to Er. Sanjay Heganna SDE for his devotion in RTDAS/RTDSS implementation & bringing out very useful report on this subject. Thanks are also due to Er. Dabhade SE , Er. Bagade Dy. Sec. Mantralaya , Mrs. Shailaja Sulakhe EE, Er. Chandure EE, Smt Sarika Kshirsagar & Smt Reshma Shinde AE-II for their whole hearted support in proper continuance of present advanced flood forecasting system.

I admire the inputs given behind the curtain by Er. Shilpa Jadhav EE who made excellent graphical representation in report. All along she had been working on scanning technical reports , editing, formatting till final submission.

Er. Manoj Mohite T.A carried out full time work of typing , editing & printing of reports & was available to chairman for almost all days. My sincere thanks to him.

Er. S.M Panure, Er. Sneha Bhosle AE-II contributed in preparation of drawings. We appreciate their work.

Er. Kumar Patil EE, Er. Rohit. Bandivdekar, Er. N.S Kare EE, Er. Abhay Ketkar EE , Er.Munjappa AE-I, Er. Shashank Shinde AE-I, Er. Shubhangi Patil JE , all stand appreciation . Shri Mhaismale , Typist dutifully discharged his duties untiringly . My thanks also goes to all supporting staff who directly or indirectly contributed to the Committee.

Lastly , as Chairman, I whole heartedly , recognise the sincere contribution made by Er. Rajendra Pawar, (Secretary CAD) Member Secretary of this Committee , who untiringly devoted full attention in Administrative & Technical issues inspite of his busy schedule. As Member Secretary, he has shouldered all responsibilities very ably & participated in all meetings, visits & deliberations. His healthy attitude & dedication helped out to present this report during current pandemic..Thanks to all again

Er. NANDKUMAR VADNERE
(Retd. Principal Secretary WRD)
CHAIRMAN EXPERT STUDY COMMITTEE

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Executive Summary

(Krishna Sub-Basin)

1. Introduction :

The ToR given to the expert study committee vide GR dated 23 August 2019 is to find out the reasons behind the flood situations in Bhima and Krishna Valley in the year 2019. Considering the voluminous work and the huge data compilation required for the studies, it was decided to prepare separate reports for each basin. Considering the severity and frequency of the floods and the extent of damages, here, initially the study report of flood in Krishna Sub Basin within the state of Maharashtra has been done by this committee and is being presented in three volumes. The study report for Bhima Sub Basin will be prepared and presented separately in due course.

During the months of July & August 2019, Sangli & Kolhapur districts in Krishna sub basin experienced extreme floods for long durations. Heavy losses to life, property & crops etc. had been reported. Different opinions at various levels were put forth concerning these flood events. Sangli & Kolhapur districts faced heavy flood situations in past also & floods of 2005 & 2006 were noteworthy. However, 2019 flood event was comparatively much more severe which lasted more than a week & losses experienced were also on higher scale.

It is, therefore, necessary to constitute a Study Committee of Experts to carry out an in-depth analysis of the reasons behind the flood situation and to suggest measures to prevent the repetition of such unfortunate events in future.

Therefore, Experts Committee of 10 members was appointed vide WRD Marathi GR No. FC-2019/291/2019/IM(M) dated on 23rd August, 2019 (**Refer Annexure 1A & 1B**) to “Study reasons of occurrence of 2019 Floods in Bhima & Krishna basin & propose technical measures to be undertaken in future”, on the following TORs.

1.1 Terms of References

1. To carry out in-depth technical analysis using modern techniques and investigate causes of 2019 flood situation occurred in Bhima and Krishna basin.
2. To specifically clarify based on hydrological studies, whether Almatti & other

reservoirs (back water effect) from Karnataka, create flood situation in Maharashtra.

3. To propose overall remedial measures to avoid such flood situation in future and /or to reduce its severity.
4. Along with policy level measures, to recommend revised plans for micro level issues, such as dam wise reservoir operation schedules, integrated arrangements for measurement & management of river discharges, control on constructions in flood zone, emergency action plans, standard operating procedures etc. and to submit report based on technical analysis.

1.2 Study groups:

The Experts study groups formed vide above GR comprises of following Institutions and Individual experts as honourable Members:

1. **Mr. Nandkumar Vadnere**, Rtd. Prin. Secretary, Water Resources Dept, Govt. of Maharashtra as **Chairman** of the Committee
2. **Mr. Vinay Kulkarni**, Member (Technical), Maharashtra Water Resources Regulatory Authority, Mumbai
3. **Mr. Sanjay Ghanekar**, Secretary (Project Co-ordination), Water Resources Department, Govt. Of Maharashtra, Mumbai
4. **Prof. Ravi Sinha**, Indian Institute of Technology, Mumbai
5. **Mr. Nitya Nand Rai**, Director, Hydrology (South), Central Water Commission, New Delhi
6. Director, Maharashtra Remote Sensing Application Centre, Nagpur
7. Dy. Director General, India Meteorological Department, Mumbai
8. Director, Indian Institute of Tropical Meteorology, Pune
9. **Mr. Pradeep Purandare,** Aurangabad.
10. **Mr. Rajendra Pawar**, Secretary (Command Area Development), Water Resources Department, GoM, Mumbai as **Member Secretary**

2. Flood Event 2019

2.1 Status and Vulnerability of Water Resources Sector in Maharashtra due to Climate Change

Out of 6 river basin systems, only 55% of the dependable yield is available in the four river basins (Godavari, Krishna, Tapi and Narmada) east of the Western ghats. The rest drains out in the westward flowing river basins into the Arabian Sea. Over dependency on ground water in a state where its recharge capacity is low, escalates vulnerability of systems dependent on the same such as irrigated agriculture, industries and drinking water.

Maharashtra is prone to drought and floods. Out of the total geographical area of Maharashtra, 40% of the area is drought prone and 7% is flood prone. Rainfall trends indicate that Maharashtra could face an increase in rainfall variability, including droughts and dry spells, as well as increased likelihood of flooding in the future. This has direct bearing on ground water as heavy intensity rainfall gets lost as runoff while low intensity rainfall which contributes to recharge decreases in frequency.

Climate change vulnerability assessments are necessary for designing targeted adaptation actions. The vulnerability analysis was carried out for different sectors at various levels of governance, on the basis of Macro level Vulnerability Indices, estimated using the Indices of Exposure, Sensitivity and adoptive capacities to climate changes. This was done by the Department of Environment, Government of Maharashtra while preparing the Maharashtra State Action Plan on Climate Change in 2014. The Action Plan has also presented climate projections for future at 25km x 25 km resolution using the Hadley Centre regional climate model.

Some of the projected changes in climate over Maharashtra are:

- Increase in mean temperature from 1.2 to 1.6 degree centigrade in 2030s.
- Rainfall is also projected to increase during the same period, with more rainfall projected as we progress from 2030 to 2050 to 2070 but it will be highly variable spatially.
- Annual rainfall shows highest increasing trend for Satara, Mumbai, Kolhapur and Sindhudurg and highest decreasing trend for Bhandara and Latur in 2030s.
- Extreme rainfall events with longer dry spells are projected to increase in all districts of Maharashtra.
- Numbers of dry days are likely to increase by minimum 3 to maximum 9 days in the state by 2030s.
- The sea level is expected to increase by 24 cm to 66 cm along the coastline in sync with the projections for global sea level rise, accompanied by an increase in wave heights, wind speeds, greater storminess and storm surges.

2.2 About the Event:

During the year 2019, eight cyclonic storms were developed over Indian Seas. Arabian Sea contributed 5 out of these 8 cyclones against the normal of 1 per year, which equals the previous record of 1902 for the highest frequency of cyclones over the Arabian Sea. Out of 5 cyclones developed over Arabian Sea 2 were very severe, one was extremely sever and one was super cyclonic storm.
Active spell of South West monsoon started from 27thJuly, 2019 and before it is fully dissipated; low pressure area was formed on the Bay of Bengal, which intensified into deep depression on 7th August. During the first spell catchment was fully saturated. Hence, during second spell almost all the rainfall converted into runoff causing severe inundation.

During South-West Monsoon season over the country, rainfall over Maharashtra shows significant spatial and temporal variability. The state experiences extremes of rainfall ranging from 6000 mm over the Ghats to less than 600 mm in the interiors. Western coast of Maharashtra, the Konkan belt is often prone to heavy to very rainfall during active monsoon conditions due to favorable orography. The Western Ghats act as obstruction to the eastward-moving Monsoon clouds forcing it to rise ultimately leading to the heavy downpour on the windward side, while the leeward side forms the rain shadow area receiving less rainfall.
Climatological records of the Satara, Sangli and Kolhapur districts indicate that, there is a large variation in space of rainfall. The western portions of these districts adjoining the Western Ghats and neighborhood get very high rainfall while near the eastern borders of the districts, receive very less rainfall.

2.3 Rainfall Pattern

Rainfall data indicates that Konkan and adjoining Madhya Maharashtra experienced very heavy rainfall. In the beginning of the flood period i.e. from 27th Jul to 3rd Aug, the heavy rainfall events were localized in the northern part of the Konkan and adjoining North Madhya Maharashtra. Many stations in Pune and Nasik districts, recorded rainfall more than 150 mm/day during the period 3rd to 5th Aug. Towards the latter part of the week, rainfall belt shifted towards south Madhya Maharashtra. Mahabaleshwar recorded highest rainfall of 380 mm on 5th Aug. 2019. It is also observed that Kolhapur district continuously experienced heavy rainfall throughout the period with highest rainfall amounts on 6th Aug. 2019. Gaganbawda recorded its highest rainfall of 340 mm rainfall on

6th Aug. It is also seen that though heavy rainfall occurred in the western part of the districts in Madhya Maharashtra, their eastern parts were devoid of rainfall.

It is further seen that during the heavy rain spell of Aug. 2019, many stations in Kolhapur district and western part of Satara district have crossed their previous record of 7 days rainfall. This indicates that compared to previous years, rainfall over the region was widespread and remained very intense for a long period during 27th July to 13th August 2019.

Sangli, Kolhapur and Satara district received very heavy rainfall of 1918 mm in comparison to 333 mm normal rainfall during 27th July to 13th August. This was about 6 times the normal and at the same time, in the free catchment, downstream of the dams, it was about 18 times the normal. Such high range of continued rainfall in short duration resulted in extreme heavy flooding mainly in Sangli, Kolhapur town and few talukas situated near Krishna and Panchganga rivers.

It is observed that, the Flood affected districts of Satara, Sangli and Kolhapur continuously received excess to large excess rainfall during the first fortnight of August.

- **Rainfall records in dam catchments**

  It was seen that the observed actual rainfall in various catchments to the upstream of dams varies from 5 to 19 times the normal. Average actual rainfall was about 6 times the normal rainfall in all these catchments bringing abnormal flood to downstream areas.

- **Rainfall records in Free catchments**

  The actual rainfall during the first 56 days of the monsoon (starting from 1st June 2019) was measured at 6 rain gauge stations, situated in the free catchments of these three districts. It is observed that the total rainfall during the peak period of 18 days (27th July to 13thAugust) measured at the same stations, was about 1.6 times the total rainfall during the previous 56 days (1st June to 26th July). Also, the actual rainfall during the event in free catchments was varied from 13 to 29 times the normal rainfall. The overall observed rainfall over the normal was about 18 times. Such abnormal high occurrence of rainfall even in free catchments also aggravated floods in Sangli & Kolhapur districts.

3. Analysis and the Reasons of Floods 2019

The analysis of the flood event was done by various renowned and expert Institutions, and also by some individuals who are the members of this expert study group, as per tasks given by Hon. Chairman. These Institutes, using their own reliable data and techniques, have done various studies independently. All the reports were discussed at length during the committee meetings and the findings were confirmed.

Various expert institutes and individuals, such as -

IMD : India Meteorological Department, Mumbai (MoES)

IITM: Indian Institute of Tropical Meteorology, Pune (MoES)

MRSAC : Maharashtra Remote Sensing Application Centre, Nagpur

CWC : Central Water Commission, New Delhi

WRD (GOM) : Water Resources Department, Government of Maharashtra

MWRRA: Maharashtra Water Resources Regulatory Authority

NGOs like Jeevit nadi and Ecological Society, Pune

A report about the Analysis of the event has been received from Central Water Commission, New Delhi. Also, some Members of the Committee, along with the WRD GOM Engineers carried out different studies, using different techniques and presented their reports.

Having considered the inputs received from various expert institutes, Individuals and the NGOs working in the field, the committee, after due deliberations in its various meetings discussed the ToR issues along with the outcome of the reports. The Reports submitted by various expert Institutes and Individuals have been appended, in original, as Appendix to this report. The Gists or extracts of their findings, as discussed and finalized in Committee meetings have been given in Chapter 4 of this report.

The overall conclusions or findings of various study reports received, in line with the Terms of References given to this Committee have been compiled concisely and are as below.

3.1 ANALYSIS & REASONS (TORNo.1): To carry out in-depth technical analysis using modern techniques and investigate causes of 2019 flood situation occurred in Bhima and Krishna basin.

The flood disasters occurred during the monsoon of the year 2019, along river Krishna was primarily due to:

a) Persistent and simultaneous occurrence over large spatial areas, of heavy precipitation in short duration

Climatological records of the Satara, Sangli and Kolhapur districts indicate that, there is a large variation in space of rainfall. The rainfall in the belt, roughly 25 to 35 kms wide parallel to the crest of the Sahyadri Range, is considerably higher than in the rest of the district. While Mahabaleshwar at an elevation of 1372 metres gets an average annual rainfall of 5886.9 mm, other stations in this belt get annual rainfall ranging between 1684 and 2195 mm. Compared to Kolhapur and Satara, district of Sangli has relatively lesser average annual rainfall of about 670mm.

The persistent intense rainfall activity over the region was in association with an active spell of monsoon started over Maharashtra from 27th July 2019 and resulted in flooding in many parts of Konkan and North Madhya Maharashtra. This was followed by another active monsoon spell from 3rd August 2019, in association with the formation of a low pressure over North East Bay of Bengal and its subsequent intensification into deep depression and westward movement in the subsequent days causing severe flood conditions in South Madhya Maharashtra.

This movement of deep depression system (an active low pressure system with wind speed ranging between 52 to 61 kmph) across central India, resulted in enhancement of rainfall over west coast and in the ghat areas of Madhya Maharashtra with heavy to very heavy rainfall and extremely heavy rainfall events for more than a week period over these places, resulting in severe flood situations.

Maharashtra State has an average annual precipitation of about 741mm.The rainfall in the state is controlled by south west and north east monsoon. About 90% of rainfall occurs during monsoon months from June to October, every year. The high intensity storms prevailing during the monsoon months result in heavy discharges in all the rivers. The continuous and heavy precipitation that occurs in the steep and undulating terrain, finds its way into the main rivers through innumerable streams and water courses. Flood event 2019 in Krishna basin is an example. Basin experienced an abnormally very high rainfall between 25th July to 13th August, resulting in severe flood in Sangli, Kolhapur, Satara districts.

Average actual rainfall was about 6 times the normal rainfall in all dam catchments bringing abnormal flood to downstream areas. The overall observed rainfall over the normal was about 18times. Such abnormal high occurrence of rainfall even in free catchments also aggravated floods in Sangli & Kolhapur districts. It can be seen that the contribution of free catchment, in the discharge observed in river Krishna, at Irwin Bridge at Sangli, was 49%. Similarly, the contribution of free catchment, in the discharge observed at Rajaram weir on river Panchganga, was 84%. The discharge from free catchment, which was substantial and had no control.

b) Typical Topographical features and river meandering

By study of topographical features of the Krishna sub-basin including its tributaries in Maharashtra, it was found that the River Krishna originates at the highest altitude of 1310 m at Mahabaleshwar, and also the Koyna at the same place, they reach at Pritisangam, the confluence of Krishna and Koyna at Karad at the altitude of about 550 m. thereafter, there is sudden change in the river bed slopes in the Krishna river upto Sangli. While negotiating the Sangli city, the rivers Yerala and River Warna meets Krishna and thereafter within few kilometers there is confluence of River Panchganga leading Krishna to state border. The Krishna river reach from Sangli city to state border is only 30-40 kms long, having very flatter bed slopes and many meanders.

This, typical topographical set-up plays major role in slowing down the flood dissipation beyond Sangli city. It was found that, in this reach of river, while there is drastic reduction in the velocities of the flow, the backwater effects of various confluences of its own tributaries further aggravate the problem. While the floods in main Krishna River are yet to dissipate this particular reach, the floods coming from tributaries by simultaneous raining cannot even enter in the main river course of Krishna. The situation was further aggravated as these tributaries could not drain out their own discharge, when river Krishna was already flooded, which was spread on the side banks of the tributaries.

This peculiar situation resulted in long term inundation alongside the flood plains of tributaries like Warna and Panchganga, while increasing the back water effect near confluences resulting in the higher flood levels at Sangli, Kurundwad and Kolhapur cities.

The problem is further aggravated by the meanders in the same reach, because of which very large flood plains are created, submerging huge areas with almost stagnant waters for prolonged periods. During this time, the flood water even tend to create shortcuts, bypassing the meanders straight into the next stretch of the Krishna river, as an effort to dissipate the floods early.

b) Large Encroachments in Flood Plains and Reduction in flood discharge capacities of the rivers, due to many Structures built across Rivers

Due to the heavy developments around the river stretches and in city areas, there have been large encroachments, of building constructions, retaining walls, roads etc. in prohibited and restricted flood plains/zones and also in the natural drainage system. At many places, the debris in huge quantum is thrown in the river courses (or its tributaries), reclamation of lands, leveling of plots have been done, thus choking the waterways of natural drains.

Many huge structures like Bridges, Barrages, weirs have been built across the main Rivers, in very unscientific way, which obstruct the flood discharges at every such location, thereby creating an afflux and reduction in the velocities. The cumulative effect of all such structures may be in increasing the flood levels and stagnation of water for prolonged period.

Poor land drainage system in urban areas :primary, secondary and tertiary levels.

- **Primary level**: The slopes of individual pots in urban areas, open grounds are very gentle. Poor planning of storm water drainage system within city areas and its maintenance leads to stagnation of floods in every part of the city instead of its quick disposal by open drains through plots, its connectivity to road side gutters leading to nearby natural nalla drains. The effective primary storm drainage system can efficiently and quickly drain out the rainfall received, and be ready to receive further rains, thereby reducing the inundation periods.
- **Secondary level**: The storm water received from the urban areas and drained into the natural nallas passing through the urban areas can be kept free of any encroachments. The bed slopes of these nallas can be maintained to quickly drain out the floods received into the main river course.
- **Tertiary level:** The main river courses nearby cities can accommodate all the flood water received from its tributaries and nallas during the heavy

rainfall. Due to the encroachments in the river itself, flooding occurs inundating large areas due to local back water effects and stagnation of waters due to poor drainage systems.

e) Sedimentation in main rivers – at confluences, along meanders

Due to sudden changes in the river bed slopes and stagnation of floods for prolonged periods, sedimentation occurs in the river courses and also in natural drainages. This results in raising the river bed levels, thereby raising the flood levels and extended inundation areas. At the meanders, the sedimentation occurs in the inner side of the curves, thus deflecting the floods further to the outer sides.

f) Absence of flood absorption capacities in reservoir planning of existing dams

At the time of project planning, the reservoirs are never planned with the provision of special cushion for flood absorption neither in consideration of river regime in downstream. The dead storage is designed to accommodate the silts and live storages are planned for complete utilization of the water stored for the objectives planned throughout the year. The Reservoir Operation Schedules are designed to assure for full storages by the end of the monsoon period to fulfill these objectives. With this background and keeping in mind the historic drought prone area, the dams are not designed for the flood mitigation too. For ungated spillways or dams having fully automated gates, there is no manual control to moderate the incoming floods. However, considering all this facts, and after reviewing the reservoir operations of all the major dams during the flood event, it can be concluded that, having many constraints and in such a severe situation, the dam operators of Maharashtra tried to mitigate the flood in proper way, by keeping the outflow from the spills on lower scale.

This year, releases from reservoirs had only minor role in flood augmentation as released volumes from the reservoirs were almost similar to inflow volumes.

Such situation happened during floods 2019 in Krishna basin and it is experienced that, the dams, because of their locations, particularly in Maharashtra, had very limited role to play.

Nevertheless, it is essential to review the Rule curves of the reservoirs of major dams. They need to be meticulously drawn to create some dynamic

cushion for moderating the floods of lower return periods particularly in the early period of monsoon.

The study report on Kerala Floods of August 2018, prepared by Hydrology (S) Directorate of Central Water Commission, New Delhi, also stated that the dams in Kerala neither added to the flood nor helped in reduction of flood, during severe situation, there where most of the dams were already at FRL or close to FRL.

3.2 ANALYSIS & CONCLUSIONS (TOR No.2): To specifically clarify based on hydrological studies, whether due to Almatti & other reservoirs (back water effect) from Karnataka, create flood situation in Maharashtra.

One dimensional unsteady flow analysis, of the River Krishna from Karad to Almatti, along with its three tributaries in the State of Maharashtra viz. Yerala, Warna and Panchganga, has been done using Version 5.0 of the HEC-RAS software. Analysis has been done for the period of 25th July, 2019, 8.0 am too 17th August, 2019, 8.0 am. Steady state analysis for PMF condition was also done.

The important conclusions of this hydrodynamic analysis are:

a) Although this mathematical model study, has certain limitations, the study indicates that Almatti and Hippargi reservoirs in Karnataka and its flood operations, during the Flood Event of 2019, has not adversely affected the flood situation in the State of Maharashtra.

b) The discharge carrying capacity of the river Krishna was inadequate to accommodate the releases of Koyna dam, the contribution of tributaries and the runoff of the free catchment

c) The river Krishna flows, a near plain land, between Sangli and the State border. It is a general phenomenon that the river takes meandering course, while traversing on a plain land. Thus, River Krishna has so many curves and meanders. The velocity of the river is comparatively less while traversing curves and meanders, causing thereby more inundation on inner as well as outer sides of the curves and meanders as compared to straight reaches of the river. The Sangli city is on one of the curve of River Krishna and Kolhapur city is on the curve of river Panchganga

d) Flow stagnation in River Krishna from Sangli city to the State border, due to confluence effect. There exists series of confluences, Yerala-Krishna, Warna-

Krishna, Panchganga-Krishna and Dudhganga-Krishna within a reach of about 50 to 55 km length. At confluence points due to formation of stagnation zone velocity is reduced.

e) Generation of backwater effect in the tributaries and nallas meeting the River Krishna. Due to comparatively higher discharge in the River Krishna, the backwater effect is generated in the tributaries. The flooding in tributaries viz. Yerala, Warna, Panchganga and other nallas like Bhilwadi, Nagthane was primarily due to backwater effect of the river Krishna. The situation was further aggravated, as these tributaries could not drain out their own discharge, till the flood in the river Krishna was receded. Water was spread on the side banks of the tributaries due to pounding effect. The river Krishna was flooded for a long duration from 5th August up to 13th of August. Consequently, the backwater spread in the tributaries and on floodplains could not return back to the river course. Thus, the floodplains were under water for a prolonged time.

f) The lateral slope of the flood plains is very gentle. The flood plains are almost flat. This has resulted into spreading of flood on larger area on both the banks of the river

4. Remedial Measures for Flood Mitigation

4.1 TOR.No.3: To propose overall remedial measures to avoid such flood situation in future and /or to reduce its severity.

and

4.2 TOR No.4:To propose along with policy level measures, to recommend revised plans based on technical analysis concerning micro level issues such as dam wise reservoir operation schedules, integrated arrangement for measurement & management for river discharges, control on constructions in flood zone, emergency operation plans, standard operating procedures etc.

The remedial measures suggested by the committee as per Terms of References nos. 3 and 4 have been enumerated here. However, these have been divided as Structural and non-structural flood mitigation measures.

4.3 Structural Measures:

1. Flood Storages

Flood storages are the capacities reserved in the dams that can be used to temporarily store the flood waters. These flood storages can either be dedicated or dynamic (temporary). Storing the flood waters in such storages during the peak floods reduces the flood discharges on the downstream of the dams. Presently none of the reservoirs in the Krishna basin in Maharashtra has a provision for flood storage. Construction of flood storages may be thought of in Koyna and Warna valley, which are major contributors to the floods.

Alternatively revising the storage planning of the existing Koyna and Warna dam may also be thought of for making a provision of flood storage in the very existing dams. However, any such provision would reduce the beneficial water use from the dam to the extent of storage so provided for the floods.

2. Flood Water Diversions:

Permanent : There is Tribunal Constraints on utilization of available waters in Krishna basin in Maharashtra. Thus there is limitation on harnessing of available waters through construction of additional dams having enhanced flood absorption capacities. A permanent infrastructure may be developed / constructed to divert part of the flood water received in large quantum into the river, to the adjacent valley or another tributary having less floods during the same period. This can be done by constructing tunnels, open channels to connecting both rivers to pass the excess flood by gravity. At times, pumping out excess water may be done, if found feasible and for less quantum. This can avoid flooding if main river course, thereby reducing flood damages.

In Krishna sub-basin, such diversion of flood water can be planned by connecting tributaries of river Panchganga to the main river, again from river Panchganga to river Warna, further towards North connecting rivers Krishna (upper reaches in satara district), river Neera and even upto Ujani dam. All these tunnels can run by gravity.

While planning such projects, detailed feasibility study as well as legal provisions for diverting water to another basin can be taken into consideration.

Temporary : for temporary diversion of the floods to lessen the impact and damages that may occur due to heavy floods, no. of small ponds may be created in low lying areas adjacent to the natural drains like tributaries or nallas. When the existing natural drainage system (primary or secondary level) is unable to accommodate the floods, the excess water can be temporarily diverted into many such small ponds (existing of newly planned). When the flood in the tertiary level drains (main river courses) starts receding, this temporary stored water can be diverted back into the nallas to pass it further into main course of the river. This system, when developed can reduce the flood hazards and inundation areas and is well known as the concept of “SPONG CITIES”. During this process, part of flood water is also conserved as ground water storage. Many such small ponds are existing in Sangli, Kolhapur and Pune cities which need rejuvenation.

3. Straightening of Meanders

The area through which river Krishna passes near the interstate border between Maharashtra and Karnataka, is very flat. In this area, there are a number of meanders to the main course of the river. Straightening the meanders in river course for increasing river flow velocity and faster evacuation of floods has been attempted at a few places in the world. Before planning of such measures, physical model study is recommended.

4. Restoration of Natural drainage Channels

It was observed by the committee during the site visit that natural drainage channels have been encroached upon and blocked or diverted in cities towns and villages alike. Even in the rural areas natural drainage has been tampered with by agricultural activities. Because of blockage of natural drainages it has been observed that it takes long time, after the floods recede, for the flood waters on the river banks to drain back into the main river channel. This in turn increases the time for which the habitations affected by the flood remain submerged. It is therefore imperative that all the encroachments in the natural drainage channels are removed to restore the efficiency of natural drainage system(primary and secondary level).

River cross sections need to be restored to its natural state by removing siltation occurred; vegetation developed and by removing or rehabilitating man made encroachments.

Increasing waterway of the river cross section by widening, along with river training works are also necessary to pass designed flood discharge.

5. Raising the River banks

Low river banks need to be raised to help confine the flood discharge within the river cross section, thereby avoiding the flood water spilling out of banks and inundating over large areas. A proper hydraulic studies and design of the cross section can be done while planning such measures.

6. Control on new developments in flood plains, restricted/prohibited zones

By establishing the flood lines (red, Blue) on the city development plans of the city, strict control over the new developments in flood plains is utmost necessary. The local government authorities like Municipal Commissioner, District Collector must exercise proper control on un-lawful activity in these flood zones.

7. Rehabilitation/removing encroachments in restricted/prohibited zones

Many encroachments are existing within these flood zones in many cities which are situated nearer to the rivers. To reduce the flood hazards and minimize the damages due to floods, all these encroachments can be either removed or rehabilitated, by enforcing stringent laws by the authorities.

8. Providing Gated spillways, raising heights of existing dams

The contribution of river Bhogavati, which is tributary of river Panchganga is substantial. The spillway of Radhanagari dam, which is situated on this river is having century old fully automated gates. Hence, there is no manual control over the gates that can be used for flood moderation purposes. If these gates be converted into manually operated gated spillway, it can help in flood mitigation.

Similarly, since the existing dams have not any provision for flood absorption capacities in their reservoirs, raising heights of the existing dams, wherever possible, can be thought of, to create additional storage space for flood absorption without affecting the planned utilization of water.

4.4 Non Structural Measures

1. Integrated ROS

The reservoirs in basin should be operated in an integrated manner to ensure that the flood discharges at vulnerable locations are minimized. This will however be possible only if the Chief Engineer in charge is granted liberty to be flexible with the reservoir operation.

Integrated operation of various reservoirs viz. Koyna, Warna and other reservoirs in Panchganga valley need to be done for proper control on flood moderation activities. The releases of Koyna are very crucial in integrated flood routing.

A synchronized operation of the reservoirs is also needed, when there are synchronized floods in adjacent sub-basins over larger spatial areas.

2. Revising ROS

Though dams in Krishna valley do not have any capacity to absorb floods, flood mitigation to a very limited extent can be achieved by bringing the reservoir operation schedules in line with present precipitation trends. The impoundment of water in reservoirs may be delayed to the maximum extent possible, without compromising the success rate of meeting the demands of users, having regard to the rainfall trends in the last 10 years.

Reservoir operation schedule should provide liberty to the Chief Engineer in charge to raise the storage level in the dam even up to the maximum water level in the event of extreme floods, in order to withhold and delay the discharge of flood waters from the dam.

Alternatively revising the storage planning of the existing Koyna and Warna dam may also be thought of for making a provision of flood storage in the very existing dams. However, any such provision would reduce the beneficial water use from the dam to the extent of storage so provided for the floods.

Considering the limited discharge carrying capacity of river Krishna, particularly near Sangli, operation of Koyna reservoir during flood is very crucial. Accordingly, the reservoir operation schedule of Koyna dam needs to be revisited. Similarly, reservoir operation schedule of Warna dam can also be updated, considering the existing water utilization pattern.

3. Revising the flood lines

It is necessary to take into account the actual observed highest flood levels and also the back water effects of river Krishna and its tributaries, while revising the flood lines to regulate the future developments accordingly.

4. Robust Rainfall forecasting

At present the rainfall forecasts disseminated by IMD are mostly qualitative in nature. IMD issues Extended Range Forecast (ERF) every Thursday for next 4 weeks at all India level which includes winds/circulations, Rainfall and maximum-minimum temperatures likely. Along with this, IMD also issues 5 days district wise weather forecast and severe weather warnings. The infrastructure of observatories of IMD, if properly upgraded to give quantitative forecast clearly indicating geographical spread and quantum of the rainfall over the next two days with a good degree of accuracy can be very much useful. Such actionable information received from IMD, along with the information and forecast can be used as the decision tool by the reservoir operators. The forecast can improve the quality and give width and depth of information about the river basin wise rainfall. There is a need of rainfall information and forecasts at block level and smaller units. Such forecasts would greatly enhance ability of the state apparatus to manage floods and limit the damages. Short range Now cast issued by IMD for next 4, 6 hrs based on radar and satellite products and NWP guidance also to be incorporated in real time operations and in DSS.

Further densification of rainfall observations in flood prone areas along with radar network jointly will improve the operational capabilities of IMD and also Users.

Along with qualitative rainfall forecast in different ranges as per SOP of IMD, the need for quantitative forecast would be added advantage for decision making of Reservoir Operations.

5. Flood forecasting model

Water resources department has established a system for real-time data acquisition and decision support. This system is not fully operational because of certain shortcomings. It is recommended that this model must be fully operationalised and integrated with rainfall forecasting model to be evolved. The capabilities in the model should be built up to make possible reasonably accurate river flow forecast two days in advance.

MoES has already developed flood forecasting system as Integrated Flood Warning System (I FLOWS) made by NCCR Chennai and IMD. Such state-of-the-art system would be beneficial as guidance for the authorities.

6. Vulnerability maps

Inundation maps for various water levels at vulnerable locations be prepared with precision to help the flood relief operations in future. Digital elevation models of sufficiently fine resolution should be used to prepare inundation maps corresponding to various flood levels in the river for all the vulnerable areas. The inundation area should be marked on village maps or town maps. These maps would help in taking advance measures like evacuation of population likely to be affected by floods.

Such real time and forecasted inundation maps at very high resolution can be generated by using flood models I FLOWS mentioned above.

7. Advanced warning mechanism & Flood zoning

Mechanism to issue advance warning be developed and installed at proper locations. The people living in flood prone areas can get timely warnings regarding flood situations, sufficiently well in advance to save their lives and the loss of the properties. Water resources department should frame a protocol for disseminating the forecast of flood to appropriate authorities for taking advance action.

Flood Zone encroachments:

It was observed by the committee during its site visit that there are large scale constructions in prohibited and restricted flood zones. The constructions and habitations in flood zones largely increase the number of persons affected in the event of high floods and put severe stress on the relief apparatus of the government during evacuation of the affected. It is recommended that flood zone regulations must be strictly enforced to remove encroachments and to ensure that there are no fresh encroachments.

HEC-RAS, capability can be used for simulating inundation with the help of reservoir releases and rainfall data (available and forecasted) and automated system to send alert signals to all concerned need to be developed.

8. Technology updation, Institutional Arrangements and Policy measures

All the latest and advance technologies available can be deployed, right from weather forecasting (NOWCAST for precision forecast), real time data acquisition systems, Real time flood forecast and decision support systems, vulnerability maps by remote sensing surveys, advanced early warning systems, wel-equipped disaster management units ready with evacuation plans, trainings and mock drills for volunteers, government staff as well as for the public.

- **General Policy Measures proposed**
  a. To Preserve, Protect, Restore Source regions of Streams, Tributaries & Rivers
  b. To Preserve and Restore Floodplains
  c. Preparedness for Floods
  d. To Prevent Adverse Impacts and Illegal Uses in the Floodplains
  e. Prevent Adverse Impacts from Developments and Encroachments

- **Policy measures for Flood Mitigation and Climate change**

  The state of Maharashtra given its large population, socio-economic and cultural diversity, long coastline and multiple agro-climatic zones, is acutely vulnerable to climate change. The state already faces threats from recurrent droughts, heavy precipitation, floods, heat waves and sea level rise. Therefore, adaptation and mitigation strategies that can address these intensified impacts and build future resilience to climate change need to form a vital part of the overall development strategy of the State. Keeping this in view, Government of Maharashtra prepared the Maharashtra State Action

Plan on Climate Change (MSAPCC) in 2014. The plan is consistent with the structure and strategies of the National Action Plan on Climate Change. A Handbook was also published in February 2018, highlighting the key portions of the Action Plan.

In concurrence with this, following specific policies are recommended as below :

1. Formulation of Basin level Flood Management Mechanism, an Authority having members of all concerned States Engineers, including a member from Central Water Commission, for accurate and fast data exchanges during the floods, so as to minimise the adverse impacts of floods.
2. Creating Hazard Mitigation Council (HMC) –A public private partnership designed to reduce economic and social costs of Natural hazards.
3. Invest in Natural hazard mitigation as a matter of policy- ongoing activity before disasters occur.
4. Support mitigation activities that will increase resilience of communities by increasing knowledge & awareness at local level for public awareness, preparedness and participation during floods.
5. Ecological Restoration of Natural water resources for mitigation of floods for longer periods.
6. A combination of measures – structured and non structured options to improve the basin's resiliency – its capacity to prepare for and recover from flooding in the future.
7. All lotic waterways – streams, Seasonal drainages, tributaries and rivers to be marked as per their natural flow and flood regimes.
8. In order to implement the strategies recommended for climate change adaptation under the Maharashtra State Action Plan and Policy on Climate Change (MSAPCC), the Government of Maharashtra has released the **Maharashtra State Climate Change Adaptation Policy** on October 25, 2017. Being the pioneering Policy, the first of its kind undertaken by a State in India, it identifies and priorities specific strategies indicated within the MSAPCC, and also mandates sector wise Specific Action Plan (including Monitoring and Evaluation) for Forests and Ecosystems, Water Resources, Agriculture, Energy, Health, Public Works, Disaster Management, Rural Development and Urban Development.

   The provisions in the policy may be implemented immediately.

5.0 Recommendations of the Committee:

With details studies done by various Institutes, Experts and after the Analysis of causes thereof, the makes following recommendations for the immediate implementation -

A. Preparedness, Robust systems for forecasting
B. Flood resilient infrastructure
C. Operational Reforms and Disaster management
D. Policy reforms/Legal provisions
E. Other Recommendations

A. Preparedness, Robust systems for forecasting

1. Advanced Weather Forecasting Systems

It would be important to include globally available numerical weather forecasts products, in the RTSF&ROS. Incorporation of appropriate weather forecast product including ensemble forecasts, in the Model available, may be necessary.

Presently various organisations and Institutes are practically working in isolation, even though all are located in Pune. A proper co-ordination system has to be set up amongst these 4/5 organisations, which are at Pune viz. IMD, IITM, CWPRS, CWC, MKVDC etc. A committee can be formed for this purpose and if the basic data is made available, it can be analyzed accurately. Establishment of one FMO in Maharashtra state can also be explored with the concerned agencies like IMD, CWC and GoM.

Now, FMO wise QPF information is available on the IMD website, 'Climate Services to Water Sector by IMD'. Water being the main sector, it is connected to other sectors like Agriculture and Food Sector, Disaster, Energy, Health, etc. IMD is generating 1 week to 4 weekly ERF of rainfall and publishing it on IMD website. The sub basins are considered as per CWC Classification and IMD website has 2 weeks extended rainfall forecast with good confidence level. But, to enhance accuracy, smaller sub-basins or catchment of a project, may be considered as a least unit.

2. Real Time Decision Support System for Flood Forecasting

Integrated Flood Management is the key word for better Flood Mitigation Measures and the Adaptation. Though complete immunity is not possible or feasible, but the impact can certainly be minimized. The IFM includes structural measures like physical restoration of waterways of all the rivers and tributaries with an aim to prevent flood waters from reaching potential damage areas. The non-structural measures can strive to keep people away from flood waters. These use knowledge, practice and agreements to reduce the risks and impacts through training, awareness and capacity building.

3. **Model study**

   This Committee has done hydrodynamic Analysis by Mathematical modeling with certain boundary conditions. But, the detailed Analysis with physical modeling of the Krishna River reaches, including all the structures like KT weirs, barrages, bridges etc. across the river need to be done, to ascertain the cumulative effect (affluxes) of such structures. There are many no. of structures, specially in Karnataka region upto Almatti dam, which may be the causes to create the flood situations locally and to large extent. A physical model study may also be required to be done to ascertain this phenomenon.

B. Flood Resilient Infrastructure

The floods are caused by various reasons such as – inadequate capacity of the rivers to contain within their banks the high flows brought down from the upper catchment areas following heavy rainfall, tendency to occupy the flood plains, varying rainfall distribution leads to flooding of non-traditional areas, areas with poor drainage facilities, excess irrigation water applied to command areas, increased ground water level etc.

The problem gets further aggravated by the factors like silting of river beds, reduction in the carrying capacity of river channels, erosion of beds and banks leading to changes in river courses, obstructions to flow due to structures built across the rivers, thrown away debris, synchronisation of floods in the main and tributary rivers and retardation at confluences.

The obstruction to the free flow of the floods through the rivers due to the big structures built across the river cross section is the major cause to create flood locally, thereby increasing the inundation areas on big scale. Though the structures have been designed to provide the waterway for regular max. discharges of the river, these structures become a blockage during the floods, by constricting the clear waterway for bigger discharges, by blocking the floating debris and consequently reducing the waterway further, by creating affluxes, by reducing the velocities, by creating the siltation of the floating sediments and thereby increasing the flood level in the locality which create larger inundation areas.

Thus, it is recommended that, the hydraulic audit of all the existing structures built across the rivers be done, urgently and appropriate actions be taken based on these reports.

The following structural measures can be taken immediately, to reduce further flood hazards.

1. **Restoration of natural waterways, desilting**

   The natural waterways of the river and tributary courses get contracted or reduced, as years pass by, due to various reasons. The natural cross-section of the river which carries maximum flood water must be restored by maintaining the river bed slopes, bottom widths as well as bank slopes. The banks may be raised to accommodate uniform discharge; slopes may be stabilized by pitching at curves. The desilting of the bed of the river can enhance to flow ability of the river, and also help in reducing the peak flood levels.

   The primary level storm water drainage system within the city area must be kept clean and maintained before every monsoon. This area can quickly drain out the storm water, and be ready to receive further rains, without any flooding or stagnation. The secondary level nalla and natural drainage system that carry storm water from urban areas upto the main river, must be free from any kind of encroachments. The tertiary level drains, i.e. main rivers can not be obstructed by the structures across it, like bridges, weirs etc., which may reduce the waterways, thereby increasing the affluxes.

2. **Encroachments removal**

   The flood plains be identified, the flood lines be clearly marked on ground demarking the prohibited and restricted zones clearly. All encroachments in the prohibited zones can be rehabilitated or removed, as far as possible.

3. **Straightening of River Meanders, direct connections**

   The synchronisation of floods in the main and tributary rivers and retardation at confluences is the serious phenomenon observed during the Floods 2019, in Krishna basin, specially between the Sangli and State border reach of the River, where the big tributaries like Warna and Panchganga meet Krishna. The existing meanders in this reach aggravated the flood situation by retarding the flood mass. During the peak period of floods, it was observed that the flood water was naturally finding its own way, through a low lying channel, diverting the meanders.

By extensive model studies and the feasibility, such projects be taken into hand to connect such meanders, which can enable to divert part of flood directly into next reach, increase velocity of flow and reduce stagnation period of flooding, thus overall reduce the flood hazards.

4. **Construction of flood dams, temporary flood storage tanks**

   The option of constructing new flood dams or by converting existing tanks alongside the rivers and tributaries for temporarily storing the excess flood helps in reducing the peak discharges into the rivers. The floods dams can be planned into the upstream catchments of existing major dams or at the downstream free catchments also.

   Open low lying areas or small tanks within urban areas can also be converted into such flood dams, which enable to have the concept of sponge cities. This can create more robust flood resilient infrastructure.

5. **Diversion of excess floods to adjacent sub-basin, Tributaries**

   On large scale, considering the basin level flood hazard potential and the available sub-basins adjacent to it, a mega project can also be planned. After geographical study of the Krishna basin, the excess flood water in river Krishna and its tributaries can be easily diverted, by gravity into the adjacent Bhima sub-basin. Both purposes can be served, of flood and drought management by planning such mega project. The techno-economical feasibility and the legal restrictions of the interstate water dispute tribunal can be taken into consideration before taking up such projects.

   On small scale, in order to reduce intensity of floods within Krishna Basin, during floods / monsoon period, priority can be given to optimum utilisation of potential envisaged on major LIS with due consideration of KWDT provision.

C. Operational Reforms and Disaster Management

1. **Integrated / Synchronised Reservoir Management(ROS)**

   Though floods are natural hazard, those are not always caused by natural reasons. A lack of effective reservoir management may increase the severity of flood damages. Apart from other natural and physical reasons to cause the floods, synchronisation of floods in the main and tributary rivers and its retardation at the confluences can be an important phenomenon, which can be talked with better IRM only.

   About the Reservoir Operation Schedules, Integrated ROS, Real Time flood forecasting, Basin simulation and synchronised reservoir management were studied in detail by the Committee set up by the WRD in 2007, under the chairmanship of Er. Vadnere, Rtd. Prin. Secretary, WRD, regarding precise determination of reservoir releases during emergency situation in the state of

Maharashtra. This committee has given 29 recommendations. These can be given due consideration for issuing the proper guidelines to effect efficient operations of the reservoirs.

When there is situation of having Major dam locations, in parallel, situated in adjacent sub-basins of the tributaries getting merged almost simultaneously into the main river (which is typical in Maharashtra), the Integrated approach with synchronized reservoir operations becomes imperative. To improve the accuracy, and availing various options for proper decision making during the floods, a software can be developed as a decision support system. The issue of guidelines for the preparation of updated Reservoir Operation Schedules, Reservoir operations, and Integrated Reservoir operations at sub-basin / basin level, Basin Simulation and Real Time Flood Forecasting can help giving confidence to Dam Operators to perform well during flood situations.

The concepts of –

a. Change over from Rigid to semi-rigid operations
b. Normal Reservoir Operations and switching over to emergency flood moderation module
c. Preparation of Regulation Schedules (ROS) as per revised guidelines
d. Revision of Dam Safety Manual
e. Integrated Reservoir Operation, Basin Simulation and Real Time Flood Forecasting Techniques
f. Marking of Blue, red and Yellow lines on maps and also on the Ground
g. Restriction of Settlements and Economic Activities in flood prone areas

must strictly be implemented.

2. Basin level Integrated Flood Management Mechanism

For this to achieve, there has to be proper synchronisation between CWC and Dam authorities/ States, at Basin level with regard to Inflow forecasting for real time reservoir management, Outflow data by Dam authorities, Rule curve updation and to follow the during dam operations.

Balancing between conflicting objectives – between conservational benefits to wider interest of the flood management, must be done. The flood management, reservoir management, flood forecasting, disaster management, dam safety: all are public interest activities. Hence all information must remain in public domain, all the time.

The human induced factors need to be addressed in management of floods – viz. Improved forecasting and timely dissemination of early warning systems, better planning, Dedicated or Dynamic water cushioning, Timings of filling of dams/release of water, co-ordination between CWC, IMD and States, Inter-state co-ordination at Basin level. Every dam can help moderate floods. In fact, this is true of every storage option including soil moisture, aquifers, local water systems, wetlands, forests as well as other storages. Every reservoir, when not operated properly, is a potential source of the flood disaster for the downstream areas.

3. **Review of Reservoir Operations**

The historical data as well as the information of decisions taken with due reasons during flood event every year, must be preserved in the form of documentation of a standard Report, at every dam location. The dam operators can submit these reports at the end of the monsoon, to the higher authorities for their review as well as to add their comments based on the typical decisions taken in particular situation. These reports can be preserved as historical data then, which can be useful to the next dam operators to understand the local geography and the hydrological variations in a better way. Annual audit of such flood reports (with key indicators) may also be done by appropriate authorities.

4. **Advanced Disaster Management Preparations**

Inundation maps for various water levels at vulnerable locations be prepared with precision to help the flood relief operations in future. Digital elevation models of sufficiently fine resolution should be used to prepare inundation maps corresponding to various flood levels in the river for all the vulnerable areas. The inundation area should be marked on village maps or town maps.

Advanced warning mechanism to issue advance warning be developed and installed at proper locations. The people living in flood prone areas can get timely warnings regarding flood situations, sufficiently well in advance to save their lives and the loss of the properties.

Peoples participation in Disaster Management need to be increased by way of Education, mock drills, volunteering etc. to control the panic among the people and also to increase their co-operation during crisis.

D. Policy Reforms / Legal Provisions

1. **Flood Plains Management Act**

National Disaster Management Authority (NDMA) has released national Guidelines on Management of Floods and Management of Urban flooding, in 2008 and 2010. As per NDMA guidelines, as appropriate legal framework to be developed by the state govts/RBAs so as to make it mandatory for obtaining clearance for the plans for constructions of the infrastructure in flood prone zones, from state WRD with respect to their safety against floods and effects thereof on the vulnerability of the area to floods and drainage congestion, who will process the cases in affixed time frame.

Although, there exist, Government Resolutions, for defining the flood lines and restricting / prohibiting development / land use, within the flood lines, it has failed to achieve the desired effect. Snap shots given by MRSAC[1], indicate that flood plains have been severely affected by the urbanization particularly in Sangli and Kolhapur districts. In last decade, unplanned urbanization, illegal and informal settlements have been developed in low laying areas which faced flood inundation problems during the Flood event. This urbanization has also blocked the natural drains. In rural areas, natural drains have been occupied by agricultural fields after necessary fillings. Encroachments on natural drains and flood plains, reduce the discharge carrying capacity of the rivers and thereby imposes additional constraints on reservoir operations. Therefore, GoM may establish Flood Zoning Authority(s) for Sangli and Kolhapur districts, with necessary powers, functions and duties, for regulating the land use in the flood plains and restricting/prohibiting/removing obstructions to the natural drainage. It is the emergent need of the hour.

2. Appropriate Amendment can be done in Maharashtra Irrigation Act, 1976, Clause 16, for the demarcation of the Flood Plain Zones.

   Frequency based Flood Inundation Maps, for various water levels be prepared by the Chief Engineers, for the flood hazardous stretches of the rivers, to evolve effective flood management strategies and planning for the developments.

   Appropriate notifications can be issued for prohibitive and restrictive zones of the development, in furtherance of the execution of these amended provisions. Accordingly, Planning Authorities and the Urban/Rural Development departments can modify their Development Control Rules.

3. **Flood and Drought Management Reforms.**

   On the lines as national level, new Organisation be formed for our State, at River Basin Level, which can be named as Flood and Drought Management

[1] Annexure D, Experts Study Reports, Report by MRSAC : Snap Shots 3.1 to 3.13.

Organisation. This FDMO may comprise of members from all water stake holders and can act as an advisory technical Institute.

4. **Hazard Mitigation Council (HMC)** –A public private partnership designed to reduce economic and social costs of Natural hazards may be created.
5. **Natural hazard mitigation** as a matter of policy –There is need to invest for flood mitigation activities before disasters occur.
6. **Support mitigation activities** that can increase resilience of communities by increasing knowledge & awareness at local level.
7. **System for public awareness**, preparedness and participation during floods can be created.

E. Other Recommendations –

1. Densification of Real Time Data Acquisition System in Dam catchments and Free catchments to the downstream.
2. The Rating curves (Stage-Discharge Curves) at various locations be calibrated with Standard Procedures. Those must be reconciled with the curves available at CWC discharge gauging stations. Also, all the existing river gauges, rain-gauge stations be planned to be upgraded properly along with new installations in upstream or downstream free catchments, if necessary so as to get accurate and more data possible, for the flood mitigation.
3. Establishment of small range Doppler Radar (X band) for forecasting of floods from a particular cloud or a cluster, of very short period of 2 to 6 hours, with high accuracy.
4. Integration of all observational networks of different stake holders on a single GIS based platform, in real time can largely improve forecast validation and also weather forecast model outputs.
5. Development of Flood Forecasting Models for important flood prone cities like Sangli, Satara, Kolhapur and Pune.
6. Upgradation of the existing RTSF & ROS (RTDSS model) in Mike-11 operations, incorporating appropriate weather forecast products.
7. After receiving Alert from IMD, help of NRSC and MRSAC to acquire microwave satellite data, for specific areas.
8. Long term Ecological measures to improve the river’s capacity to manage her floods.
9. Long term Agri- Environment Measures in Rural areas

10. Comprehensive Landscape Management Plan, by Mapping the natural drainage system in the flood prone areas, with Ecological View in a Drainage Basin for Mitigation of Excessive Flooding in Low Laying Areas
11. To create a series of wetlands / detention basins in the ideal location of hydrologic floodplains and topographic floodplains to mitigate floods.
12. Flood risk assessment and Drainage strategy for all major development proposals.
13. A combination of measures – structured and non structured options will improve the basin's resiliency – its capacity to prepare for and recover from flooding in the future.
14. Upgraded Reservoir Operation Schedules for Existing dams
15. A detailed record of all the rainfall data collected as well as the details of all Reservoir Operations done by the Dam Operators, along with the decisions taken must be kept and be submitted to the higher authorities, by the end of Monsoon. The Chief Engineer, as a Basin Authority can go through the operations made, may point out some suggestions for betterment. The record can be preserved permanently, for any future reference. It can also be useful for the new Dam operator, as a standard operating procedure for that particular reservoir.
16. A detailed survey of all the main Rivers and its Tributaries be done immediately, with L-sections, cross sections so as to ascertain the real flood carrying capacities of the rivers.
17. A pre-monsoon Inspection of the rivers and tributaries, along the alignments from its origin to end, be done by a committee comprising the members of WRD, Revenue, Local bodies, Developmental authorities, members of Disaster Management Cell along with peoples participation.
18. A master plan for the Major cities, which are flood prone and more vulnerable viz. Sangli, Kolhapur and many others be prepared after the detailed city specific surveys, considering the reasons and the solutions thereof.

Chapter 1 – Introduction

The ToR given to the expert study committee vide GR dated 23 August 2019 is to find out the reasons behind the flood situations in Bhima and Krishna Valley in the year 2019. Considering the voluminous work and the huge data compilation required for the studies, it was decided to prepare separate reports for each basin. Considering the severity and frequency of the floods and the extent of damages, here, initially the study report of flood in Krishna Sub Basin within the state of Maharashtra has been done by this committee and is being presented in three volumes. The study report for Bhima Sub Basin will be prepared and presented separately in due course.

During month of July & August 2019, heavy flood situation was created in Maharashtra due to occurrence of intense rainfall. Especially Sangli & Kolhapur district in Krishna sub basin experienced extreme floods of long durations. Heavy losses to life, property & crops etc had been reported. Different opinion at various levels were put forth concerning these flood events. Sangli & Kolhapur districts faced heavy flood situations in past also & floods of 2005 & 2006 were noteworthy. However, 2019 flood event was comparatively much more severe which lasted more than a week & losses experienced were also on higher scale. For optimal use of water in Krishna basin & to mitigate flood situations, state of Andhra, Karnataka & Maharashtra have constructed number of major projects. Adjacent to Maharashtra, just below state boundary, Karnataka constructed Almatti dam & Hippargi barrage. Almatti dam is 269 km downstream of Sangli city & 211 Km form Rajapur KT weir at State border. Present FRL of Almatti dam is 519.6 m which is allowed to be raised to 524.25 m by KWDT, in future. Number of technical issues surfaced out concerning height of Almatti dam & possible submergence in Maharashtra during flood situations in the recent years. Flood management have been facing many challenges presently due to frequent occurrence of floods with greater intensities & growth of rapid urbanization surrounding rivers. Though Maharashtra is not known as flood prone area, recent frequent floods & consequent losses necessitated to acquire better technical knowledge & work out remedial measures with the help of modernized techniques to mitigate floods.

It is thus necessary to find out causes of occurrence of floods with detailed technical analysis & overall in-depth studies, to ultimately forecast & mitigate floods impinging on different valleys. As such various experts related to this issue need to study and work out overall strategies, decide workable policies & pragmatic remedial measures in flood management. The group of experts from various organizations like Maharashtra Remote Sensing Application Centre (MRSAC), Indian Meteorological Department (IMD), Indian Institute of Tropical Meteorology (IITM), Indian Institute of Technology (IIT), Maharashtra Water Resources Regulatory Authority (MWRRA), Water Resources Department (WRD), Hydrologists, experienced flood management personnel and experienced officials executing Real Time Flood Forecasting System can certainly shoulder this responsibility.

1.1 Formation of Expert Committee

Hon. Chief Minister of Maharashtra during Cabinet Sub Committee meeting, while discussing flood disaster mitigation plan & overall issues concerning emergent situation caused by floods & related measures thereof, decided to appoint Expert Committee. As such Expert Committee of 11 members was appointed on 23rd August, 2019 to “Study reasons of occurrence of 2019 Floods in Bhima & Krishna basin & propose technical measures to be undertaken in future” (**Annexure -1A;1B in Marathi & English**).

1.2 Terms of References

The Terms of References (TOR) given to Committee is as follows:

1. To carry out in-depth technical analysis and investigate causes of 2019 flood situation experienced in Bhima and Krishna basin using modern techniques.
2. To specifically clarify based on hydrological studies, whether due to Almatti & other reservoirs (back water effect) from Karnataka, create flood situation in Maharashtra.
3. To propose strong overall remedial measures to avoid such flood situation in future and or to reduce its severity.
4. To propose along with policy level measures, to recommend revised plans based on technical analysis concerning micro level issues such as dam wise reservoir operation schedules, integrated arrangement for measurement & management for river discharges, control on constructions in flood zone, emergency operation plans, standard operating procedures etc.

1.3 Details of Krishna Basin

Krishna River is the second largest river in Peninsular India. It originates in the Mahadev range of the Western Ghats near Mahabaleshwar, at an altitude of 1337 m above msl (13°.7'/N to 19°20 '/N and 73° 22'/E to 81°10'/E) and flows through Maharashtra, Karnataka, Andhra Pradesh and drops into the Bay of Bengal in Andhra Pradesh state. Krishna Basin is having a total area of 2.59 Lakh sq. km, which is nearly 8% of the total geographical area of the country. The total length of river is about 1400 km. The basin is roughly triangular in shape and is bounded by Balaghat range on the north, by the Eastern Ghats on the south and the east by the Western Ghats on the west. Map of Krishna Basin of India is shown in Annexure "maps" (Map No.1)

Krishna Godavari Commission had divided entire Krishna basin into 12 sub divisions & they are designated as K1 to K12 spread in all states i.e. Maharashtra, Karnataka & Andhra Pradesh (now divided into Andhra & Telangana). 5 basins K1,K2,K3, K5 & K6 are spread in Maharashtra state.

Water availability and allocation:

Krishna Water Dispute Tribunal (KWDT-1) distributed 75% dependable basin flow of 58328 Mm3 (2060TMC) among three riparian states namely Maharashtra, Karnataka & Andhra. Later Krishna Water Disputes Tribunal (KWDT-II) reviewed & distributed 75% and 65% dependable flows among three states (Final decision of KWDT-II has not been notified in gazette since the reports are challenged in Supreme Court by Andhra, Karnataka & newly formed Telangana state). State wise initially Maharashtra was allocated 16564 Mm3 (585 TMC) water at 75% dependability by KWDT-I. Later KWDT-II reallocated Average 18858 Mm3 (666TMC) water with consideration of 65% dependability. Certain restrictions were imposed in Ghataprabha, Bhima & Krishna basin on use & diversion for power etc. in Maharashtra.

Table 1.1: State wise Allocation of water as per KWDT-II

Dependability	75% (KWDT-I) in Mm3 (TMC)	65% Dependable Flow in Mm3 (TMC)			Average (KWDT) in Mm3 (TMC)	
		Additional	Minimum	Total	Additional	Total
Maharashtra	16564 (585)	1218 (43)	85 (3)	17867 (631)	991 (35)	18858 (666)
Karnataka	20783 (734)	1727 (61)	198 (7)	22708 (802)	2973 (105)	25682 (907)
Andhra Pradesh	22963 (811)	1217 (43)	170 (6)	24238 (856)	4106 (145)	28457 (1005)
Total	60310 (2130)	4162 (147)	453 (16)	64926 (2293)	8070 (285)	72996 (2578)

Maharashtra being upper riparian state, maximum rainfall / runoff occurs (due to Sahyadri Ghat area) in comparison to other D/S States. Total available water at 75% dependability in the entire Krishna basin in India is around 58,328 Mm^3 (2060TMC), whereas Maharashtra generate around 29,300 Mm^3 (1035 TMC) and yearly allocation to Maharashtra is 15,856 Mm^3 (560 TMC). It is thus pertinent to note that out of total available water nearly 50% volume of water is alone produced in Maharashtra. After using 27% share allocation, remaining 23% quantum i.e. 13,444 Mm^3 (475 TMC) normally flow down to Kolhapur & Sangli town situated near state border causing occurrence of frequent floods around state border of Maharashtra. Intermittent very high rainfall of short durations also cause extreme floods.

1.4 Krishna basin in Maharashtra

In Maharashtra Krishna basin drains 69,425 sq.km which is 28.61% of total Krishna basin. There are 5 basins in Maharashtra i.e. Godavari, Krishna, Narmada, Tapi & Western flowing rivers. The drainage area of Krishna sub basin in Maharashtra is 69,425 sq.km which is 26.81% of area of Maharashtra. It is the second largest basin in Maharashtra. The river traverses through length of 301 km in this state & rises in the Mahadev range of the Western ghats near Mahabaleshwar at an altitude of 1395 m above sea level. Krishna Basin map of Maharashtra state is shown in Annexure “maps” **(Map No. 3)**

1.4.1 Krishna & Bhima sub basin in Maharashtra

Krishna basin is having two major broad sub basins namely Krishna & Bhima. They are further divided in 5 sub basins as K1, K2, K3, K5 & K6. Following table 1.2 shows details of areas.

Table 1.2: Sub basin wise area in Maharashtra.

Broad sub basin	Sub Basin	Sub Basin-Wise Area in Sq. Km	Area Sq. Km.
Krishna	Upper Krishna sub basin (K-1)	17128	20526
	Middle Krishna Sub basin (K-2)	1388	
	Ghatprabha (K-3)	2010	
Bhima	Upper Bhima Sub basin (K-5)	45335	48999
	Lower Bhima Sub basin (K-6)	3564	
	Total	**69425**	**69425**

It is seen that Krishna sub basin comprise of K1, K2 & K3 having total 20,526 sq km area, while as Bhima sub basin comprise of K5 & K6 having total 48,899 sq km area. In Maharashtra Krishna basin is spread over in four administrative regions viz. Pune, Nashik, Aurangabad & Kokan (negligible) as seen below in table 1.3.

Table 1.3: Region wise Area

Sr. No	Sub Basin	Total Area in Sq.Km.	Region wise area in Sq. Km			
			Pune	Nasik	Aurangabad	Kokan
1	Upper Krishna sub basin (K-1)	17128	17128	0	0	0
2	Middle Krishna Sub basin (K-2)	1388	1388	0	0	0
3	Ghatprabha (K-3)	2010	1893	0	0	117
4	Upper Bhima Sub basin (K-5)	45335	34598	6180	4557	0
5	Lower Bhima Sub basin (K-6)	3564	1997	0	1567	0
	Total	**69425**	**57004**	**6180**	**6124**	**117**

Major area of Krishna basin in Maharashtra (82%) is spread over in Pune, Satara, Sangli, Kolhapur & Solapur district. Remaining 18% is spreaded in Ahmednagar, Osmanabad, Beed & negligible in Sindhudurg district.

1.4.2 Status of Irrigation Projects in Maharashtra.

To create ultimate irrigation potential in Krishna basin from allocated water to Maharashtra, the details of completed & ongoing state sector projects in shown in Table 1.4 below.

Table 1.4: Status of State sector Irrigation Projects

Sr. No.	Sub Basin	No. of Completed Projects				No. of Projects under construction				Total No of Projects
		Major	Med	Minor	LIS	Major	Med	Minor	LIS	
1	**K-1**	04	04	101	00	05	11	23	07	155
2	**K-2**	00	02	18	00	00	00	00	02	22
3	**K-3**	00	03	21	00	00	03	06	00	33
4	**K-5**	07	32	651	01	08	04	15	13	731
5	**K-6**	00	07	112	00	00	00	09	00	128
	Total	**11**	**48**	**903**	**01**	**13**	**18**	**53**	**22**	**1069**

Thus, in all 210 Major / Medium / Minor / L.S. Projects are in place in Krishna sub basin and 859 projects are constructed in Bhima Sub Basin.

1.5 Krishna sub-basin system

Terms of reference (TOR) relates to Krishna & Bhima sub basins in Maharashtra. Initially, entire study is proposed to be undertaken for Krishna sub basin (K-1, K-2, K-3) as such all further chapters relates to Krishna sub basin.

Krishna sub Basin in Maharashtra (K-1, K-2, K-3) having 20,526 Sq.km catchment area has following broad features.

Broad features of Krishna sub basin are shown below:

Upper Krishna K-1- The river Krishna from source to the confluence with it of the Dudhganga ; the sub-basin includes the catchment area of the river Krishna and of all its tributaries which fall into the Krishna in this reach up to and including the Dudhganga. Area of sub basin is 17128 Sq.km. Total area of this sub basin lies in Pune region.

Middle Krishna K-2- The river Krishna, from its confluence with the Dudhganga to its confluence with the Bhima; the sub-basin includes the direct catchment of the Krishna in this reach as well as of all its tributaries out falling in this reach, except that of the Ghatprabha and of the Malaprabha (K-3 & K-4). Area of sub basin is 1388 Sq.km. Total area of this sub basin lies in Pune region.

Ghataprabha K-3- The entire catchment of the Ghataprabha from source to its confluence with the Krishna, including the Catchment area of the Hiranyakeshi, the Markandeya and other tributaries of the Ghataprabha. Area of sub basin is 2010 Sq.km. Major area of this sub basin is from Pune region. Due to restriction on water use, major flow goes down to Karnataka state leaving hardly any scope for flood mitigation in Maharashtra.

1.5.1 Krishna River System in the Maharashtra and Karnataka State

From its source, the Krishna speeds south-wards skirting the eastern spurs of the hills through the districts of Satara, Sangli & Kolhapur in Maharashtra. After passing the dam sites for the Krishna Projects at Dhom & Borkhal, the Krishna receives the water of the Venna on the right bank, 72 km from its source at Mahuli near Satara city. Lower down, the river is joined by the Urmodi and the Tarali on the right bank.

Flowing past the Khodshi weir from which the Krishna canal takes off, the Krishna is joined on the right bank by the Koyna of which the Wang is a tributary, at km 37 at an elevation of 760 mtrs Lower down, the Krishna receives the waters of the Yerala from the left About 244 km from its source near Sangli, the Krishna receives on the right bank the waters of the Warna of which the Kadvi is tributary.

Near Kurundvad, at about 295 km, the Krishna receives on its right bank the united water of the Panchaganga, that is from tributaries namely Kasari, Kumbhi, Bhogavathi, Tulshi and the Dhamni. At about 306 km, the Krishna is joined on the right bank by the Dudhganga of which the Vedganga is a tributary. About 306 km from its source and at an altitude of about 519 mtrs, the Krishna enters Karnataka State. L- section of Krishna river showing various tributaries in Maharashtra is shown in Annexure “Maps” (**Map no.7**)

The river leaves the heavy rainfall zone and turns east. In the run of 300 km within Maharashtra, the bed fall is 2.86 m per km, the fall up to km 137 being steeper at the rate of 4.20 m per km. After flowing for some distance in Karnataka, the Krishna is joined by the Agrani on the left bank, the Ghataprabha on the right bank at 507 km and the Malaprabha on the right bank at 542 km.

The Junction of the Malaprabha is between Almatti and narayanpur, the dam sites of the Upper Krishna Project. At Jaldurga falls below narayanpur, the Krishna drops about 121.95m in about 5 km from the table land of the Deccan platue to the alluvial lands of Raichur district Lower down, the Krishna receives the waters of the Don on the left bank and at about 788 km, the waters of the Bhima on the left bank at an altitude of 342.98 m. in the run of 773 kms within Karnataka, the bed fall is 0.40 m per km. Annexure “Maps” (**Map no. 4**) shows Krishna river from origin (Maharashtra) upto Almatti dam (Karnataka) which is 211 km D/S of Maharashtra State border. Geography of river system of Krishna sub basin in Maharashtra is shown in **Annexure no. 2A**. **Map no. 7** shows elevation of all river/ tributaries at their origin and confluence points. It is seen that Krishna, Koyna, Warna, Panchganga, Dudhganga, Vedganga & Ghatprabha rivers along with 22 tributaries are spread over in K-1, K-2, K-3 sub basins. Entire river system flows down from origin at 1395 m level till 519 m level at state border of Maharashtra. General slope Map (**Map no.5**) of Krishna sub basin prepared using remote sensing techniques is also annexed for overall elevation view.

1.6 Flood Mitigation in Krishna Sub Basin

In Krishna Sub basin, Satara, Sangli & Kolhapur districts normally experience floods since their location is adjacent to Krishna & Panchaganga rivers, which carry sizable flows due to frequent very high & intense rainfall in ghat areas above series of dams constructed. The rise in the flood is also attributed by heavy rainfall in free catchment below dams and U/s of these districts. The dam constructed in upper Krishna sub Basin (K1) and middle Krishna (K2) mainly play role for flood mitigation in Maharashtra while as flows from Ghatprabha basin (K3) mostly drains in Karnataka, hence not considered in this report.

The list of 9 major and 13 medium Irrigation Projects serving for flood mitigation measures in the Krishna Sub Basin ((K-1, K-2) are listed in **Annexure 2B**. The list of these projects is shown in table 1.5.

Table 1.5: Details of Irrigation Projects

Type	Name of the Projects	River / Tributaries	Total Capacity (Mm3)	Spilling range m3/ see
Major	9 major- Koyna, Dhom, Urmodi Kanher, Dhom balkawadi, Tarali, Tulshi, Warna, Radhanagari.	Krishna, Koyna, Warna, Urmodi, yerala, Tarali, Uttarmand, Wang	Major 5520.57 Mm^3 194.97 (TMC)	Major 640 to 5743 M^3/sec
Medium	13 medium - Yeralwadi Ner, Wang, Morna(G), Uttarmand, Nagewadi, Mahu,Hatageghar, Yevti, Masoli, Kadvi, Morna, Kumbhi, Kasari.	Nandani, Panchganga, Kasari, Kumbhi, Jambhli, Tulshi, Bhogavati	Medium 448.76 Mm^3 17.15(TMC)	Medium 150 to 2083 M^3/sec

Line diagram of above major /medium projects in Krishna sub basin is shown in **map no.6**. Apart from conservation /power generation major dams namely, Koyna & Warna especially play major role during flood moderation process though other gated dams help to reduce intensity of flood in smaller proportion.

Comprehensive report on various Terms of Reference is included in foregoing chapters. Detailed & Technical analysis, findings/ observation & proposed remedial measures / recommendations is presented by experts & by the Committee.

Chapter 2 – Floods in Krishna Basin & Event 2019

2.1 Rainfall Pattern in Krishna Basin

Basin receives 85% of its annual rainfall during south-west monsoon. The rainfall, unevenly distributed varies temporally and spatially across the basin. Upper reaches of Krishna Basin lie in the 25 Km wide crest zone of Western Ghats which is the belt of heaviest rainfall region. Annual rainfall varies from 1000 to 3000 mm in this reach. River further enters in the region falling in the rain shadow area of the Western Ghat which receives less than 600 mm annual rainfall. Rainfall gradually increases to about 900 mm towards the East coast. The average annual rainfall in the basin is 1096.92mm. The sub-basin wise average annual rainfall, annual maximum and minimum rainfall are evaluated for all three sub-basins and presented in Table 2.1.

Table 2.1: Annual observed station Rainfall in Krishna sub-basins

Sr. No.	Sub- basin	Geographic Area (Sq.km..)	Max mm	Station	Min mm	Station	Mean mm
1	2	3	4	5	6	7	8
1	Upper Krishna. (K1)	17128	6208	Mahabaleshwar	500	Khanapur	1300
2	Middle Krishna. (Agrani) (K2)	1388	814.22	Pandegaon	209.42	Pandegaon	481.30
3	Ghatprabha (K3)	2010	8997.70	Kitwade	407.00	Kadal	1300
	TOTAL	**20526**					

The seasonal average rainfall over Krishna basin in Maharashtra during rainy period June to October in Satara, Sangli and Kolhapur districts is 1129.40mm, 499mm and 1841mm respectively. Rests of the months are mostly dry.

Average Rainfall in Dam catchment area of Satara, Sangli and Kolhapur District are 6208 mm, 4500mm and 8997mm respectively. The same in free catchment area below Koyna, Dhom, Kanher, Tarali and Urmodi dam up to Sangli is 600 to 800 mm and average rainfall in Free catchment area below Dams in Kolhapur Districts is 800mm to 900mm. Highest Average rainfall during rainfall period is in July. The same for Satara, Kolhapur and Sangli district are 370mm, 338mm and 108mm respectively.

2.2 Existing Floods Mitigation infrastructure in Krishna Basin

a. General:

Flood is a natural phenomenon, which occurs due to prolonged high intensity of rain. Flooding occurs when an extreme volume of water is carried by rivers, creeks and many other geographical features into areas where the water cannot be drained adequately. Often during times of heavy rainfall, drainage systems in residential areas are not adequate, or unchecked civil development severely impedes the functionality of an otherwise acceptable drainage system. This situation becomes hazardous when it causes colossal loss to human lives and property.

Due to their peculiar locations Satara, Sangli and Kolhapur districts normally experience floods in upper Krishna Basin. Floods experienced in recent periods i.e. year 2005, 2006, and 2019 are noteworthy. During last 5-6 decades various major and medium projects were completed in Krishna basins which help to mitigate floods to certain extent. The list of these projects is appended in Annexure-3. The list of the projects above Sangli (Irwin Bridge) and Kolhapur (Rajaram Weir) is shown in Table 2.2 and 2.3 respectively.

Table 2.2: Details of Reservoirs on upstream of Sangli (Irwin Bridge)

Main River	Major tributaries	Districts covered	Major and Medium Project	Storage Capacity MM^3 (TMC)
Krishna	Koyna, Warna, Venna, Urmodi, Tarali, Yerala, Uttarmand, Wang, Nandani,	Sangli, Satara	7 Major- Koyna, Dhom, Kanher, Urmodi, Tarali, Dhom balkawadi,Warna 10 Medium -, Mahu, Nagewadi, Uttarmand, Morna (G), Wang, Yeralawadi, Ner, Hatgeghar, Yevtimhasoli, Kadvi	5185 (183.14) 309.988 (10.95)

For flood mitigation all above major gated dams and medium gated dams namely Wang, Morna, Uttarmand, and Mahu need to be carefully operated. Koyna dam play very important role since has large spillway capacity. Warna dam just downstream of Sangli is also vulnerable as far as inundation is concerned.

Table 2.3: Details of Reservoirs on upstream of Kolhapur City (Rajaram Bandhara)

Main River	Major tributaries	Districts covered	Major and Medium Project	Storage Capacity MM^3 (TMC)
Krishna	Kasari, Dhamni Jambhali, Kumbhi,Tulashi, Bhogawati, Panchganga.	Kolhapur	2 Major- Radhanagari, Tulashi. 3 Medium- Kasari, Kumbhi, Morna(Shirala)	334.96 (11.83) 175.56 (6.20)

As far as floods of Kolhapur city is concerned, though Radhanagari dam is having high spillway capacity, there are automatic gates installed and comparatively some partial relief is only possible through operation of Tulshi, Kasari and Kumbhi reservoirs.

b. Catchment Area

Catchment area above all dam in Krishna basin is 3317 Sq.km. and the same above Sangli town is 2710 Sq.km. and 197.2 Sq.km. above Kolhapur town. The free catchment area below dams at main river gauging stations is shown below in table no 2.4

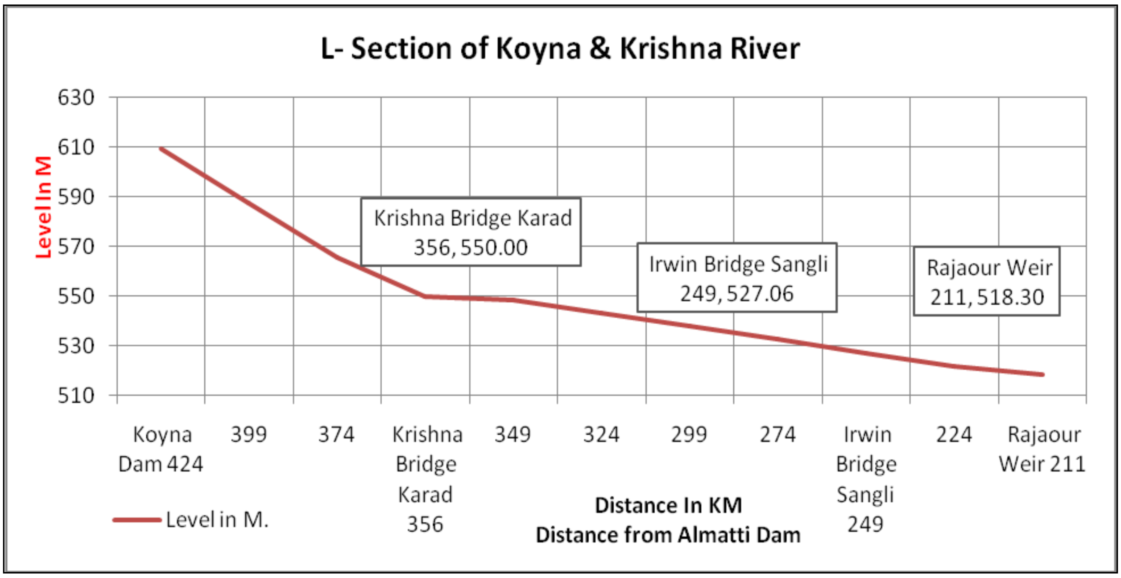
Table 2.4: Details of Free Catchment area

Sr. No	Free Catchment Area up to	River Name	Area in Sq Km
1	Krishna Bridge, Karad	Krishna	3131
2	Irwin Bridge, Sangli	Krishna	6646
3	Ankali Bridge, Sangli	Krishna	8252
4	Rajaram Weir, Kolhapur	Panchganga	1409
5	Rajapur Weir, State border.	Krishna	10953

2.2.1 Flood Prone Area

Krishna River originates in the Western Ghats near Mahabaleshwar at an elevation of about 1,395 meters, in the state of Maharashtra. Koyna River is a tributary of the Krishna River which originates in Mahabaleshwar, Satara district, Maharashtra. It rises near Mahabaleshwar, a famous hill station in the Western Ghats. Unlike most of the other rivers in Maharashtra which flows East-West direction, the Koyna River flows in North-South direction. The river meets Krishna River at Pritisangam, Karad. Krishna river average bed slope up to Karad is 1: 880 and from Karad to Bhilawadi Village average river bed slope is 1: 4672 and Bhilawadi to Irwin Bridge Sangli average river bed slope is 1: 4113. And Further Irwin Bridge, Sangli to Rajapur village, average river bed slope is gentle i.e. 1: 4338. L-section of river Krishna from Koyna Dam up to Rajapur weir (State border) is shown below in Graph 2.1.

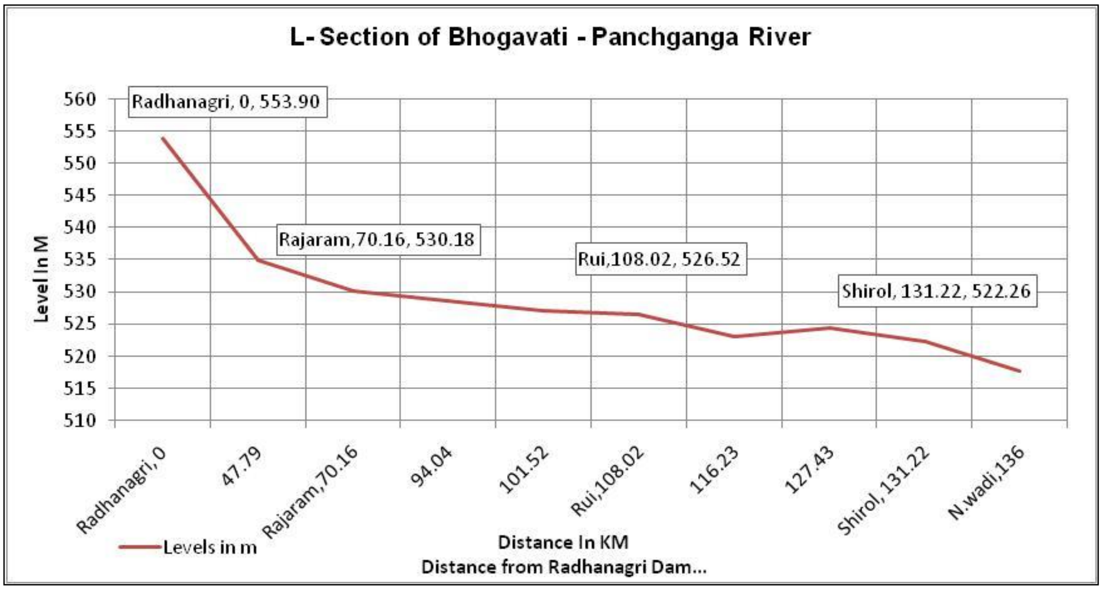
Graph 2.1: L -Section of Koyna & Krishna River



Bhogavati River originates in the western ghats near Asane (Dajipur) in the Kolhapur district of Maharashtra state. At Radhanagari dam elevation is 553.90 m. Bhogavati river flows in south-north direction approximate 40 Km where river Tulashi meets at Bid (Tal. Karveer). Then, at Bahireshwar (Tal. Karveer) Kumbhi & Dhamani river meets and Kasari river meets at Prayag Chikhali. Then, after it runs as Panchganga river till confluence to Krishna river.

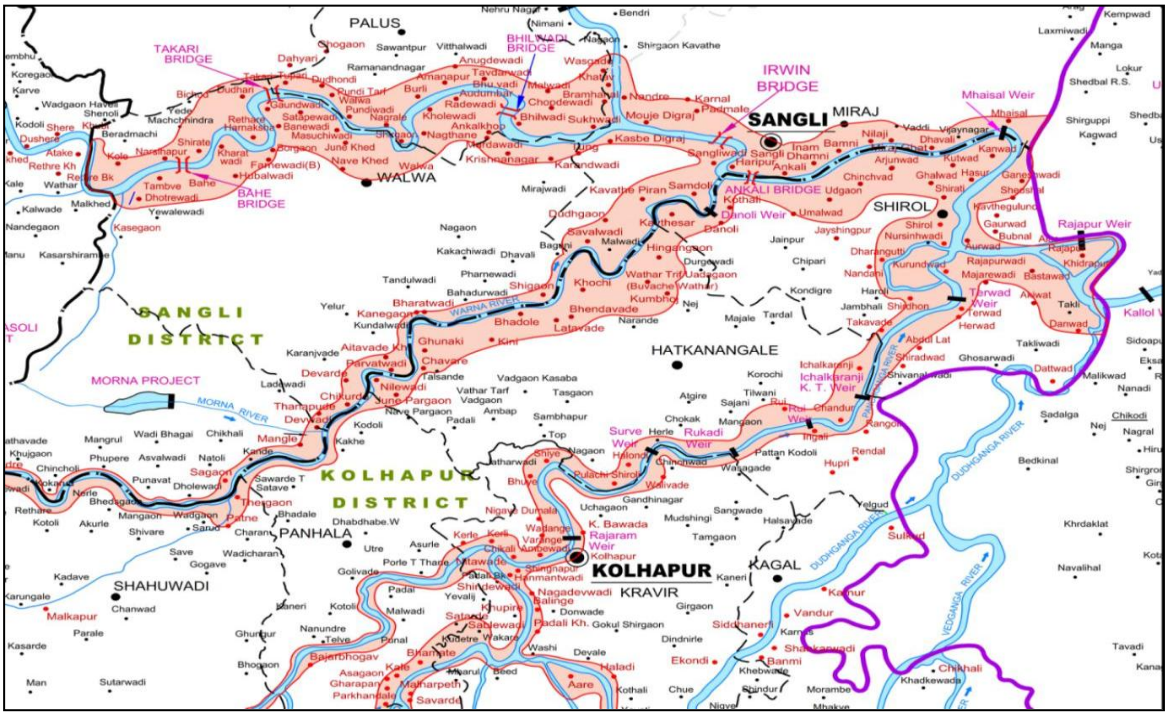
Average bed slope of Bhogavati River from Radhanagari to Prayag Chikhali is 1: 2529. Then, Prayag Chikhali to Rajaram K.T.weir average bed slope is 1:4641. Similarly, from Rajaram K.T.weir to shirol K.T.weir it is 1:7700. L-section of Bhogavati and Panchganga River is as shown Graph 2.2 below.

Graph 2.2: L -Section of River Panchaganga



Flood is a natural disaster that often occurs in Satara, Kolhapur and Sangli districts during the rainy season. There are some areas always flooded but there are also new areas affected by floods so that citizens and local governments were unprepared for the disaster. This caused a great impact and losses including flooded settlements, destruction of agricultural land, damage to infrastructure, significantly disrupt economic activity, and the impact on public health. This aims to identify flood prone areas for natural disaster mitigation. This integrates geospatial data analysis including topography, rainfall data, mapping of land cover and field survey. The results show that generally flood-prone areas are distributed in low land, around Krishna river flows both in the middle and downstream of rivers, landform of floodplains and alluvial plains, and generally in areas with extensive settlements and farmland. The (Fig 2.1) shows typical inundation area of Sangli and Kolhapur districts as below.

Figure 2.1: flood Spread area of Sangli and Kolhapur District



a. Flood Prone area in Satara District

- **Topography**

Upper part of Satara district is situated on Sahyadri hill ranges. It normally receives heaviest rainfall in Krishna valley. All rivers flowing in direction of Sangli town originates between altitude of 1290m to 1310m. River bed gradient in this region is 1:880 for length of 68km from Koyna Dam.

- **Dams**

Koyna Major dam is largest dam in Satara district. There are in all 6 major dams and 8 medium dams are located in Satara district.

- **Catchment Area**

Free catchment area up to Karad city including free catchment area up to khodashi river is 3131 Sq.km... and total catchment area upto karad is 4686 Sq.km.

- **Flood Prone Area**

With South West monsoon currents normally being very active during July - August months over parts of Maharashtra (entire west coast and Madhya Maharashtra) receive incessant rains. Two major rivers flowing in Satara district are Krishna and Koyna which originate from Sahyadri hill ranges where average rainfall is nearly 5000 to 6000 mm. In monsoon season, normally heavy to very heavy rainfall is observed over dam catchment while as reduced intensities are experienced in downstream catchment area. During heavy flooding Patan, Wai & Karad tahsils are generally affected by floods.

b. Flood Prone area in Sangli District

- **Topography**

Sangli City situated on both banks of Krishna river bank between altitude 535.0 m to 550.0m. Krishna and Warana are main rivers flowing in Sangli district. Confluence of Krishna and Warana river is near Sangli City. River Bed level at Irwin bridge is 527.06 m. The length of Krishna river in the district is 105 kms. The bed gradient of river is 1in 4577 from Karad to Irwin bridge, Sangli. Since there is flatter slope and existence of number of meanders, velocity get reduced around Sangli. In addition, due to confluence of Krishna and Warna and also of Krishna and Yerala river, inundation effect around Sangli town is experienced during heavy floods.

- **Dams**

There are 7 major reservoirs upstream of Sangli City namely Dhom, Kanher, Urmodi, Tarali, Koyna, Dhom Balkawadi, and Warana dam is responsible for flood around Sangli. In addition, there are 10 medium dams.

- **Catchment Area**

Free catchment area up to Sangli city is 6646 Sq.km... Free catchment area including Warana river is 8252 Sq Km. and total catchment area up to the Sangli city is 9357 Sq. Km.

- **Flood Prone Area**

Two major rivers flowing in Sangli district are Krishna and Warna. Both catchments normally receive high rainfall in July and August in comparison to free catchment downstream. Inspite of proper reservoir operations, many times, heavy discharges are required to be let out from major dams namely Koyna, Warana, dhom, and many medium dams during heavy floods. Sometimes heavy down pour is also experienced in free catchment. This situation collectively results in huge inflow in the river and area get flooded around Sangli town and few talukas near river mainly Shirala, Walwa, Miraj, Palus & Tasgaon tahsil.

c. Flood Prone area in Kolhapur District

- **Topography**

Kolhapur City situated on the bank of Panchganga river which is a major tributary of Krishna river. Average altitude of Kolhapur city is 540.00m at the river bank and 560.00m on the hilly portion.

- **Dams**

There are 2 major dams namely, Radhanagari (automatic gates) and Tulshi and 3 medium dams viz. Kasari, Kumbhi & Morna (Shirala) are constructed in Panchganga sub basin.

- **Catchment Area**

The Free catchment area of Panchganga river up to Kolhapur City (Rajaram weir) is 1409 Sq.km and Total catchment up to the Kolhapur City (Rajaram Weir) is 1606 Sq.km inclusive of dam catchments.

- **Flood Prone Area**

Three major rivers flowing in Kolhapur district namely Krishna, Warna and Panchganga receive very high rainfall in upper catchment. Practically all dams including Radhanagari (automatic gates) normally release higher outflows during heavy floods. Kolhapur city, parts of tahsils namely Karveer, Panhala, Radhanagari, Hatkangale, Shirol, Bhadargad, Gadhinglaj & Ajra are normally flooded during very heavy inflows.

2.2.2 Flood Forecasting Regulation and Monitoring

The main Krishna River Basin in the state of Maharashtra comprises of two sub basins namely Krishna and Bhima. The river Bhima is a major tributary of main Krishna. River Bhima has its confluence with Krishna in Karnataka state. As such these two river basins, for simplicity may be treated as separate sub basins. Flood regulation and monitoring mechanism is shown below.

a. Basin Flood Control

In Krishna Basin arranged Flood Control Unit for easy forecasting and regulation of flood. In this unit, Chief Engineer, WRD, Pune working as Basin flood control officer.

b. Sub Basin Flood Control

Superintending Engineer, Pune Irrigation Circle, Pune working as Bhima Sub basin flood control officer under Basin Flood Control Officer and Superintending Engineer, Sangli Irrigation Circle, Sangli working as Krishna Sub basin flood control officer under Basin Flood Control Officer. Krishna is

Interstate river, hence for regulation and co-ordination of flood Superintending Engineer, Sangli Irrigation Circle, Sangli is appointed as a Inter State Flood Control Coordinating Officer (Almatti).

c. Central Flood Control cell

For Bhima and Krishna basins, the Krishna Basin Flood Control Cell is established, which collects the reservoir levels, rainfall and spillway discharge for each of the reservoirs twice a day (07:00 hrs and 17:00 hrs) in normal circumstances and hourly in flood like situation.

The data is received by any available means viz. Cell Phones, Wireless, Land Line, e-mail etc. The Flood Control Cell is under the Executive Engineer, Khadakwasla Irrigation Division. During monsoon (from June to October) three Sectional Engineers along with four wireless operators manage the cell 24X7 in three shifts. During non-monsoon periods, the wireless operators collect the data.

d. Flood Control Cell at District Level-

For Sangli, Satara, and Kolhapur we establish District level Flood Control Cell. Which collects the reservoir levels, rainfall and spillway discharge for each of the reservoirs twice a day (07:00 hrs and 17:00 hrs) in normal circumstances and hourly in flood like situation.

The data is received by any available means viz. Cell Phones, Wireless, Land Line, e-mail etc. The Flood Control Cell is under the Executive Engineer, Sangli Irrigation Division. Sangli for Sangli District during monsoon (from June to October) and For the Kolhapur District Flood Control Cell is under the Executive Engineer, Kolhapur Irrigation Division. Kolhapur. And for the Satara District Flood Control Cell is under the Executive Engineer, Satara Sinchan Division, Satara. Sectional Engineers along with four wireless operators manage the cell 24X7 in three shifts. During non-monsoon periods, the wireless operators collect the data.

2.2.3 Real Time Data Acquisition System (RTDAS)

The Krishna Bhīma sub basin area experienced flashy floods in year 2005 and 2006 which caused heavy damages to the lives and properties in the basins. Therefore, under World Bank Aided HP-II project, Government of Maharashtra developed Real Time Stream flow forecasting (RTSF) and Reservoir Operation System (ROS) with Real Time Data Acquisition System (RTDAS).

Objective of the system

To equip Water resources department with a web-based real time stream- flow forecasting (RTSF) and reservoir operation system (ROS) for Krishna & Bhima basin.

- System should be used to optimize releases from reservoirs for multiple uses throughout the year.
- Provide a system to better manage floods.
- Develop a system to optimize coordinated and integrated reservoir operations.
- To build upon existing HIS on a RTDAS telemetry network.

For development purpose, system is principally divided in to two components i.e. Real Time Stream Flow Forecasting (RTSF) and Reservoir Operation System (ROS) and Real Time Data Acquisition System (RTDAS

2.2.4 Standard Operation Procedure (SOP)

Systematic Operation Plans are prepared for Flood forecasting and flood monitoring in Krishna basin. Communication protocol for water resources projects for various trigger mechanisms for various discharge levels, like preparedness level, normal level, alert level, danger level and disaster level has been already established vide standard operating procedure (SOP) for flood control issued by Government of Maharashtra in June 2014. This communication protocol includes communication mechanism from dam in-charge sectional engineer to Chief Secretary/ Chief Minister/ Central government.

Pre-monsoon and flood review are taken at secretary, Water resources department with nodal officers appointed by the department for flood management. As per provisions of disaster management act, 2005 meetings are conducted before monsoon season in order to assess flood preparedness at various levels like at district collector level to Chief Secretary level.

The SOP is to take timely action to prevent or minimize damage to human and animal life and to property and environment in and around the river. This will be done by improving systematic coordination among Departments and stakeholders, and streamlining communication at all levels of decision making.

- **Objectives are as follows**

Identify hazard potential on downstream of the dam. Monitor flood situation. Warning about probable floods in advance, to have advance action. Monitor and protect irrigation as well as all other infrastructures created and protection of Human lives & live stocks. Restore damaged infrastructure due to floods.

- **The SOPs are prepared for**

All gated dams with manual, electrical or mechanical power operated gates, irrespective of the storage capacity or spillway discharging capacity. All dams with gross storage capacity of 150 Mcum or more. All dams with a spillway discharging capacity more than 3000 Cumecs irrespective of storage.

- **Components of the Standard Operating Procedure**

Pre monsoon preparations. -
Appointment of flood control officers.
Fixing duties & responsibilities of flood control officers.
Operating procedure during the monsoon.
Flood control measures to be taken at dam site.
Action to be taken by Water Resources Department.
Actions to be taken by other Departments of the Government.
Post monsoon inspections of damage and distress in dam and in river.
All Dams are inspected by the responsible relevant officer as instructed in Government Circulars and manuals, the works related with Dam Safety and flood control are taken on priority & completed by WRD officers.

2.2.4.1 Systematic Operation Plan (SOP) for Interstate and Coordination with Almatti Reservoir Administration.

Standard Operating Procedure (SOP) has been prepared for operation of Almatti reservoir with the consent of both the states and it is approved by both states. Krishna is Interstate River hence for regulation and co-ordination of flood Basin flood control officer appoint Superintending Engineer, Sangli Irrigation Circle, Sangli as a Inter State Flood Control Coordinating Officer (Almatti). Duties of Inter State Flood Control and Liaising Officer are as follows

Inter State flood control & Liaising Officer are appointed for Flood control of dams having impacts of floods on Maharashtra and Karnataka states. In order to co-ordination the flow control operation pertaining to dams/reservoirs which have implication as far flood management is concerned a co-ordination meeting presided over by Chief Secretary GoM at Mumbai with the Almatti Dam officials. Aspects discussed in meeting are modality of exchange of hydrological & climatological data, standard operating procedure & reservoir regulation procedure. According to SOP it is decided to maintain reservoir level at 518.00 to 518.50 m. up to 15th August.

Communicate with officers of concerned State and exchange hydro-logical data with them on a daily basis. During critical flood situations exchange of above data is made hourly or at closer intervals as necessitated. Coordinating officer is communicate discharge in river flowing at Rajapur weir (State border) for easy operation of Almatti reservoir. Almatti reservoir officer communicate inflow, outflow, water storage and water level. The data is received by any available means viz. Cell Phones, Wireless, Land Line, e-mail etc. This data is communicated to Basin flood control officer for reference and decision making.

This communication protocol includes communication mechanism from dam in-charge sectional engineer to Chief Secretary/ Chief Minister/ Central government

2.2.4.2 Co-ordination Methodology

For Bhima and Krishna basins, the Krishna Basin Flood Control Cell is established, which collects the reservoir levels, rainfall and spillway discharge for each of the reservoirs twice a day (0700 hrs and 1700 hrs) in normal circumstances and hourly in flood like situation.

In the control room, the staff from the Police Department is also deployed round the clock to communicate the flood situation to respective police commands in the districts. In Collector Office also one disaster management cell established for monitoring rainfall and flood. They collect data from District flood control cell. The collected data is entered into computers and every day at 0800 hrs. reports are generated and send to the Chief Engineer, Water Resources Pune. The Chief Engineer (SP), Water Resources, Pune and the Superintending Engineer, Pune Irrigation Circle.

The copy is also sent to Mantralaya (Ministry of Water Resources) in Mumbai Flood Unit, Minister of Water Resources, Divisional Commissioner, Pune, SE (CADA), Solapur and Baramati Hostel (Members of Parliament: On demand). All the information on spillway discharges are given to the Police Department. The updated data is also published daily on the website (http://www. punefloodcontrol.com). In case of high releases from dams, the information is provided to concerned corporations/municipal authorizes as well as to the Police for evacuation from low lying areas.

2.2.5 Reservoir Operation and Flood Routing

The primary requirement of dams is to plan the outflows in such a manner that by the end of the monsoons the reservoirs are filled to 100% capacity, subject to the adequacy of rainfall. The operation of each gated dam is regulated through a Reservoir Operation Schedule (ROS) that lays out the schedule of releases. The ROS is prepared on the basis of fortnightly reports 95% dependable yields during the monsoon. As per the ROS the reservoirs are kept at lower levels in order to have absorption capacity to accommodate flood water and thereby enable controlled discharges through spillway gates.

Water managers face the challenge of controlling the outflows while at the same time maintaining reservoir levels such that they are filled by the end of the monsoon. The increasingly inconsistent pattern of rainfall and its uneven distribution due to climate change have increased the difficulty of this task.

A large number of major, medium and minor water resources projects have been constructed in the Krishna basin of Maharashtra. There are about 22 major and medium multipurpose reservoirs which includes hydro power, irrigation, domestic and industrial uses. Due to an increase in urbanization and industrialization, the reservoirs which were built for providing water for Irrigation as a major purpose, now majority of volume of water is required to be diverted for catering the needs of municipal population and industrial needs. The all projects in Upper reach of Sangli and Rajapur K T weir are appended in the Annexure - 3. Some major projects are listed in below table 2.5.

Table 2.5: Gauging Locations for flood forecasting

Sr No	Name of Dam	Name of River	Reservoir Capacity in MM3(TMC)
1	Koyna	Koyna	2980.68(105.25)
2	Dhom	Krishna	382.32(13.50)
3	Kanher	Venna	285.98(10.10)
4	Tarali	Tarali	165.67(5.85)
5	Urmodi	Urmodi	282.06(9.96)
6	Warana	Warana	974.04(34.40)
7	Radhanagari	Panchganga	236.71 (8.36) Auto gates

2.3 Flood Event of 2019

2.3.1 Rainfall

It can be seen that the flood effected districts of Satara, Sangli and Kolhapur continuously received excess to large excess rainfall during 25 July 2019 to 13 Aug 2019. During the first week of August starting from 1st August; districts of Satara, Sangli and Kolhapur received large excess rainfall with a departure of +431%, +406% and +344% respectively of their long period average. Though there was reduction in the rainfall intensity, during the second week also these districts continued to receive excess rainfall with a departure of +89% of its normal over +152% over Kolhapur, + 89% over Satara and +59% over Sangli districts.

Monsoon remained widespread with active to vigorous rainfall activity over meteorological subdivisions of Madhya Maharashtra during period. Heavy to very heavy rainfall with isolated extremely heavy falls occurred over many places over Madhya Maharashtra during the first week of August with a peak activity during 3-9thAugust 2019. Though intense rainfall activity over ghat areas of Madhya Maharashtra reduced from 9th August, the effect of large-scale flooding continued over the area.

According to the rainfall data available (as per IMD Departmental, Part Time and DRMS – District Rainfall Monitoring System network) in Kolhapur district 11 stations received heavy rainfall during the period 3rd to 9th August on different days. Few stations like Ajra, Chandgad, Radhanagari, Panhala and Gaganbawda reported heavy to very heavy rainfall with isolated extremely heavy falls continuously for a week. The highest rainfall was reported at Gaganbawda, 335 mm on 6th August 2019.

Similarly, in Satara district 6 stations received heavy rainfall during the period 3rd to 9th August on different days. All stations received continuous rainfall with stations like Mahabaleshwar, Javali, Medha and Patan receiving continuous heavy to very heavy rainfall during the entire period. The highest rainfall was reported at Mahabaleshwar, 289 mm on 5th August 2019.

Dam catchment

Total rainfall observed during 25/7/19 till 13/8/19 in Koyna, Dhom, Urmodi, and Tarali dam catchments and there for Radhanagari was 3229, 606, 1168, 1593, and 3074 respectively. This rainfall was almost 5 to 19 times over normal rainfall in that period. Maximum rainfall observed at Navja in (Koyna Catchment) was 420 mm and the same at Radhanagari was 445 mm.

Free catchment

Total rainfall observed during 25/7/19 till 13/8/19 below dam at Kolhapur, Karad, Islampur, Palus, Tasgaon, Sangli, Shirala, Miraj, Vita and Kadegaon in free catchment was 921, 707, 414, 211, 279, 268, 814, 232, 333, and 545 respectively. This rainfall was 13 to 28 times over normal rainfall during this period.

2.3.2 Flood

The flood level in the Krishna River Observation point at Irwin Bridge, Sangli on 05/08/2019 exceeded the danger level 540.77m. The maximum flood level observed at this observation point was 544.57m (56 feet 5 inches) on 09/08/2019. The water level started receding below danger level from14/08/19.

The flood level in Panchaganga river observation point at Rajaram K.T. weir Kolhapur had exceeded danger level i.e. 543.30 m on 03/08/2019. The Maximum level reached up to 546.97 m on 07/08/2019.The level started to recede from 14/8/19.

2.4 Inundation / Flood damage

Satara, Sangli & Kolhapur districts suffered due to intense rainfall between 2nd August to 15th August. Major flood situated was experienced in Sangli town between 5th August to 14th August when flood crossed danger levels. The similar situation was observed in Kolhapur Town between 3rd August to 14th August when flood levels were above danger level. Maximum submergence (inundation) took place on 9th August in Sangli town & nearby tahsils. In case of Kolhapur, maximum submergence was observed on 7th August when few talukas were hit by flood.it is estimated that in Satara, Sangli & Kolhapur districts about 1.66 lakh ha seasonal & perennial crops were affected in total 28 talukas. In addition, loss of Infrastructure, Animals, Artisans, Commercial properties, Houses etc were reported. Total losses suffered for above 3 districts were estimated to Rs.3475 crore.

Chapter 3 – Approach to Study

In order to carry out in depth analysis of 2019 flood situation experienced in Bhima and Krishna basin and causes thereof using modernized technique, clarifying whether back water effect due to Almatti and other reservoirs of Karnataka create flood situation in Maharashtra and in addition to propose various remedial measures to avoid such situation in future and or to reduce its severity, input from various related fields / departments / authorities / institutes viz. Meteorology, tropical meteorology, remote sensing, hydrology, hydrodynamics, Water Resources Regularity Authority, Indian Institute of Technology, mathematical modelers and Water Resources Department was necessary. Accordingly, the Government inducted experts from such fields in this Committee.

3.1 Induction of Experts

Preparation of comprehensive technical report was a team work. Initially in preliminary meeting, Chairman while discussing TOR[s], requested all members for their specific contribution expected to carry out work related to their fields. All expert members freely opined to participate, contribute and prepared chapters of their interest.

Water Resources Department (WRD) is shouldering the overall responsibility of measurement and acquisition of hydrological data, investigation to construction of dams & management of flood opted to present study on total flood analysis based on observation & also flood forecasting based on RTDAS / RTDSS through specially created Hydrology Project Wing. WRD being closely associated with exchange of information with Almatti authorities in Karnataka ventured to report on back water studies with technical support by Maharashtra Water Resources Regulatory Authority (MWRRA) & IIT. Indian Meteorology Department (IMD) and Indian Institute of Tropical Meteorology (IITM) readily consented to carry out storm analysis study based on climatological observations. MRSAC being an authority on remote sensing, carried out GIS based Analysis of floods in Krishna basin, inundation studies, encroachment etc. using latest technology of Remote Sensing & Geographic information system. Central Water Commission offered to carry out back water studies using mathematical modeling. All experts inducted for flood studies assured to study, discuss & presented their reports.

3.2 Study Groups

Since subject matter of different TORs need detailed study from specific fields / experts, it was decided then to form various study group of experts to study and prepare report for each TOR. as follow.

TOR- I: **To carry out in-depth technical analysis and investigate causes of 2019 flood situation experienced in Bhima and Krishna basin using modern techniques.**

Shall be studied and report to be prepared by Water Resources Department / IMD/ IITM/ RTDS (Hydrology Project). **Engr. R.D. Mohite, Chief Engineer (WRD) would be Team Leader.** Superintending Engineer namely H.V.Gunale, Sangli Irrigation Circle Sangli, S. L. Doiphode, Satara Irrigation Circle Satara, R.M.Sankpal, Kolhapur Irrigation Circle Kolhapur, S.D.Chopde, Pune Irrigation Circle Pune, Pravin Kolhe, Pune irrigation Project Circle Pune, S.M.Sangle, Koyna Design Circle Pune would assist this group. They are expected to analyze all observed data available with them and should prepare report duly supported by maps, graphs etc. concerning analysis of flood situation created in Sangli and Kolhapur districts. Hydrology Project Organization should prepare their report based on information received through RTDAS including simulation and compare them with actual observation and reasons for variation, if any. IMD/IITM should bring out meteorological report of this flood event duly supported by analysis (storm analysis).

Apart from above technical analysis based on modernized techniques, various reasons for occurrence of such flood event should be incorporated. For this job the requisite information / opinion from other members / experts should be taken.

TOR- II: To specifically clarify based on hydrological studies, whether due to Almatti& other reservoirs (back water effect) from Karnataka, create flood situation in Maharashtra.

Shall be studied by CWC and WRD and IIT. They would present their findings concerning back water effect caused due to Almatti and other reservoirs from Karnataka. Shri. NityanandRai (Director, CWC, New Delhi) and Er. H. T. Dhumal have undertaken this study with application of different mathematical models. **Er. V. M. Kulkarni (Technical Member, MWRRA) would be the Team Leader.**This group would prepare complete Chapter on this TOR with the assistance from Er. S.M.Sangle, Superintending Engineer, Koyna Design Circle, Pune & Er. Anjali Kakade. The team is expected to incorporate earlier studies made on back water effects & their present findings. This issue is very much volatile, delicate and debatable at all levels as such clear and frank technical findings are expected.

TOR- III: To propose strong overall remedial measures to avoid such flood situation in future and or to reduce its severity.

Shall be studied and prepared by WRD and MRSAC. This TOR is of great importance since strong remedial measures are to be proposed so as to avoid such flood situation in future and or to reduce its severity. **Er. Sanjay Ghanekar (Secretary, WRD) would be the Team Leader.** Mr. S. N. Das, Director, MRSAC and Mr. Vivekanand Ghare, Senior Consultant, MRSAC would study and present all possible inputs available through their Remote Sensing and GIS techniques. They are expected to study and prepare flood plain maps of study area, inundation maps on various dates during flood events, geographical features, slopes, terrain, DEM, rainfall status on daily basis etc. WRD officers would keep liaison with them to prepare ground mapping etc.

Since all possible remedial measures are to be proposed by this group, they need to keep liaison with all other group / invitees viz. CWPRS, CDO, MERI, DSO, NCCR, NGOs, CE(CWC) etc. and present major remedial measures may be structural and nonstructural.

TOR- IV: To propose along with policy level measures, to recommend revised plans based on technical analysis concerning micro level issues such as dam wise reservoir operation schedules, integrated arrangement for measurement & management for river discharges, control on constructions in flood zone, emergency operation plans, standard operating procedures etc.

Shall be studied and prepared by Hydrology Expert and WRD. This TOR is also important since many activities related to flood operation and mitigation procedures on micro level need to be addressed afresh wherever needed and requisite remedial measures thereof will have to be proposed. Revised 1) reservoir operation schedule, 2) Emergency operation plan (EOP), 3) Standard operations procedure (SOP), 4) Integrated arrangement for measurement and management for river discharges, 5) Control on construction in flood zone areas etc. **Prof. Pradeep Purandare would be Team Leader**. Er. V.G.Rajput, Chief Engineer (S.P.), Pune, Er. S. L. Doiphode, Satara Irrigation Circle, Satara, Er. Prasad Narvekar, Superintending Engineer, e-Governance, Er. S. D. Chopde, Superintending Engineer, Pune Irrigation Circle, Pune and other officers from WRD would assist.

Accordingly, all Team Leaders studied concerned issues and prepared their reports.

3.3 Committee discussions

Though all above groups are formed for undertaking studies on various TOR^{s}, team leaders are expected to keep liaison with all members, invitees, outside experts, NGOs to prepare and bring out very useful and updated comprehensive information, working procedures, pragmatic solutions and finally workable remedial short term and long term measures using modernized techniques in overall flood management issues. Accordingly, Committee discussed& finalized report.

3.4 Visits & Experiences

1) Flood Prone areas:

Field visits to flood prone areas was extremely important to understand overall geographical features of such areas, mainly causing flood situation in cities and other places. Visits to various locations of Krishna and Panchganga rivers / tributaries, confluence points, inundated areas in cities & other locations, bridges, culverts, K.T. weirs, barrages, ghats, river gauging stations, meandering and loop forming locations etc. were given by the concerned committee members.

2) Revenue / Municipal Authorities:

Meetings with Collector & Municipal Commissioners of Sangli & Kolhapur were held to know their present role of flood disaster mitigation & difficulties faced in flood operation as per government directives. Their views / recommendations were found to be of prime importance since after dissemination of flood warnings these authorities have to ultimately face flood situation & has to actual work in flooded areas for evacuation & all activities concerning flood mitigation.

3) IMD / IITM:

To know the role of forecasting by Meteorology Department, the meeting was convened with IMD officials at IMD Pune along with various experts/ Retired Director General IMD for their valuable contribution and advising on useful short term & long-term strategies in Meteorology and to hold proper dialogue to strengthen co-ordination between various Departments / authorities / Institutes / organizations,

4) Almatti Dam:

As per TOR-II, Committee desire to visit Almatti dam authorities to hold relative discussions with them and to know methodology adopted for reservoir operation for effective flood moderation. Coordination between two states is extremely important not for only exchange of data but to effect reduction in intensity of floods & so damages.

5) CWPRS:

CWPRS Pune is the major research organization in the field of hydraulics & allied research as subordinate office of the MOWR, River Development & Ganga Rejuvenation, GOI and deals with planning, organizing & undertaking specific research & development studies related to optimizing designs of river, coastal, water storage & conveyance hydraulic structures. More importantly physical modeling of various issues related to water sector where especially mathematical modeling may not give reasonable solutions, is successfully carried out in CWPRS. Committee visited CWPRS. It was gathered that certain problems encountered during flood studies viz. inundation effect caused due to confluence of rivers & overall terrain in flood prone area of Sangli & Kolhapur districts can be addressed to. CWPRS.

6) Forum of Experts:

Discussions with Forum of Experienced personnel in flood water management, Technocrats, Research Workers, Scientists, Professionals from various organizations / institutes, Hydrologists, retired high level officers from WRD & experts in field of Meteorology, Hydraulics, Modelers may be working in Govt. or Private sector were useful.

7) Aerial Inspection in Krishna Basin during Flood:

Arial inspection of Krishna river by low aircraft was done by 5 Engineers from Maharashtra & Karnataka on 9^{th} August 2019 i.e. during peak flood in Panchganga & Krishna . Through such Arial Reconnaissance Survey report, valuable findings came out concerning 2019 flood event.

3.5 Reporting

Before preparation of final report & recommendations, outcome of group presentation on different TOR^{S} by specified groups were reviewed by the Committee & further refinements viz. changes / modifications / additions were made by holding frequent discussions. Findings in field visits & remedial measures were proposed by committee members to bring out pragmatic solutions. Suggestions from Revenue authorities, Corporations, different Institutes / organizations / centers / professionals was again discussed. In short, all the inputs received by the committee were properly & carefully scanned & put up for final reporting & recommendations.

Committee also laid hand on all available publications, BSI, literature, reports concerning flood management. It also studied various case studies prepared by known experts in this field.

Chapter 4–Analysis of the Event-FLOOD 2019 and the Causes/Reasoning of the floods (TOR1)

A. Gists of the Analysis (Study Reports) done by various Organizations

The detailed analysis of the flood event has been done by various renowned and expert Institutions, and also by some individuals who are the members of this expert study group. The detailed studies have been done independently by the Institutes/ Individuals, using their own reliable data and techniques. All the reports were discussed at length during the committee meetings and the findings were confirmed. The study reports received, as they are, have been appended along with this report in Annexures.

The gists of findings, of these study reports received, are given as below:

4.1 Study Report by India Meteorological Department and IITM Pune under MoES / (Storm Analysis: July - Aug 2019)

4.1.1 Introduction:

During South-West Monsoon season over the country, rainfall over Maharashtra shows significant spatial and temporal variability. The state experiences extremes of rainfall ranging from 6000 mm over the Ghats to less than 600 mm in the interiors. Western coast of Maharashtra, the Konkan belt is often prone to heavy to very rainfall during active monsoon conditions due to favorable orography. The Western Ghats act as obstruction to the eastward-moving Monsoon clouds forcing it to rise ultimately leading to the heavy downpour on the windward side, while the leeward side forms the rain shadow area receiving less rainfall.

Climatological records of the districts of Satara, Sangli and Kolhapur indicate that, there is a large spatial variation in rainfall in these districts. The western portions of these districts adjoining the Western Ghats and neighborhood get very high rainfall while near the eastern borders of the district, receives very less rainfall.

The average annual rainfall of Kolhapur is 1718.8 mm. The rainfall decreases from west to east. Gaganbawada on the Western Ghats receives as much as 5860.4 mm in a year while Kurundwad near the eastern border gets only 607.5 mm annually. The annual rainfall over Satara is 828.4mm excluding hill stations.

The rainfall in the belt, roughly 25 to 35 kms wide parallel to the crest of the Sahyadri Range, is considerably higher than in the rest of the district. While Mahabaleshwar at an elevation of 1372 meters gets an average annual rainfall of 5886.9 mm, other stations in this belt get annual rainfall ranging between 1684 and 2195 mm. Compared to Kolhapur and Satara, district of Sangli has relatively lesser average annual rainfall ranging about 670mm (Climate of Maharashtra state, 2005).

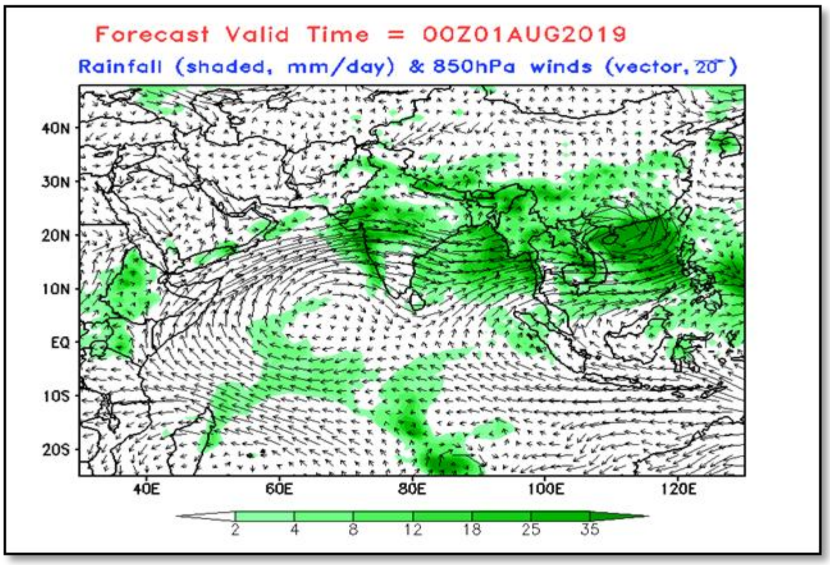
4.1.2 Weather System leading & Extreme Heavy Rainfall:

The persistent intense rainfall activity over the region was in association with an active spell of monsoon started over Maharashtra from 27th July 2019 and resulted in flooding many parts of Konkan and North Madhya Maharashtra. This was followed by another active monsoon spell from 3rd August 2019, in association with the formation of a low pressure over North-East Bay of Bengal and its subsequent intensification into deep depression and westward movement in the subsequent days causing severe flood conditions in South Madhya Maharashtra.

In association with the formation of low-pressure area over the Bay of Bengal, the monsoon currents over the west coast strengthened from 3rd Aug 2019. Low Pressure Area formed over North Bay of Bengal and adjoining coastal areas of Bangladesh & West Bengal on 5th August 2019. It concentrated into a Depression over Northwest Bay of Bengal off north Odisha-West Bengal coasts on 6th August and intensified into a Deep Depression on 7th August 2019. During 7-9 August, it rapidly moved across Odisha, north Chhattisgarh to over southeast Rajasthan and weakened into well marked low, maintaining its intensity as well marked low, it lay over northern parts of Gujarat region adjoining south Rajasthan on 10th Aug, weakened further into low pressure and became less marked on 13th Aug 2019.

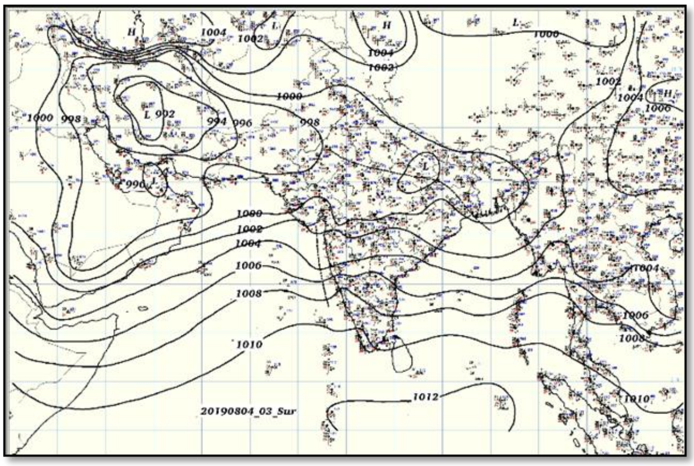
This movement of deep depression system (an active low pressure system with wind speed ranging between 52 to 61 kmph) across central India, resulted in enhancement of rainfall over west coast and in the ghat areas of Madhya Maharashtra with heavy to very heavy rainfall (~120 to 200 mm) and extremely heavy rainfall (more than 200 mm) events for more than a week period over these places, resulting in severe flood situations.

Figure 4.1.1: Extended forecast showing Low Pressure area formation over Bay of Bengal and strengthened wind speed over west coast



Another favorable weather system was the presence of **Off Shore Trough** (formation low pressure area during monsoon) along the west coast during the period and the **East-West Shear Zone** observed at mid and upper tropospheric levels during the period provided all supporting dynamical conditions across the region leading to very high number of extreme rainfall events with longer spell period.

Figure 4.1.2: Formation of off shore trough along west coast during the period as seen from surface weather chart

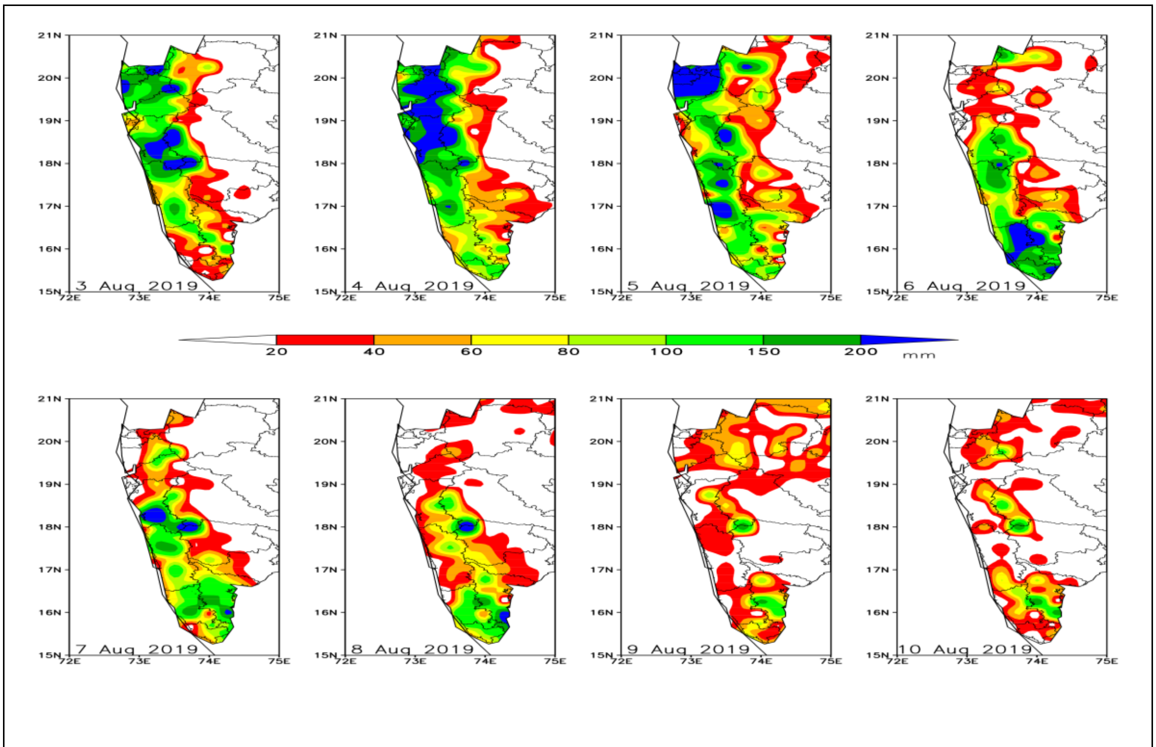


4.1.3 Rainfall Pattern - 3rd & 10th August, 2019:

Daily rainfall distribution during the flood period is shown in Figure 4.1.3. Figure indicates that during the entire period, Konkan and adjoining Madhya

Maharashtra experienced very heavy rainfall. In the beginning of the flood period from 27 July to 3rd Aug, the heavy rainfall events were localized in the northern part of the Konkan and adjoining North Madhya Maharashtra. Many stations in Pune and Nasik districts, recorded rainfall more than 150 mm/day during the period 3-5 Aug. Towards the latter part of the week, rainfall belt shifted towards south Madhya Maharashtra. Mahabaleshwar recorded highest rainfall of 380 mm on 5th Aug. 2019. It is also observed that Kolhapur district continuously experienced heavy rainfall heavy rainfall throughout the period with highest rainfall amounts on 6th Aug. 2019. Gaganbawda recorded its highest rainfall of 340 mm rainfall on 6th Aug. It is also seen that though heavy rainfall occurred in the western part of the districts in Madhya Maharashtra, their eastern parts were devoid of rainfall.

Figure 4.1.3: Spatial distribution of rainfall over the region during peak rainfall activity period of 3-10 Aug 2019

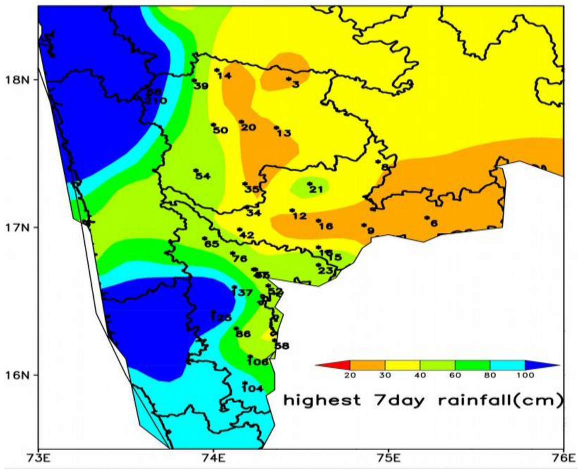


4.1.4 Wide Spread Heavy Rainfall (7 day):

Figure below shows spatial pattern of Highest ever recorded 7-days rainfall based on the daily rainfall data during 1951-2018 and values recorded at stations in the year 2019 are displayed in black figures (in cm).

(Highest recorded Rainfall 1951 – 2018)

Figure 4.1.4: Spatial Pattern of highest ever recorded 7 days' rainfall (cm) during 1951-2018 (indicated in different colors).

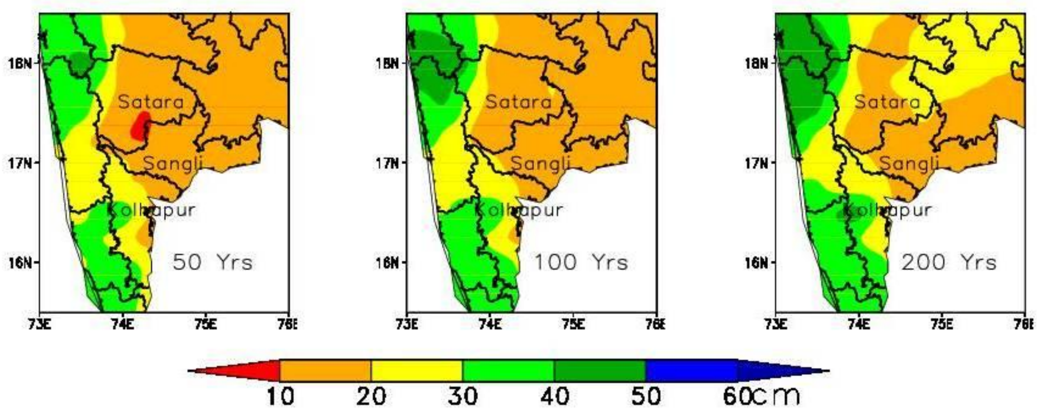


It is seen that during the heavy rain spell of Aug. 2019, many stations in Kolhapur district and western part of Satara district have crossed their previous record of 7 days rainfall. This indicates that compared to previous years, rainfall over the region was widespread and remained very intense for a long period during 27th July to 12th August 2019.

4.1.5 Return Period Analysis:

Return period analysis for the districts of Satara, Sangli and Kolhapur have been carried out. Figure 4.1.5 shows 50, 100 and 200 years return period values for 1-day rainfall in the three districts. Estimation is based on daily rainfall data during the period 1951-2019 for 53 stations of these districts.

Figure 4.1.5: Return period analysis for the districts of Satara, Sangli and Kolhapur



It is seen that except for the extreme western part of Satara district, 50-200 year return period values ranges from 10-30 cm. For Sangli district, these values ranges from 10-20 cm, which is nearly same as that of extreme rainfall. This is due to less variability in the extreme 1-day rainfall. For the ghat areas of Satara, where Mahabaleshwar is situated these values cross 40 cm. For Kolhapur district,50-200 year return period values are more than 20 cm. During the period 4-6 August 2019, many stations in Kolhapur district recorded rainfall of the order of 100-year return period.

4.1.6 Comparison with Storms of 2005:

Earlier records of flooding over the same region happened in 2005. In the year 2005, Krishna River in Maharashtra experienced high floods in July –August due to heavy rainfall in the catchment area of Krishna and its tributaries in western Maharashtra. The sustained flood continued over a period of two weeks from 26th July to 07th August 2005. Due to this flood, the towns of Karad, Arjunwad, Kurundwad experienced sustained flooding. Major part of the Sangli Township was inundated.

In the present analysis, a comparison of 2019 rainfall with that observed during 2005 has been carried out. Analysis of the rainfall data (from State Govt) over different catchments of Koyna, Dharan, Krishna and Warna during the flood years of 2005 and 2019 has been carried out.

The analysis indicates that, the highest rainfall amounts received during both the years were comparable. During 2019 the heavy rainfall spell was rather continuous with peak activity during period 3^{rd} – 9^{th} Aug 2019 lasting for a week. During 2005 floods, peak heavy rainfall activity occurred in two spells, first spell during 23-28 July 2005 and another spell during 31^{st} July- 4 August 2005. Thus, the total number of torrential rainfall days was more during 2005 floods. From the available rainfall data, the highest rainfall amount received during 2019 floods were over Radhanagari (445 mm) and Koyna-Nawja (420 mm) whereas during 2005 floods were over Savale (478 mm) and over Mahabaleshwar (462.2 mm).

4.1.7 Climate Change Projections:

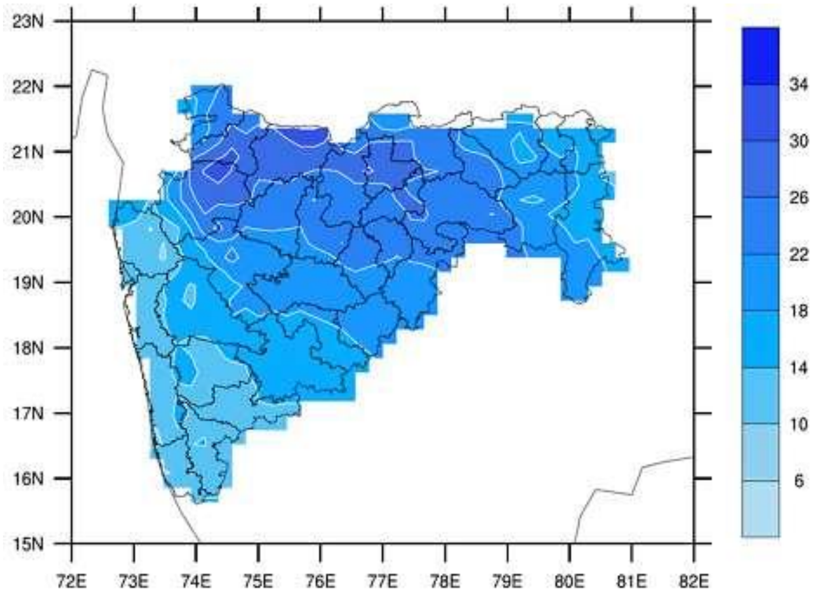
The climate modeling results show that temperature and rainfall are projected to increase all over the state though there are regional variations. Over time, the projected rise in mean temperature is greater for the 2070s compared to the 2050s and the 2030s. Amravati and Aurangabad divisions may experience a greater rise in annual mean temperature than other parts of the state. The projected increase in monsoon rainfall by the 2030s and 2050s is relatively more for Amravati and Nashik divisions, though divisions like Konkan and Nagpur receive, and are projected to receive more rainfall.

This overall increase in monsoon rainfall for the state is consistent with the findings of the Fifth Assessment Report of the Intergovernmental Panel on Climate Change (IPCC). On the basis of the vulnerability index Nandurbar ranked as the most vulnerable district, followed by Dhule and Buldhana. Satara was regarded as the least vulnerable district, whereas Ratnagiri and Sindhudurg were also considered less vulnerable as compared to other districts.

4.1.8 Impact on Extreme Rainfall events in 2030s:

A warmer atmosphere has a higher capacity to hold water. This is likely to produce more intense rainfall events with longer dry or low rainfall spells between these events. Figure 4.1.6 show that, extreme rainfall is projected to increase in all regions with greater increases in the northern parts of the state (Aurangabad and northern regions of Nasik division).

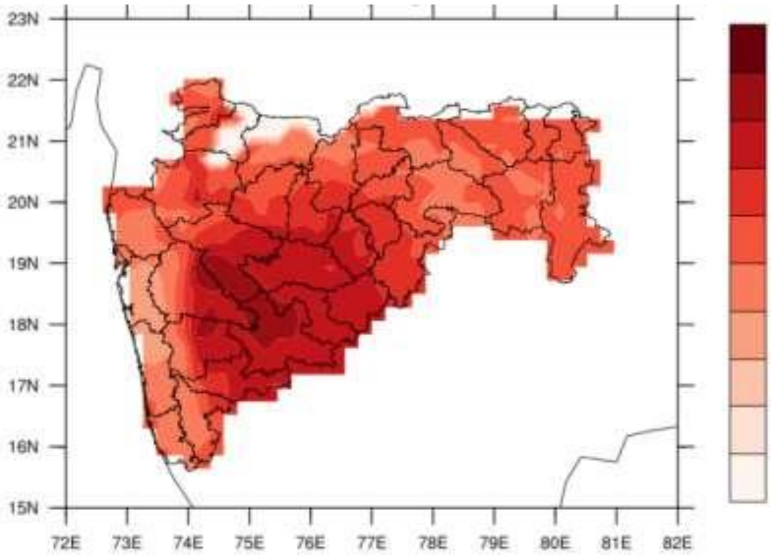
Figure 4.1.6: Increase in extreme rainfall in 2030s relative to baseline (in %) (Source: TERI report, 2014)



4.1.9 Scenarios in 2030s of Low Rainfall Events:

Figure 4.1.7 indicates that Parts of south-central Maharashtra are projected to experience more dry days in the 2030s relative to the baseline for this region. This is a matter of concern for agriculture and water resources in this region.

Figure 4.1.7: Number of low rainfall days in the 2030s relative to baseline.(Sources: TERI report, 2014)



4.1.10 Water Resources Impacts:

Projected increase in rainfall in the form of heavy precipitation events may increase the runoff in the rivers. Increase in surface run off is projected in certain catchments e.g. sub-catchments of Godavari are expected to have increased runoff in the month of July. Increase in the number of low rainfall days in south central Maharashtra indicates dry periods in this region which is likely to have an impact on water resources as well agricultural patterns.

4.2 Study Report by Maharashtra Remote Sensing Application Centre (MRSAC)

(Remote Sensing and GIS based Analysis of Flood Event 2019)

4.2.1 Introduction

MRSAC, an autonomous body of Planning Department, Government of Maharashtra, being a member of Experts Study committee for Krishna Flood, carried out study in the flood affected area of Krishna basin (excluding Bhima sub-basin) of Maharashtra using latest technologies of Remote Sensing and Geographic Information System. The study area, under this report, is limited to Krishna sub-basin of Maharashtra.

4.2.2 Analysis & Reasoning of 2019 Floods

MRSAC has analysed the flooding 2019 situation, using Remote Sensing and GIS, in Krishna sub-basin, based on various Geo-spatial databases layers such as Digital Elevation Model (30m), Slope, Drainage, Watershed, Administrative layers, Transport layers, Rainfall data, Inundation map, Flood line 2019, etc. available from different sources such as National Remote Sensing Agency (NRSA), MRSAC, WRD GoM & freely available dataset.

A detail report is attached with the main report of Flood Committee. The major observations on terrain analysis, Rainfall, Estimated Runoff, Inundation in Krishna river basin are:

- The study area experienced an unprecedented& unexpected extremely heavy rainfall (+200 mm) per day continuously for 6-7 days within a span of 20 days duration (26th July to 15th of August, 2019). Such excessive rains caused flood inundation in Kolhapur and Sangli.
- The daily rainfall analysis done in GIS platform to work out Iso-hytel curves and distribution of daily rainfall using Spatial Deterministic methods of interpolation – Inverse Distance Method (IDW)for a period from 25th July 2019 to 15th August, 2019 vis-à-vis flood information in Krishna sub-basin reveals that –
  - The Satara, Sangli, Kolhapur districts received more than 100 mm of rainfall on 26th to 30th July and after a break of rainfall (less intensity) on 31st July to 2nd August, there was continuous rainfall of more than 100 mm from 3rd of August to 8th August.
  - Extremely heavy rainfall of + 250 mm occurred for three to four days from 4th August 2019 onwards, in reservoir catchment as well as in free catchment caused floods on the banks of Krishna, Warna, Panchganga, Dudhganga rivers in Sangli and Kolhapur districts in Maharashtra.
  - The Panchganga, Dudhganga, Ghatprabha, Vedganga, and small river further southern part in Kolhapur district experienced very heavy rainfall (200 mm and more) on 6th and 7th August, 2019.) After 9th of August, 2019 onwards, the rainfall intensity got reduced.
  - The area on the left bank of Krishna and further east to it had always experienced less intensity of rainfall (not more than 50 mm). Area further east of this in Agrani and Yerala or in Godavari basin experienced drought like conditions.

- The analysis is further extended to estimate volume of runoff in various sub-basins of Krishna river, limited to Maharashtra State. The volume of runoff is estimated with runoff coefficient of 0.78, considering fully saturated soils and due to non-availability of all other parameters for working out surface runoff. The results have been worked on daily basis and shown on maps and charts in the report. This estimation is a rapid analysis, done in absence of availability of all parameters required as per Soil Conservation Services and Curve Number (SCS–CN) method. The cumulative runoff so estimated at Irwin bridge for the period of 30/7/2019 to 14/8/2019 is 5214 MCM as against actual measured runoff 5418 MCM. This reveals that GIS Platform can be effectively used to estimate the runoff.

4.2.3 Flood inundation maps based on the interpretation of Radar Sat 2 Satellite data–

The flooding in Krishna basin started from 5th of August, 2019, however, for ascertaining inundation, there was no satellite data available with ISRO, in optical and microwave region of Electro-Magnetic Radiations (EMR). The Disaster Management Support Division of National Remote Sensing Center (NRSC), ISRO, Govt. of India could acquire microwave satellite data for 9th August, 2019 (18.00 Hrs. IST), 13th & 15th of August 2019and generated near real time flood inundation map. The reported inundation includes wet area/ rain water accumulation and water in low-lying area on 9th, 13th and 15th August, 2019. The inundation map along with area statistics for 9th, 13th & 15th August2019, as referred in report as figure 4.2.65, 4.2.66, 4.2.67 and Table no.4.2.6; 4.2.7,4.2.8 reveals that-

a) Although, both the banks of Krishna, Warna, Panchganga, Dudhganga rivers of Sangli and Kolhapur districts were got flooded from 5th of August, 2019 onwards, there was still extensive inundation over large area on9th of August, 2019(as seen in inundation map) and slowly receded till 15th of August.

b) Inundation Area Observations :(Refer report Table no. 4.2.9; 4.2.10&4.2.11 in Annexure 4.2.1)

Inundation on 9th of August 2019 -

- In Kolhapur District, a total of 215 Villages got affected in 9 Talukas, encompassing an area of 332.30 sq.kms, (21.20% of area) in these villages.
- In Sangli District, a total of 58 Villages got affected in 3 Talukas, encompassing an area of 105.05 sq.kms, (15.17% of area) in these villages.

Inundation on 13th of August 2019 –

- In Kolhapur District, a total of 78 Villages got affected in 4 Talukas, encompassing an area of 207.91 sq.kms, (24.47 % of area) in these villages.
- In Sangli District, a total of 30 Villages got affected in 2 Talukas, encompassing an area of 36.11 sq.kms, (9.56% of area) in these villages.

Inundation on 15th of August 2019 –

- In Kolhapur District, a total of 68 Villages got affected in 4 Talukas, encompassing an area of 133.59 sq.kms, (almost 19.28% of area) in these villages.
- In Sangli District, a total of 4 Villages got affected in 1 Taluka, encompassing an area of 2.20sq.kms. (1.40% of area) in these villages.

c) Limitations: The flooding in City area of Sangli and Kolhapur could not be picked up due to mixing of digital signatures of urban area with water spread and course resolution (+50 to 100 m) of RadarSat 2.

4.2.4 Inundation up to Almatti

In order to study, the inundation up to Almatti reservoir of Karnataka, MRSAC has examined the web site of National Remote Sensing Agency (NRSA), Indian Space Research Organisation (ISRO), Government of India - https://bhuvan-app1.nrsc.gov.in/bhuvandisaster/#flood where Disaster Management Support Division of NRSA has published inundation in Karnataka and Maharashtra for different period from 9th to 15th August, 2019.

A combined snap shot of inundation in Karnataka and Maharashtra for 9th, 13th and 15th August 2019 have been captured and are attached with this report. The combined snap shots comparing with daily rainfall data indicate that even though the daily rainfall in Sahyadri hill range got receded from 8th of August, 2019, there was extensive inundation upto 15th of August, 2019 on either side of Warna, Panchganga in Maharashtra and Krishna river in Maharashtra and Karnataka too.

4.2.5 Observed flood line 2019 for Krishna River and Flood Vulnerability map

To decipher precise flood lines of2019, a detail survey has been done by WRD officials by collecting observed flood line 2019 from the field level information and plotting it on the Village level map. These maps have been imported in GIS environment using Geo-referencing technique. Similarly, MRSAC has also attempted to prepare a **Flood Vulnerability map** using Digital Elevation Model

(DEM) by extrapolating observed flood lines and corresponding elevation (m) in DEM. The Flood vulnerability map has layers like Observed flood 2019 lines (of WRD), flood inundation for 9th August, 2019 of NRSA, ISRO, Habitations/ Villages/ Taluka/ District boundaries, Transport network& Flood Vulnerable zone.

4.2.6 Encroachment/ Blocking in Rivers

For identification of encroachment / blocking in Rivers, namely Krishna, Warna, Panchganga and its tributaries (limited to those tributaries which are very nearer to main river), a visual scanning of satellite data available on Google site have been carried out in a "Google Map", a browser-based application. A detailed examination is a laborious& human intensive work& hence a rapid analysis has been done in a limited area to identify / demonstrate Encroachment pattern in the Krishna basin.

a) Encroachment/ Blocking in River & Tributaries in Rural Area

- Blocking / Choking of Tributaries due to weed/ grass, mainly due to very flat surface, clayey and impervious soil.
- Blocking of Tributaries due to encroachment by farm lands
- Meandering rivers affecting flow of water.

b) Encroachment/ Blocking in River & Tributaries in Sangli City

Krishna river flows from western side of Sangli city, the terrain analysis reveals that the central part of Sangli (east -west trending) has higher elevation than northern and southern part of Sangli city. The flood inundation is observed in low lying areas in northern &southern part of Sangli, having elevation less than high flood level. A tributary in northern part of Sangli city flows over a wide and flat topography, before meeting Krishna river and in this area, Sangli city has extensively urbanised in last 10 years. This area got affected in 2019 floods.

Similarly, the southern part of Sangli city having south to south -south-east (SSE) slope, got inundation in 2019 floods. This area is also urbanised a lot in last 10 years. Few snap shots, attached in the report, shows changes in urbanisation and flood inundation 2019, particularly in Venkatesh Nagar, Kalanagar, Krushnanagar, Sanjay Nagar, Ram nagar& Indira nagar localities. Local Urban body of Sangli need to carry out a detail ground survey of such areas to remove encroachment in tributaries / storm water drains.

c) Encroachment/ Blocking in River & Tributaries in Kolhapur City

Panchganga River surrounds Kolhapur, forming an inverted “U” shape. Rajaram barrage Panchganga has river-gauging mechanism. The terrain analysis reveals that the central part of Kolhapur city is at higher elevation than western, northern & eastern part of the city. Lot of urbanisation has done in the western, northern & eastern part of the city, where the terrain is almost flat and tributaries are wide. Such areas are observed to be at lower elevation than high flood level of 2019, particularly in low-lying areas in western, northern & eastern part of Kolhapur city. These areas have been extensively urbanised in last 10 years.

Few snap shots, attached in the report (see Annexures), show changes in urbanisation and flood inundation 2019, particularly in Padalkar colony, east of Nejdar Colony, KasbaBawda main road to Padalkar colony, Ranmala & new Shahupuri, Kadamwadi,Jadhavwadi, Kannagar, Mahavir College locality, etc. Local Urban body of Kolhapur need to carry out a detail ground survey of such areas to remove encroachment in tributaries / storm water drains.

The main observations on encroachments/ blocking in the rivers are –

- In rural area, tributaries are blocked due to extended agriculture activity in tributaries.
- Tributaries are blocked / narrowed down due to presence of grass / weed in the river channels, thereby reducing water flow.
- Blocking of tributaries and ingress of urbanisation on the flood plain in Cities in un-controlled manner in Sangli and Kolhapur cities.

4.3 Study Report by Water Resources Department, GoM (Flood event 2019)

Krishna sub-Basin authority (WRD) which is mandated for creating Irrigation potential and overall water management actively, played its role in flood moderation process during 2019 flood event.

There are 9 major & 13 medium projects in upper & middle Krishna sub basin, out of which 9 gated projects are crucial for flood moderation namely Koyna, Dhom, Urmodi, Kanher, Tarali, Tulashi, Warna as major and Kumbhi, Kasari as medium projects. Radhanagari, though a major project, is having automatic gates, which provides no manual control for flood moderation.

4.3.1 Rainfall Analysis

To analyze the precipitation, 127 automated rain gauge stations are established in dam catchments and free catchments of Krishna basin. Also 39 fully automated climate stations established to forecast the weather. This data is used to analyze the rainfall and runoff trends in the basin in addition to 22 dams above. Maharashtra State has an average annual precipitation of about 741mm.The rainfall in the state is controlled by southwest and northeast monsoon. About 90% of rainfall occurs during monsoon months from June to October, every year. The high intensity storms prevailing during the monsoon months result in heavy discharges in all the rivers. The continuous and heavy precipitation that occurs in the steep and undulating terrain, finds its way into the main rivers through innumerable streams and watercourses. Flood event 2019 in Krishna basin is an example. Basin experienced an abnormally very high rainfall between 25th July to 13th August, resulting in severe flood in Sangli, Kolhapur, and Satara districts.

4.3.2 Analysis of rainfall records in dam catchments

After the analysis of overall actual rainfall during heavy floods especially in Sangli, Kolhapur, Satara districts situated in Upper and middle Krishna sub-basin, covering dam catchments, it was seen that the observed actual rainfall in various catchments to the upstream of dams varies from 5 to 19 times the normal. Average actual rainfall was about 6 times the normal rainfall in all these catchments bringing abnormal flood to downstream areas.

4.3.3 Analysis of rainfall records in free catchments

The actual rainfall during the first 56 days of the monsoon was measured in 6 rain gauge stations, situated in the free catchments of these three districts. It is observed that the total rainfall during the peak period of 18 days (27.7 to 13.8) measured at the same stations was about 1.6 times the total rainfall during the previous 56 days (1st June to 26.7). Also, the actual rainfall during the event in free catchments was varied from 13 to 29 times the normal rainfall. The overall observed rainfall over the normal was about 18 times. Such abnormal high occurrence of rainfall even in free catchments also aggravated floods in Sangli & Kolhapur districts.

4.3.4 2019 Flood Moderation in Krishna Basin

At the outset, it is to understand that the Dams NEVER create floods, if no dam situation is considered, if they are handled properly during such events. Moreover, the Dams in the Krishna basin have not planned for flood control purpose. No specific flood cushion is kept while designing the storage capacities of the reservoirs. As such, rather than flood control, flood moderation is only possible. Further, even for the flood moderation, it has limitations due to limited channel capacities at downstream created later due to encroachments.

For example, the designed spillway capacity of the Koyna dam is 5743 cumecs, whereas the channel capacity downstream of the dam is much less. In 2019, max discharge of 2971 cumecs(52% of the designed capacity) was released. If the discharge released through Koyna dam is more than 2548 cumecs, the Patan town gets affected. In the flood event 2019, as the free catchment also contributed substantial runoff, there were additional restrictions on dam operators in moderating incoming floods. However, dam operators have moderated the incoming floods to the extent possible, and in no case the flood released from the dam was more than the natural (nodam) condition.

The first challenge in efficient & effective ROS implementation is to achieve good flood moderation in the absence of specific provision of flood absorption space in reservoirs. During normal floods, it may not pose issues but, in emergent situations when very heavy inflows (like 2019 flood event) are experienced continuously for longer durations & high intensities, overall control on flood operation through dams becomes extremely difficult.

The dam operators have to operate reservoirs skillfully. They have to seek the safety of the dams first, by releasing max water as per the inflow received, as well as limiting the max discharge through its spillway upto its design capacity. Normally, the dam operators now a days cannot release the water upto its designed capacity as the waterways in the downstream reaches of the rivers have been drastically reduced due to encroachments in the river portions. Thus, by taking the safety of the dam at his own risk and keeping the probable losses of the properties and lives at the downstream in mind, they have to take the calculated risk by temporarily creating the dynamic flood cushioning, and encroaching upon the FRL or even upto MWL.

Flood moderation within certain limits can be achieved through proper reservoir operation on gated dams. In Krishna valley on all gated dams, reservoir operation schedules (ROS) are updated as per guidelines issued by WRD in Krishna valley. There are 118 numbers of Minor Irrigation dams (3% of total capacity of dams) provided with ungated free open spillways. As stated above there are 9 gated reservoirs in Krishna basin which are flood moderation tools in the hands of dam operators in Krishna basin (K1,K2) having spilling range between 5743 m^3/sec (2.02 lakh cusecs) to 640 m^3/sec (0.23 lakh cusecs) on major dams and 1722 m^3/sec (0.6 lakh cusecs) to 416 m^3/sec (0.15 lakh cusecs) on medium dams.

The principal function of a reservoir is the regulation of natural stream flow for surplus water in wet season & releasing the stored water in the future during dry season for utilization or to supplement the reduction in river flow. In short, the reservoir has to play role of equalizing the natural stream flow & changing the temporal & spatial availability of water. In this process, the conservation demands are best served when reservoir is as much full as possible at the end of filling period.

On the other hand, for the flood control purpose, which is also of a great importance, requires empty storage space to accommodate incoming floods to reduce its severity to downstream areas. Such conflicts need to be resolved through proper ROS. Especially on large basin like Krishna having number of dams constructed on tributaries / main river, integrated reservoir operation approach is essential, where hydrological behavior is altogether different over the basin.

Overall possible efforts were made in Krishna valley to reduce the severity of floods, to the downstream areas through gated dams by keeping out flows within lower ranges than the inflows at the same time & reducing peak outflows. It may be noted that 2019 floods were abnormal & lasted for longer periods, bringing the flood volume almost four times the total storage capacities of all dams together, while one-time flood quantum could only be stored causing unavoidable spills.

Increased generation of runoff from free catchments further downstream aggregated flood intensities in the areas in Sangli, Kolhapur districts. The dams like Koyna, Warna play important roles in Krishna valley owing to their size & surplus capacities in flood moderation process. In addition, dams namely Dhom, Tarali, Kanher, Urmodi, Tulshi, Kasari and Kumbhi also contributed to reduce severity of flood in Krishna valley. However, it shall be noted that, all the dams in Maharashtra have been situated in Sahyadri mountains along the West coast, having very less period (sometimes in hours) to get filled and also for flood forecasting. The efforts made to the extent possible in all these reservoirs operations for three major dams are discussed below.

a. Koyna Dam

Storage capacity at FRL is 2960 Mm^3(105.2 TMC). The catchment area of the ShivajiSagarLake is about 891.78 Sq. km. It has six radial gates of size 12m x 8m. The discharging capacity @ F.R.L (2163'-6") 5083 Cumecs (1.79 lakh cusecs) & MWL (2165'-00") is 5465 Cumecs (1.93 lakh cusecs). The inflow received at Koyna Dam is mainly dependent on the, rainfall magnitude at Koyna, Navja, and Mahabaleshwar. As per experience approximately 25mm (1 inch) of rainfall renders 17 to 20 Mcm (0.6 to 0.7 TMC) of water in the Koyna reservoir.

ROS is prepared for 90% dependability. When heavy rainfall started on 25 July, storage in dam was 1503.54 Mcm (53 TMC), well below crest level of dam (Crest level storage 73.18 TMC). The reservoir was nearly 50% full in terms of storage. Storage in the dam reached at crest level on 31st July.

Releases were started immediately from 1st August initially through power house and then through spillway, as inflow in the dam increased. Maximum inflow in the dam was 5167.26 Cumecs (1.82 lakh Cusec) on 5th August. On 6/8/2019, inflow received in one day at Koyna dam was 343.68 Mcm (12.14 TMC) which was the highest flood inflow received at Koyna dam in the history since year 1961. As a result, the water level in the reservoir kept on rising. Though inflow in the dam was much higher, to mitigate the flood it was decided to encroach guide curves and to keep outflow at minimum level of 3506.83Cumecs (1.23 lakh Cusecs). Maximum difference in outflow and inflow during this period was about 1660.52 Cumecs (58633 Cusecs).

Total inflow in dam during this period was 3343.17 Mcm (118.05 TMC)while total outflow was 1939.35 Mcm (68.48 TMC).Thus, flood of 1403.82 Mcm (49.57 TMC) was mitigated in the dam. Encroachment in ROS, water level in dam was raised above ROS, due to which 13.98 TMC flood could be mitigated when there was severe flood situation in Sangli.Koyna dam was mainly instrumental in moderation of flood.

b. Warna Dam

Storage capacity at FRL is 974 Mm^3(34.40 TMC). The catchment area is 301 sq. kms. Spillway capacity is 2135 cumecs (0.75 lakh cusecs). It has 4 radial gates of 12m x 8 m size.

The inflow intensities at Warna Dam are mainly dependent on very high range of rainfall in western ghat @Patharpunj, Dhangarwada, Nivali and Warnavati. Average rainfall at Patharpunj rain gauge station is around 7000 to 8000 mm. This monsoon Patharpunj rain gauge station recorded highest rainfall of 9872 mm.

ROS is prepared for 90% dependability. This year, when heavy rainfall started on 30th July, storage in dam was 779.76Mcum (27.5 TMC) nearly at crest.Water was released through power house from 24.07.2019 and through spillway from 01.08.2019. The water level on 25th July was 612.70 m and was 6.09 m below the upper guide curve level.Maximum inflow in dam was on 4th and 5th of August 2019. Patharpunj rain gauge station recorded rainfall of 515.00 mm on 4th of August 2019 and further inflow on 5th august 2019 was as high as 1193 m3/sec, hence, outflow was increased to 970 m3/sec. As per further reduction in inflow, outflow was also brought down, which was on lower range.

Total inflow in dam during this period was 829.77 Mcum (29.30 TMC) while total outflow was 527.88 Mcum (18.64 TMC). Thus, flood of 301.89 Mcum (10.66 TMC) was mitigated in the dam.

c. Radhanagari Dam

This old dam having fully automatic gates, though couldn't play role in ROS for flood regulation, absorbed initial heavy flood to the extent of its FRL capacity of 236 Mm^3(8.3 TMC). It has 109 Sq Km catchment areas. Spillway capacity of 7 Automatic gates is 297 cumecs (10500 cusecs). In addition, through 3 old service gates, around 678 cumecs (24,000 cusecs) maximum discharge can be managed. Powerhouse release is 40 cumecs (1400 cusecs). Thus, total spilling capacity can be raised to 1015 cumecs (36,100 cusecs) in emergency. Old service gates cannot be easily operated.

Powerhouse discharge of about 40 cumecs was going on from 10th July 2019. Due to increased inflow from 31st July automatic gates started opening one by one. On 5th August, all 7 gates got opened to its full capacity, hence service gates were opened & total combined discharge was raised from 322 cumecs to maximum of 493 cumecs (17400 Cusecs) on 6th August. Due to reduction of inflow the service gates were closed and automatic gates gradually started closing till 12th August.

4.3.5 Critical Flood Period

Due to simultaneous occurrence of very high range of rainfall over different sub basins in Krishna valley, especially in short period from 27th July till 10th August 2019 caused serious flooding in Krishna, Warna, Panchganga and other rivers and their tributaries in upper and middle Krishna basin. Sangli and Kolhapur districts suffered from severe floods.

Above analysis of flood on various dams clearly show that, higher magnitude of inflows was noticed between 30th July till 10th August & maximum (peak) runoffs from dam catchment were noticed during 3rd August to 6th August. Consequent unavoidable moderated releases from all such dams and further generation of heavy runoff on larger free catchments aggravated flood magnitude over downstream areas, during 3rd August to 14th August.

It was also seen that, in Krishna valley on above 10 crucial dams, considering flood mitigation and safety of dams, during flood events, maximum outflows were reduced, ranging from 50% to 90% of maximum inflows viz in Koyna, Urmodi, Tarali, Warna& Kasari dams. Maximum outflows had also to be kept on higher proportions over maximum inflows in case of Kanher, Tulshi & Kumbhi Dams. Radhanagari being provided with automatic gates, outflows and inflows were nearly same.

It is pertinent to note that during initial period of flood event, in the first instance, heavy floods were mitigated on all major / medium dams upto crest levels, which contribute nearly 2/3 storage capacity to their full capacity. In the year 2019, the dams in Krishna valley were having hardly 14.40 % minimum storages before onset of monsoon as such purpose of flood mitigation as well as conservation had been achieved through dams.

4.3.6 Knowing Flood

In Krishna basin, to ascertain the magnitude of floods in Sangli, Kolhapur & state border, Irwin bridge (Sangli), Rajaram weir (Kolhapur) & Rajapur weir (Maharashtra state border) are being used as gauge discharge measurement location where stage discharge (SD) curves prepared are being used for ascertaining flood values. Overall analysis of 2019 flood event at such location is given below. During abnormal flood event like 2019, the flood levels had reached their historical maximum levels; in such conditions extrapolating SD curve derived values of floods.

a. Shigaon R. G. Station

It is established since more than 2 decade on Warna river u/s of confluence withKrishna. From the inflow data during 30th July to 17th August, the Warna dam outflow & free catchment runoff brought peak discharge of 2252 cumecs on 7th August 2019.

b. Irwin bridge, Sangli

The flood level in the Krishna River Observation point at Irwin Bridge, Sangli on 05/08/2019 at 11.00 am exceeded the Danger level 540.77m. The maximum flood level observed at this observation point was 544.57m (57 feet 5 inches) on 09/08/2019. And the water level started to recedes below danger level on 14/08/19. Sangli city area from 05th August to 13th Aug (9days) was inundated. Maximum Flood discharge observed was 6324 Cumecs on 09/08/19 and out of this discharge released from above dams is 3967 Cumecs.

At flood observation location at Irwin Bridge, Sangli, Total 9356 Sq. Km catchment area out of which 6646 Sq. Km (71%) is of free catchment and 2711 Sq Km (29%) Dam catchment area. Total runoff water received from rainfall was 5435 Mm3 (191.95 TMC) out of which 2677 Mm3 (94.56TMC) (51%) was received from dam catchment area, while 2757 Mm3 (97.39TMC) (49%) was received in free catchment. Free catchment received 1.1 times more runoff outflow than dam catchment. 49 % runoff water is flown from 29 % of dam catchment.

c. Rajaram Weir

The flood level in Panchganga river Observation point at Rajaram K.T. weir Kolhapur had exceeded danger level i.e. 543.30 m on 03/08/2019. The Maximum level reached on 07/08/2019 at 11.00 amwas 546.97 m. The level started to recede below danger level from 14/8/19. Kolhapur city area from 03 August to 14 Aug (11 days) inundated. Maximum Flood discharge observed was 2111 Cumecs on 08/08/19 and out of this discharge released from above dams is 728 Cumecs.

At flood observation location at Rajaram weir Kolhapur, Total 1606.2 Sq.Km. total catchment area out of which 1409 Sq. Km (87.72%) is of free catchment area and 197.2 Sq. Km (12.28%) Dam catchment area. Total runoff water received from rainfall was 2656 Mm3 (93.80TMC) out of which 424 Mm3 (15 TMC) (16%) was received from dam catchment area while 2231 Mm3 (78.80 TMC) (84%) was received in free catchment. Free catchment received 4.25 times more runoff outflow than dam catchment. 16% runoff water is flown from 12.28 % of dam catchment. The flood mitigation in Panchganga basin is difficult since free catchment contribute large area nearing to 88% of total catchment up to Rajaram weir (Kolhapur) where there is no control over the free runoff. In remaining 12% area above dams, out of 4 dams Radhanagari is major dam where also practically no flood mitigation is possible except initial filling upto crest level of automatic gates. It is therefore necessary to convert automatic gates into radial gates. This is all the more necessary because presently provided service outlet gates are quite old and need to be disfunctioned.

d. Rajapur

The flood level in Krishna river at Observation point of Rajapur weir (State border) had exceeded danger level i.e. 536.73 m on 07/08/2019. The Maximum level reached up to 538.10 m on 11/08/2019. The level started to recede below danger level from 15/8/19. Shiroltahsil area from 05 August to 15 Aug (11 days) was inundated. Maximum Flood discharge observed was 9735Cumecs on 11/08/19 and out of this; discharge released from above dams was 3508 Cumecs.

At flood observation location at Rajapur weir State border, out of total 14270 Sq. Km catchment areas, 10953 Sq. Km (76.76%) is free catchment area and only 3317 Sq Km (23.24%)is dam catchment area. Total runoff water received from rainfall was 9344 Mm3 (330 TMC), out of which 3681 Mm3 (130 TMC) (39%) was received from dam catchment area while 5663 Mm3 (200 TMC) (61%) was received from free catchment. Free catchment received 1.54 times more runoff outflow than dam catchment. Thus, 39% runoff water is flown from 23.24% of dam catchment.

It is to note that, the yield at all dams above Rajapur (State Boundary) during flood event was 6630 Mm3 (234 TMC), out of which 2949 Mm3 (104 TMC) was mitigated in dams as storages and remaining 6381 Mm3 (130 TMC) was the downstream outflow from dams. Further the yield from free catchment was around 5663 Mm3 (200 TMC). The measured outflow at Rajapur weir to downstream state was 9344 Mm3 (330 TMC). Thus, it can be seen that, out of total yield of 12293 Mm3 (434 TMC) at Rajapur (Maharashtra), around 2949 Mm3 (104 TMC) could only be mitigated in such a severe 2019 flood event with available flood mitigation infrastructure in Maharashtra.

4.3.7 Flood Vulnerability

Overall picture of high range of inflows of various dams / R.G. stations has been discussed in detail in above sections. Out of 10 dams, Koyna, Dhom, Tarali, Kanher, Urmodi & to certain extent Warna dam (Inundation) is situated in vulnerable catchment area zone above Sangli town. Similarly, Tulshi, Kasari, Kumbhi & Radhanagari dams occupy some vulnerable catchment area in Panchganga basin above Kolhapur Town. High range of inflows especially peaks were observed in all dams above between 3rd – 6thAugust.

Consequently, R.G. stations received high range of inflows, with certain lag time. Peak observations at Irwin bridge (Sangli), Rajaram weir (Kolhapur) & Rajapur weir (State border) were observed during 7th – 11th August.
Inspite the efforts of moderation of floods on various dams, due to high ranges of abnormal inflows over continuous period, it couldn't be avoided the flooding / inundation in Sangli city for 9 days and Kolhapur city for 11 days.

4.3.8 Observations

❖ **Flood event 2019**

Maharashtra experienced abnormally very high rainfall during 25th July to 13th July 2019, which resulted in heavy flooding in upper Krishna sub basin especially in Sangli and Kolhapur districts. Flood mitigation efforts were made by WRD through Reservoir Operation on 9 major gated dams (U/S of Sangli and Kolhapur), which contribute 23% of catchment area of Krishna sub basin

- **Rainfall**: - Sangli, Kolhapur and Satara district received very heavy rainfall of 1918 mm in comparison to 333 mm normal average rainfall during 27th July to 13th August. This was about 6 times of the normal. Further it was observed that, rainfall in dam catchments in above period varied from 5 to 23 times the normal rainfall, and the same in the free catchment was about 12 times. Such high range of continued rainfall in short duration resulted in extreme heavy flooding mainly in Sangli, Kolhapur town and few talukas situated near Krishna and Panchganga river.
- **Runoff :-** in short span, about 6630 Mm^3 (234 TMC) yield was generated in all dams U/S of state border, out of which 2949 Mm^3 in (104 TMC) could be absorbed in dams & remaining 6381 Mm^3 (130 TMC) quantum had to be released D/S in free catchment area which itself generated 5663 Mm^3 (200

TMC) runoff. As such out of total yield 12,293 Mm^3 (434 TMC) generated at state border, hardly 2949 Mm^3 (104 TMC) could be effectively mitigated in such abnormally high flood event of short span. In fact, more than 3 times quantum has flown down in consideration of safety storage built up in dams. It is thus observed that, huge quantity of 9344 Mm^3 (330 TMC) runoff caused serious flood situation in Krishna Valley in Maharashtra and also further in Karnataka.

- **Historical floods: -** Historical flood level data is available at Kurundwad R. G. Station (CWC) near state border from 1914 till 2019 (>100 years). 2019 maximum flood estimated was of highest order nearing 10040 m^3 /sec (3.54 lakh cusecs) at 541.35 m level. Second highest similar flood levels of 539.76 m were observed in 1914 & 2005 year when estimated discharge was about 8361 m^3/sec (2.94 lakh cusecs measured at Rajapur close to Kurundwad). Thus, it can safely be concluded that 2019 flood level was highest in last 100 years (rather > 100 yrs.) & 2019 flood intensity was 1.2 times that of earlier observed floods.

- **Reservoir Operations: -** Excluding Radhanagari dam (automatic gates), there are 8 major gated dams (5284 Mm3 / 186 TMC storage), out of which Koyna &Warna dam having total capacity of 3955 Mm3 (140 TMC storage) play major role in mitigation of flood through proper Reservoir Operation. In all major projects like Dhom, Tarali, Kanher, Urmodi, Tulshi, Kasari, and Kumbhi for effective flood control, outflows were guarded to minimum possible ranges.

  In spite of all efforts, considering the extreme (abnormal) higher floods, limited mitigation capacity of dams, absence of flood absorption capacity, reduced channel capacities of river D/S, Sangli and Kolhapur district had to face severe flood during this year owing to their peculiar locations near rivers.

❖ **Flood in Sangli& Kolhapur town**

- **Sangli town** – Flood impinge in Sangli town when water level cross danger level at 540.77 m with discharge at about 3874 m3 /sec (1,36,755 cusecs). Flood level crossed danger level on 5th August. It went on increasing to maximum 544.57 m (+ 3.8m) on 9th August when highest flood of 6324 m3 /sec (2, 23,399 Cusecs) had passed over Irwin bridge, which was highest in known history.

  Then after reduction in runoff from U/S, it touched back danger level on 14th August. Thus, Sangli town was continuously in flood for nearly 9 days. It is noticed that 5832 Mm3 (206 TMC) was flown over Irwin Bridge, which was maximum up till now. During peak discharge about 40 % was from U/S dams it is noticed that 2019 flood level was higher by 1.3 m compared to 2005 flood.

- **Kolhapur town** – Danger level for Kolhapur town (Rajaram Bandhara) is 543.30m with discharge at about 1778 m3 /sec (62755 cusecs). Flood level crossed danger level on 3rd August. Maximum water level reached was 546.97 m (+3.67m) On 7th August when peak discharge was 2111 m3 /sec (74560 cusecs). It is to note that, there is practically very little flood mitigation possible in Panchganga basin because 88 % catchment is free & within remaining 12 % dam catchment area, the Radhanagari dam, which is a major one, is having automatic gates. Tulshi, Kumbhi, Kasari dam have very low mitigation capacities.

  Flood again touched back danger level on 14th August. Thus, Kolhapur city was flooded for nearly 11 days. It is noticed that due to continued rain, higher range of flood was observed till 13th August in Panchganga basin. It is noticed that 2019 flood level was 1.8m higher than 2005 flood. It will be prudent to convert automatic gates into radial gates on Radhanagari dam.

- **General -**

  From above, it can be concluded that 2019 flood occurred was abnormally higher flood experienced due to higher intensity of rainfall over larger areas within short durations, it's continuity over extended period and simultaneously on different sub valleys also enhanced its severity. During this severe flood event, reservoir operation on various major / medium dams could not play effective role in prevention of heavy flood in Sangli & Kolhapur towns & their talukas adjacent to Krishna river & it's tributaries, in consideration of their locations, size, dynamic flood storage space, reduced river channel capacities, bigger size free catchments receiving simultaneous heavy rainfall of longer periods.

  Floods are natural phenomena and are inevitable. In consideration of broad causes of flood in order viz meteorological, topographical and human factors, the last factor need to be studied seriously and concerned measures should be adopted carefully. Damming and diversion of rivers is one of the human factors. Here, the scientific reservoir operation by dam operators is expected which is one of the important non structural measures to control flood in given situation. Well prepared ROS is the basic tool available on all gated dams with him thus broadly rule curve are relied upon by him during flood mitigation. Characteristics of particular river basin should also be known to him as such feedback of information of past reservoir operations carried out in different hydro meteorological events should be also available with him as experience. It is seen that such past details of hydrological data, flood event and consequent efforts made the then in RO is not readily available with dam operators who are not permanently working and transferable in WRD. It is therefore necessary to introduce the system of preparation of Reservoir Operations Report for each year by Dam operators to apex basin level office (Regional Chief Engineer) who should compile such reports at the end o rainy season and with due modifications arrange to be disseminate at EE, SE for their past record.

River channel capacity d/s of dams and around cites have been seriously lowered due to encroachment in river banks, natural sedimentation etc. since few decades, which bring limitations in flood mitigation to dam operators. However, present accurate details of such river channel capacities at various vulnerable reaches/locations is hardly available to him, so as to control releases appropriately. it is therefore proposed to carry out detailed afresh survey on all river reaches below major / medium dams, vulnerable locations near cities, major bridges and heavy cross drainage works in time bound programme of 5 years on priority. Regional Chief Engineer (apex basin level authorities) should ascertain present river channel capacities basin on such survey.

Presently exchange of data like water levels, inflow/ out flow, storage on various dams / weirs in Maharashtra to Karnataka is given & information of only Almatti is dam is received from Karnataka to Maharashtra during flood. It is felt necessary to get the information from various big structure like Hippargi barrage, Galgali & Padsalgi weirs etc which are nearer to boundary of two states, for proper & effective monitoring of flood, especially during severe floods.

Individual ROS for dams was carried out in early monsoon period during July. However, in first week of August abnormal rainfall caused emergency situation in which integrated reservoir operation had to be started which happened to be a difficult task to dam operators so also to basin authorities. It needs continuous monitoring and analysis of hydrological Information and many times it had to be done on experiences gained. Though basin simulation is an answer, its scientific implementation in practice is difficult in emergent situation. So best scientific alternative for precise determination of reservoirs releases during emergency situations can be effectively implemented adopting

Real Time Data Acquisition System (RTDAS) and Real Time Decision Support System (RTDSS). In Krishna basin this system is established, which needs further updation and setting up of RTDSS.

4.4 Study Report by Water Resources Department, GoM (Real Time Decision Support System for Krishna Sub Basin)

4.4.1 Background

Hydrology Project was initiated by Ministry of Water Resources, (MoWR) Government of India in the year 1995.The Government of Maharashtra was also one of the implementing agencies of Hydrology Project. The project was carried out through two phases-Hydrology Project Phase-I (1995-2003) and Phase –II (2006-2014). As MoWR and most of the states were aware of the importance of Hydrological Information System (HIS) and it use in water sector, MOWR became instrumental in introducing Real Time hydro-met monitoring.

In response to the objectives of MoWR, Government of Maharashtra had proposed to upgrade the existing Hydrological Information System (HIS) developed under HP-I (built on conventional network) with real time data acquisition system (RTDAS) for Krishna and Bhima basins. Simultaneously, it was proposed to develop a real time streamflow forecasting (RTSF) and reservoir operation system (ROS) in Krishna and Bhima river basins to manage the floods and operate reservoirs optimally for multiple uses.

After 2005 and 2006 floods, the technical committee constituted (Vadnere committee) in January 2007 by Government of Maharashtra, for suggesting precise determination of reservoir releases during emergency situation in the state of Maharashtra also strongly recommended adoption of Real Time Flood Forecasting System.

Based on the objectives, set forth by MoWR and based on the recommendations of the Committee, development of Real Time Decision Support System (RTDSS) for Krishna & Bhima Sub Basins was taken up through Chief Engineer, Planning and Hydrology, under WRD Maharashtra, in October 2009 under Hydrology Project Phase-II and the task of model development was completed in October 2013 and the system was made functional from monsoon 2014

.

4.4.2 Preamble and Data used

The RTDSS model was intended to help WRD officers to better manage floods in the Krishna Bhima sub basin by giving two day's advance forecast of water levels and discharges at reservoir locations and at various downstream forecasting points on the rivers.

Real Time Decision Support System (RTDSS) comprises of two main parts (A) Real Time Data Acquisition System (RTDAS) and (B) Real Time Stream Flow Forecasting & Reservoir Operation System (RTSF&ROS).

a. RTDAS

The hydro-meteorological data observation network, known as Real Time Data Acquisition System (RTDAS) for 249 stations was established in Krishna Bhima sub basin to observe rainfall, reservoir and river water levels, spillway gate opening in case of gated reservoirs and to observe some meteorological parameters like maximum temperature, minimum temperature, relative humidity, solar radiations, wind direction, wind velocity atmospheric pressure etc. The data thus observed at various remote locations which are equipped with un-interrupted solar power supply, is transmitted to the server through GSM/GPRS/VSAT telemetry system to the Data Centre situated at Sinchan Bhavan, Pune. This hydro-meteorological data along with the rainfall forecast issued by India Meteorological Department (IMD) are the main input to the RTSF&ROS Modeling system.

The real time data and the results of Real Time Stream flow forecast are made available to all stakeholders of the basin on website; http://www.rtsfros.com/mahakrishna, for viewing/use.

b. RTSF&ROS Modeling System

The total 69,967 sq.km area of Krishna and Bhima sub basin in Maharashtra was sub divided in 122 small sub catchments to simulate the spatial variation in the lateral inflow to the river system for the purpose of model development. The NAM (Rainfall- runoff) model is used and coupled with the MIKE models forming part of the integrated modeling system. NAM model simulates the runoff from the various sub catchments in the system on the basis of an input of historical and predicted precipitation and potential evapo-transpiration.

The MIKE 11 hydrodynamic model is used for the two combined basins. Based on the rainfall- runoff from the NAM, a continuous routing of the flows and flood waves through the main rivers and reservoirs of the basin was carried out by the MIKE 11 hydrodynamic model. It describes the propagation of flood waves through the river and reservoir system and after simulation gives the forecasted discharges and water levels throughout, for application to short term forecasting.

Flow forecasting involves the use of hydrological and hydraulic models to transform measured and predicted rainfall in a catchment to a forecast time series of flows and water levels in a river system. They are typically used to provide warnings to residents at risk during times of flood, but can also be applied to predict inflows to reservoirs to optimize the operations and hydropower production. All results from the forecast simulations were presented on a WEB portal http://www.rtsfros.com/mahakrishna. The web page has provision for display of four different data types: discharge, water level precipitation and data from reservoir (water level, inflow and outflow).

c. Model Used For 2019 Flood

During the 2019 flood, the forecasted water level and discharge bulletins were attempted to be generated as the outputs of the RTSF&ROS model on trial basis. River water levels at 94 locations in the main rivers and reservoirs were obtained through www.rtsfros.com/mahakrishna. Moreover, a separate WhatsApp group named “rtsfros model users” was formed to disseminate the bulletins for discharges, water levels, daily and current rainfalls of the basin. Real Time Rainfall Data statistics Bulletin was made available to all. However, the model has certain limitations as indicated in this report, thus needs some updating to match the outcome of this exercise to the actual observed data.

4.4.3 Method of Study

The NAM (Rainfall- runoff) model is used and coupled with the MIKE models forming part of the integrated modeling system. The hydrological model maintains a quantitative memory of the water accumulated in the catchments in the form of soil moisture, and ground water. This accumulated water volume will be released as runoff to the main rivers during the succeeding periods, simulated by the hydrological model. Converting the predicted precipitation to runoff hydrographs, the model provides a quantitative response to the predicted weather forecast.

4.4.4 Conclusions of Study

Basin Simulation Division, under Water Resources Department, Pune based on their observations has drawn following conclusions.

Analysis of 2019 flood event for Krishna sub basin was carried out using the results of RTDSS model at three gauge and discharge stations, namely Irwin bridge (Sangli), Ankali Bridge and Kurundwad (state border) all on river Krishna, Irwin bridge station being the upstream most, then Ankali bridge in between and Kurundwad being downstream most. The period for analysis is taken as 25th July 2019 to 13th August 2019 during which there were continuous heavy rainfall spells. It was tried to compare at all the above-mentioned gauges and discharge stations, the observed and simulated discharges and water levels.

4.5 Reports by Central Water Commission, New Delhi

4.5.1 Krishna River System:

The Krishna river system is the second largest east flowing river in Peninsular India, draining into the Bay of Bengal. The source region of the river is the Mahadev ranges of western ghats near Mahabaleshwar, about 64 kms from the Arabian Sea at an altitude of 1337 m above mean sea level and flows through the states of Maharashtra, Karnataka and Andhra Pradesh. Details of state wise drainage areas are as below:

State	Length (km)	Drainage area (sq.km)	Percentage
Maharashtra	306	69,425	27
Karnataka	483	1,13,271	44
Andhra Pradesh	612	76,252	29
Total	**1401**	**2,58,948**	**100**

4.5.2 Flood forecasting setup

At present there are 24 flood-forecasting stations under Krishna basin, out of which 5 are in Maharashtra, 10 in Karnataka, 2 in Telangana and 7 in Andhra Pradesh.

4.5.3 Meteorological Situation

During the period ending 1st June to 7th August 2019, the upper Krishna and Bhima sub-basins have received large excess and excess rainfall respectively, and in the week period from 1st August to 7th August entire Krishna basin has received large excess rainfall with a max excess of 396% in upper Krishna basin. Mahabaleshwar in the source region of Krishna has recorded a cumulative rainfall of 5740 mm upto 7th August as compared to a normal of 3600 mm.

4.5.4 Flood Situation

In association with such heavy rainfall scenario, Krishna basin started experiencing floods with first flood forecast exceeding the criteria of 1425 cumecs at Almatti dam from 7th July onwards and continued upto 17th July 2019. After a low flow period, again the flood started picking up from 28th July onwards and continued with very heavy inflows of more than 8000 cumecs from 5th August onwards with a forecast of around 12000 cumecs for 8th august 2019. During the period from 28th July to 7th August, Almatti dam reached a peak level of 519.35 m compared to its FRL of 519.60 m on 28th July 2019. After that as the inflow started picking up, the dam started releasing more than the incoming floods and reduced the level from 519.35 m on 28th July to 517.18 m.

First spell of flood forecast in respect of Koyna dam started from 7th July and ended on 13th July and again another spell started from 29th July onwards. Koyna dam reached very near to its FRL on 4th August and it started releasing water with a peak discharge of 1,20,000 cusecs on 7th August.

4.5.5 Problems faced by state governments

While Maharashtra government had requested the Government of Karnataka to reduce the Almatti dam levels to 517.00 m, GoK started reducing the dam levels from 519.35 m from 28th July onwards to 517.18 m.

Further, there is likelihood of flood problems in downstream reaches of Almatti dam, if they resort to sudden releases to bring the reservoir level down drastically at short notice, as Narayanpur dam and PD Jurala project in the downstream of Almatti dam have less capacities which will in turn release these waters leading to submergence of large tracks of lands. Hence, Almatti has started reducing the levels slowly from 519.35 m on 28th July when the first rain started to 517.18 m, a drop of around 2.17m.

Since, there was extremely heavy rainfall in source region of Krishna and its tributaries such as Bhima, Koyna, Panchganga, Dudhganga, Warna etc. most of the dams in Maharashtra were forced to release the excess flow as most of the dams have reached near FRL by the time most severe rainfall occurred.

As such it is seen that, heavy downpour of rain in Western Ghats in Maharashtra and Karnataka is the main reason for floods along riverbanks. It is seen that flood discharges released on 7th August are about 4,00,000 cusecs in Almatti dam which is less than 40 % of its design discharge of 12,00,000 cusecs.

4.6 Study Report by Jeevit Nadi and Ecological Society, NGO, Pune, Maharashtra

4.6.1 Rationale of this report

Floods bring devastation to settlements, misery to human lives, and disruption to the city and a financial loss to the nation. All of these were evident in the months of August and September 2019, in whole Krishna Basin. The Districts of Pune, Sangli, Kolhapur & parts of Karnataka state too experienced unprecedented flooding.

A cursory analysis of these floods revealed that it was not just the fury of nature that wreaked the havoc, but a large part of the problem can be attributed to human interventions in the Riverine and stream ecosystems. Indiscriminate construction in the floodplains, blocking and choking of river channels, straightening the natural flows, removal of bank vegetation and many other actions have added to the crisis. Much of the pillage could have been avoided, had the river been left in its natural state.

4.6.2 Introduction

The flow and flood of each river depends upon the rainfall, soil composition, climate and the landscape. Intensity of floods also varies. The rivers of Maharashtra receive heavy rainfall in the source regions, medium to high to the east of the source regions and scanty rains in the rain shadow belts of the floodplains.

The floods of 2019 have had a devastating effect on many districts of Maharashtra, which has led to many questions being raised. This year's rainfall not only broke records of recent times but also brought devastating situations in Satara, Sangli, Kolhapur and Pune districts. The two key differences this time were the high intensity rains and their duration.

4.6.3 About floods

There are three types of flooding that a river can experience.

- ❖ **Fluvial flooding** – Excessive rainfall over long periods can exceed the capacity of the river to hold and transport water. This causes swelling in the rivers and water spills out from the channel.

- **Pluvial flooding** – This flood can be away from rivers. In this situation, the land where excessive rainfall occurs is unable to effectively drain or absorb the rainwater. This leads to inundation of low-lying areas.
- **Coastal flooding** – This type of flood occurs near the seacoast at the mouth of rivers.

Many urban areas experience both fluvial and pluvial floods, because locations next to rivers are preferred settlement areas and urban hardscapes reduce water percolation, which results in surface inundation. Rivers and streams may flood when

A. Prolonged rainfall over the course of several days.
B. Intense rainfall over a short period of time
C. A debris, encroachment or Solid waste causes a river or stream to spill over its banks and inundate the surrounding area.
D. Natural streams & seasonal storm water drains get diverted, blocked or buried.
E. Since Waterways and their structure is not defined properly, and rules for securing their flow and flood regimen irrespective of Urban or Rural land use, the same are encroached, intervened or diverted due to lack of regulations for streams, Seasonal Storm drains, tributaries and Rivers. Many rivers are yet to be mapped for flood lines. Landowners must be guided by rules about Water bodies.
F. Lack Of awareness among common. No one anticipated the loss and damage because No one knew the degree and type of rainfall due to climate change impact.

Though point A & B are the fury of Nature and nothing can be done about the same, point C, D, E, F is in our hands to manage the floods.

Conditions prior to a rainfall event influence the amount of storm water runoff into waterways. Thus, topography, soil conditions, and ground cover play very important roles. Dry soil accommodates greater infiltration of rainfall and reduces the amount of runoff entering streams. Conversely, soil that is saturated as a result of previous rains has a lower capacity for infiltration, and results in higher rates of surface water runoff.

4.6.4 Some common problems in Krishna Basin

1. Modifications and alterations in land use pattern: Due to urbanization, the agriculture or horticulture once existed is not there anymore in peri urban areas. Farmers have sold the lands to building Industry and rampant and uncontrollable construction activity is the current scenario.

2. Soil, Grass cover and Green belts are replaced. Buildings, parking lots and concrete roads have replaced the green and soil cover reducing the percolating areas. Under normal circumstances the green belt work as sponges. But due to development of urban areas, rain water flows through storm drainage pipes and sewers and ditches which get choked with solid waste in them.
3. Large scale excavations being carried out for large construction projects generate huge amount of debris which is conveniently dumped along the banks of Rivers. New Bridge constructions or Drainage works also heavily obstruct the flow of rivers, particularly all happening simultaneously.
4. The debris along the banks also raises the bed level of river and the flow is often obstructed during heavy rains.
5. During expansion of city, the width of the river gets reduced drastically. This obstructs the flood carrying capacity of the river.
6. Construction of retaining walls along the channel banks may create problem of flood in monsoon due to reclamation of land in flood zone by builders or societies.
7. The springs along the river act as hyporheic zones for the river, which recharge the ground water and replenish the catchment of river. Constructions happening without mapping of these sources have choked all the live springs.
8. The embankments along the banks change the hydrology of the river.

4.6.5 Urban, Rural and peri-urban areas

- **Urban**: Pluvial flood risk can be heavily mitigated in new developments through a combination of avoiding the highest risk locations, investments in drainage systems, flood proof building designs and surface water management involving green and blue space and blue corridors.

  Policies and Rules shall be different for new constructions, old constructions and Institutions and Govt. buildings, retrofit designs for old structures to manage storm drains, upgradation of existing storm drainage systems.

  A key challenge remains for existing built-up areas at high risks, although gradual upgrading of drainage systems and surface water management can reduce risk.

- **Rural / Agriculture:** Management of floods in Rural / Agricultural lands require different planning and policy decisions. All these lands in flood plain region could have a balance of mosaic type land use. Combination of turning fallow lands in grass lands, Agro-forestry along the banks, land acquisition for protecting banks could be the part of policies, rules and laws.
- **Peri-urban:** Management of floods in these areas are critical and play a major role. Due to improper and insufficient planning, lack of infrastructure and absence of co-ordination between Gram Panchayats and Municipal Corporations are the

consequences of floods. Since all peri urban areas are spread around the city developments irrespective of source regions of flood plains, a sustainable urban design and management of Natural resources will play a key role in this sector.

4.6.6 Proposed solutions:

A holistic approach to flood mitigation includes the following measures -

1. Long term Ecological measures to improve the river's capacity to manage her floods.
2. Identifying and understanding the waterways, securing their flow and flood regimen. Protection and restoration of all waterways closer to their original state. With help of experts in Ecology, Hydrogeology, Geography etc.
3. Policy measures to improve current situations and ensure further degradation of river is stopped and frequent Assessment of the same.
4. Immediate actions to be taken -Technological measures to improve constructed storm drain network & up gradation & management of the same, measures controlling pollution and solid waste.
5. Long term Agri- Environment Measures in Rural areas
6. River Bank Management after mapping Geomorphologic characters in a given area

    a. Riparian vegetation,
    b. series of wetlands,
    c. Retention & Detention ponds

4.6.7 Tributary streams

Tributary streams contribute to large quantities of water Restoring them closer to their original state is essential. They also bring in polluted water in the urban areas. Ecological treatment like preserving natural geomorphic features and bank vegetation is necessary. Reducing pollution by bio filtration in all the tributary streams is essential so that river will receive relatively clean water.

4.6.8 Agri- Environment Measures in Rural areas

a. Utilization of forests to optimum
b. Using fallow lands for flood control measures
c. Use of retention and detention ponds
d. Public awareness and active participation in managing floods
e. Retaining water through natural media must have priority over soft run off. – Combination of use of detention ponds, retention ponds, wetlands, soil, and vegetation as per the local conditions should be used.

f. In event of heavy and lasting rainfall measures to reduce sediment yield could be more beneficial-creating mosaic of forests in a landscape could reduce sediments. – Riparian woodlands and meadows should be encouraged
g. Restore River's natural flood zones

4.7 Report of Aerial Inspection of Krishna River, during floods 2019

4.7.1 Preamble

In Maharashtra, two weeks of heavy rainfall in July-August 2019 flooded many western districts of state such as Pune, Kolhapur, Satara& Sangli. Upstream dams in hilly portion of Krishna valley were overflowing, at the same time downstream plains were experiencing heavy to very heavy rainfall. Rivers were flowing out of their banks. Communication links, such as roads, railways were disrupted. Large track of plains in Sangli& Kolhapur districts were under water including Sangli& Kolhapur city.

Assessment of situation from ground was almost impossible. It was imperative to have birds-eye-view. It was a time to correctly judge the situation, analyze it technically & find out the reasons thereof.

Accurate information relating to impact of flooding causes and mechanism is of great value while managing and understanding flood risks. Better flood records help us for future planning and development and can assist emergency response preparation. Data collected is also useful to calibrate hydraulic model for flood studies. Information captured during or after floods events is used to help, protect communities, homes, and business and save lives.

In a high-level meeting under Chief Secretary, Govt. of Maharashtra, it was decided to carry out Aerial survey of flood-ravaged districts by the team of Engineers. It was decided to carry out the arial inspection by low flying aircraft.

Accordingly, four engineers from state of Maharashtra (some of them were also the members of this study group) were sent on this mission. Government of Karnataka also deputed one of their engineers to join this mission.

4.7.2 Path Followed by Aircraft

Aircraft started on 9th August 2019, early morning from Mumbai to Belgavi (Karnataka). After having discussions with the Principal Secretary & Chief Engineer, WRD Govt. of Karnataka at Belgavi airport, the team took off to inspect the flood-affected area.

Aerial survey started from Koyna dam in Maharashtra upto Almatti dam in Karnataka, following the Krishna river main course, confluences inundated banks. The parts of Panchganga River, major cities like Sangli, Kurundwad and adjoining areas were also covered. Major portion of valley was surveyed, video film& digital photographs were taken from the aircraft at an altitude of around 1000 to 3000 feet. The overall inspection of the inundation area, assessment of extent of flood affected zones, severity and reasons of the flood situation could very well understood after this visual survey.

4.7.3 Observations

1. After the visual inspection as birds eye view, one could get the real severity of the flood situation, its large extent of inundation area due to pooling of impounded water, creating a huge flood plain starting from upstream of Sangli city and extended upto state border (around 40 km of river length of Krishna). Before the nature, all man-made activities seem to be very small, is the first lesson learnt.
2. Video footages revealed how water interacted with the landscape, the speed of flow and the paths followed by water.
3. In upper reaches of Panchganga & Krishna Rivers, velocity of flood flow was seen as normal and rivers were flowing within their main courses due to high banks. But, in lower reaches of Panchganga& Krishna Rivers, velocity of water was very low. Rivers were flooded out of their banks, particularly extending the legs of the flood plains at the confluences of tributaries, since the rivers Warna, Panchganga and Dudhganga meet river Krishna as almost head on. Water was spread over the farmlands and adjacent villages & towns.
4. Farmlands, along both sides of the riverbanks, mainly consisting of dense Sugarcane crops, were under water. Dense cultivations were obstructing flow of water, like arresting the floods within farms, thus reducing its velocity.
5. It is observed that many villages and towns were under submergence. It's mainly due to developments within flood plain zones. Encroachments by constructing homes seem to be main cause of griming the situation.

6. Meandering of river in plains area was seen at 4/5 places. These meanders were in close vicinity of each other. At some places, the floodwater spread was so much that the water followed the shorter path, instead of negotiating through the curves. Some curves were located about 1 to 2 km from each other. Spread of water of one curve was touching to banks of subsequent curve. Curved length of river and the siltation in the riverbeds has reduced velocities of the flow, ultimately resulting in increase in spread area by raising the flood levels. It changed river paths in due course of time showing dynamic change in flood prone area. Suitable solution to connect curves of river to reduce flow path seem to be necessary.
7. Contraction and expansion of flows were observed near bridges e.g. Irwin Bridge Sangli, Sangam Bridge Karad. This was due to reduced flow area(waterways) available at these sites. It is observed that the width of the river changed rapidly within city areas, which created rapids and vortexes in river reaches dominated by a funnel effect.
8. It is observed in aerial survey that, many of public utility structures like weirs, roads, causeways, bridges even national highways were submerged. The location and design of these structures seem to be without considering flood plain zones. They shall be seriously revisited in this context.

   From the aerial inspection of the floods, the causes of the flooding and its stagnation for prolonged period were concluded as:

- Stagnation due to meandering of the river course
- Stagnation of water at the confluences due to pooling of impounded water and the local back water effects
- Due to obstructions created by weirs, bridges built across the main rivers, reducing the waterways.

B. Causes / Reasoning of the Event

a) Climatological Changes and Abnormal Rainfall Pattern

Formation of severe cyclones over Arabian sea leading to persistent and simultaneousoccurrence over large spatial areas, of heavy precipitation in short duration, in the catchments of river Krishna and its tributaries, both in dam and free catchments.

Climatological records of the Satara, Sangli and Kolhapur districts indicate that, there is a large variation in space of rainfall. The rainfall in the belt, roughly 25 to 35 kms wide parallel to the crest of the Sahyadri Range, is considerably higher than in the rest of the district. While Mahabaleshwar at an elevation of 1372 metres gets an average annual rainfall of 5886.9 mm, other stations in this belt get annual rainfall ranging between 1684 and 2195 mm. Compared to Kolhapur and Satara, district of Sangli has relatively lesser average annual rainfall of about 670mm.

The persistent intense rainfall activity over the region was in association with an active spell of monsoon started over Maharashtra from 27th July 2019 and resulted in flooding in many parts of Konkan and North Madhya Maharashtra. This was followed by another active monsoon spell from 3rd August 2019, in association with the formation of a low pressure over North East Bay of Bengal and its subsequent intensification into deep depression and westward movement in the subsequent days causing severe flood conditions in South Madhya Maharashtra.

This movement of deep depression system (an active low pressure system with wind speed ranging between 52 to 61 kmph) across central India, resulted in enhancement of rainfall over west coast and in the ghat areas of Madhya Maharashtra with heavy to very heavy rainfall and extremely heavy rainfall events for more than a week period over these places, resulting in severe flood situations.

Dam catchments: After the analysis of overall actual rainfall during heavy floods especially in Sangli, Kolhapur, Satara districts situated in Upper and middle Krishna sub-basin, covering dam catchments, it was seen that the observed actual rainfall in various catchments to the upstream of dams varies from 5 to 19 times the normal. Average actual rainfall was about 6 times the normal rainfall in all these catchments bringing abnormal flood to downstream areas.

Free catchments: The actual rainfall during the first 56 days of the monsoon was measured in 6 rain gauge stations, situated in the free catchments of these three districts. It is observed that the total rainfall during the peak period of 18 days (27.7 to 13.8) measured at the same stations, was about 1.6 times the total rainfall during the previous 56 days (1st June to 26.7). Also, the actual rainfall during the event in free catchments was varied from 13 to 29 times the normal rainfall. The overall observed rainfall over the normal was about 18times. Such abnormal high occurrence of rainfall even in free catchments also aggravated floods in Sangli & Kolhapur districts.

b) Typical Topographical features and river meandering

By study of topographical features of the Krishna sub-basin including its tributaries in Maharashtra, it was found that the River Krishna originates at the highest altitude of 1310 m at Mahabaleshwar, and also the Koyna at the same place, they reach at Pritisangam, the confluence of Krishna and Koyna at Karad at the altitude of about 550 m. thereafter, there is sudden change in the river bed slopes in the Krishna river upto Sangli. While negotiating the Sangli city, the rivers Yerala and River Warna meets Krishna and thereafter within few kilometers there is confluence of River Panchganga leading Krishna to state border. The Krishna river reach from Sangli city to state border is only 30-40 kms long, having very flatter bed slopes and many meanders.

This, typical topographical phenomenon plays major role in slowing down the flood dissipation beyond Sangli city. It was found that, in this reach of river, while there is drastic reduction in the velocities of the flow, the backwater effects of various confluences of its own tributaries further aggravated the problem. While the floods in main Krishna River were yet to dissipate this particular reach, the floods coming from tributaries by simultaneous raining could not even enter in the main river course of Krishna. The situation was worsened the tributaries could not drain out their own discharge, when river Krishna was already flooded, which was spread on the side banks of the tributaries for prolonged period.

This peculiar situation resulted in long term inundation alongside the flood plains of tributaries like Warna and Panchganga, while increasing the back-water effect near confluences resulting in the higher flood levels at Sangli, Kurundwad and Kolhapur cities.

The problem was further aggravated by the meanders in the same reach, because of which very large flood plains were created, submerging huge areas with almost stagnant waters for prolonged periods. During this time, the floodwater created shortcuts, bypassing the meanders straight into the next stretch of the Krishna river, as an effort to dissipate the floods early.

c) Large Encroachments in natural drainage system, reducing flood discharge capacities of the rivers.

Due to the heavy developments around the river stretches and in city areas, there have been large encroachments in the natural drainage system. At many places, the debris in huge quantum is thrown in the river courses (or its tributaries), reclamation of lands are done, leveling of plots, construction of retaining walls by encroaching the river banks etc. due to such activities, the natural waterway of every natural drain (from small nalla, tributary to main river course is encroached upon resulting is less flood discharge carrying capacities at all places.

Contraction and expansion of flows were observed near bridges e.g. Irwin Bridge Sangli, Sangam Bridge Karad. This was due to reduced flow area (waterways) available at these sites. It is observed that the width of the river changed rapidly within city areas, which created rapids and vortexes in river reaches dominated by a funnel effect.

It is observed in aerial survey that, many of public utility structures like weirs, roads, causeways, bridges even national highways were submerged. The location and design of these structures seem to be without considering flood plain zones. They shall be seriously revisited in this context.

d) Poor land drainage system in urban areas: primary, secondary and tertiary levels.

Primary level: The slopes of individual pots in urban areas, open grounds are very gentle. Poor planning of storm water drainage system within city areas and its maintenance leads to stagnation of floods in every part of the city instead of its quick disposal by open drains through plots, its connectivity to road side gutters leading to nearby natural nalla drains. The effective primary storm drainage system shall efficiently and quickly drain out the rainfall received, and be ready to receive further rains, thereby reducing the inundation periods.

Secondary level: The storm water received from the urban areas and drained into the natural nallas passing through the urban areas shall be kept free of any encroachments. The bed slopes of these nallas shall be maintained to quickly drain out the floods received into the main river course.

Tertiary level: The main river courses nearby cities shall accommodate all the flood water received from its tributaries and nallas during the heavy rainfall. Due to the encroachments in the river itself, flooding occurs inundating large areas due to local back water effects and stagnation of waters due to poor drainage systems.

e) Sedimentation in main rivers – at confluences, along meanders

Due to sudden changes in the riverbed slopes and stagnation of floods for prolonged periods, sedimentation occurs in the river courses and also in natural drainages. This results in raising the riverbed levels, thereby raising the flood levels and extended inundation areas. At the meanders, the sedimentation occurs in the inner side of the curves, thus deflecting the floods further to the outer sides.

f) Absence of flood absorption capacities in reservoir planning of existing dams

At the time of project planning, the reservoirs are never planned with the provision of special cushion for flood absorption. The dead storage is designed to accommodate the silts and live storages are planned for complete utilization of the water stored. The Reservoir Operation Schedules are designed to assure for full storages by the end of the monsoon period. For ungated spillways or dams having fully automated gates, there is no manual control to moderate the incoming floods.

g) Other reasons

The floating debris coming along with the floods got stuck to the walls of weirs, bridge piers and parapets, thus obstructing the flow resulting in temporarily increasing the flood levels in the reach. Regular maintenance of these structures shall also be done.

Chapter 5 – Hydrodynamic Analysis of Flood Event 2019 (A Study Report on Effect of Almatti backwater in Maharashtra) (TOR2)

5.1 Review of Earlier Studies

5.1.1 Rama Prasad (2001):

I. **Scope of study:** The study is focused on backwater and sediment effect of Almatti dam on river Krishna.

II. **Methodology used:** Standard step method is used to calculate back-water effect. Only 26 cross sections out of surveyed 40 were used. Rugosity coefficient has been computed for bank portion i.e. the limits within which the river flows normally every year, by method described in Ven Te Chow's Open Channel Hydraulics, using discharge at three gauging stations Galgali, Kurundwad and Arjunwad. Manning's "n" for high flows with respect to land use and land cover are also computed. HEC-6, application is used for sediment analysis. Velocity is assumed to be constant along the cross section.

III. **Findings:**

   I. Expected cumulative changes in the bed level are very small. Sediment deposition will not be a serious problem.

   II. Backwater profiles for Almatti dam for return period of 25, 50, 100, 1000 and PMF merges with the normal water profiles downstream of Hippargi barrage and backwater of Hippargi barrage merges with normal water profiles well downstream of the Maharashtra State border.

IV. **Observations / Inputs for further studies:**

   I. Rugosity coefficients computed seems to be on scientific basis which can be used for further studies with interpolation for other cross sections.

5.1.2 S. V. Vedula and Ranga Raju:

I. **Scope of the Study:**

   I. Backwater calculations, without siltation for post dam condition, for FRL of 519.6 meter and 524.256 meters, for 5 year flood, 25 year flood and PMF.

   II. Sedimentation in reservoir for post-dam condition for FRLs 519.6 meters and 524.256 meters.

II. **Methodology used:** HEC-6 application is used which has capability of modelling structures. Out of 40 surveyed cross sections 29 cross sections were used. Manning's "n" values of 0.025 for bed and 0.05 for banks are used. Hippargi barrage is taken as internal boundary with crest level at 516.64 m.

III. Findings:

I. In unsilted condition the backwater is confined in Karnataka territory only.

II. Rise in bed level of river Krishna, due to siltation, occurs in Karnataka only. It requires at least 290 years of operation of the reservoir for siltation to occur in Maharashtra territory.

III. Even after siltation over a period of 46 years, the backwater of Almatti does not extend into the territory of Maharashtra.

5.1.3 Central Water Commission Feb. 2006 (F.E. & AS.A Directorate):

I. Scope of the Study: The scope of the study was limited to assessment of likely submergence in the State of Maharashtra due to Almatti dam and the Hippargi barrage constructed in the State of Karnataka.

II. Methodology used: CWC has carried out study with MIKE-11 model, using finite difference solution of St. Venant's equations of conservation of mass and momentum. In the model 29 cross sections in 265 km river length are modeled. The study was carried out for 5 different scenarios.

III. Conclusions: The study concluded that:

I. The backwater profile of Almatti dam for FRL at 524.25 m and PMF discharge of 31,000 cumecs at Almatti and 24, 609 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs, merges with water profile in natural condition of the river Krishna at 221km from Almatti dam.

II. The backwater profile of Almatti dam for FRL at 524.25 m and PMF discharge of 31,000 cumecs at Almatti and 24, 609 cumecs at Hippargi, considering Almatti reservoir only, merges with water profile in natural condition of the river Krishna at 145 km from Almatti dam.

III. The backwater profile of Almatti dam for FRL at 519.60 m and PMF discharge of 31,000 cumecs at Almatti and 25845 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs, merges with water profile in natural condition of the river Krishna at 221km from Almatti dam.

IV. The backwater profile of Almatti dam for FRL at 519.60 m and PMF discharge of 31,000 cumecs at Almatti and 25,845 cumecs at Hippargi, considering Almatti reservoir only, merges with water profile in natural condition of the river Krishna, at 126 km from Almatti dam.

V. The backwater profile of Almatti dam for FRL at519.60 m and discharge of 10,100 cumecs at Almatti considering Almatti reservoir only, merges with water profile in natural condition of the river Krishna at 175 km from Almatti dam.

IV. **Observations:** It is seen that the observed water levels during 2005 flood are considerably higher than that simulated for the scenario (iii) above viz. at Rajapur simulated backwater profile elevation is 534.26 m whereas observed value is 536.65 m.

5.1.4 Study for Prediction of Bed Levels of the River Krishna and Back Water Profiles on Upstream of Almatti Dam (Prof. R. G. Garde, Pune March, 2007):

I. Scope: The scope of this study was to predict bed levels of river Krishna, in a reach of about 270 km, from Almatti dam up to Sangli, taking into consideration sedimentation that likely to occur, after 10, 20, 30 years of operation of the Almatti reservoir and to predict the backwater profiles for flood discharge of different return periods viz. 25, 50 years and PMF, in the river Krishna and its three tributaries viz. Warna, Dudhaganga and Panchganga.

II. Methodology used: One dimensional model, capable of simulating tributaries and local inflows was considered to be appropriate for this study. Accordingly, HEC-6 Version 4.1 (August 1993), the application developed by Hydrology Engineering Center (HEC), USA was used. Along with the river Krishna, some reaches of Warna, Panchganga and Dudhaganga rivers near its confluences with river Krishna were also simulated. Cross sections modeled were 40 in 270 km length of the river Krishna, 3 in 32.5 km reach of the river Warna, 14 in 71.25 km reach of the river Panchganga and 6 in 64.3 km river reach of the river Dudhaganga. However, K.T. weirs and other low-level hydraulic structurers were not simulated, on the assumption that these structures will be fully submerged and will be of less significance during periods of floods. Hippargi barrage was simulated as an internal boundary with appropriate waterway and weir crest level for the condition of fully opened gates.

III. Findings:

I. Predicted bed levels are higher by 0.1 to 0.4 m along the river Krishna in the reach of Maharashtra. However, at the confluence with Panchganga and Dudhaganga, rise in bed levels are substantially higher (2 to 4 meters). This may be due to relatively flatter bed slope of river Krishna as compared to steeper bed slopes of the tributaries. Predicted rise in bed levels after 10 years at confluences of Panchganga and Warna are 1.15 m and 0.87 m respectively.

II. Back water effect due to Almatti near the State boundary is marginal. (0.02 m for 25-year returns period and 0.05 m for PMF)

III. Predicted water levels at Sangli-Wadi bridge for return floods of 25 years (9061 cumecs), 100 years (12,183 cumecs) and PMF (16,318 cumecs) are 546.50 m, 548.75 and 551.94 m respectively.

IV. Observations/ Inputs for further studies:

I. It is necessary to verify silted level of river Krishna on downstream of confluences with Warna, Panchganga and Dudhaganga. If there is significant accumulation of silt in last 12 years, the dredging of this may facilitate smooth passage of flood.

II. Study of historical flood events reveals that the flood peaks of river Krishna and its tributaries occurred on same day in most of the flood events. The same scenario is expected in future also. Hence, integrated reservoir operation is of paramount importance.

III. Almatti discharge in the year 2019 is 16,143 cumecs which is near to 25-year return flood of 16,968 cumecs as estimated in this study. However, observed flood at Sangli Bridge is 6361 cumecs which is very much less than 25 year estimated flood of 9061 cumecs at Sangli Bridge.

5.1.5 Effect of storage of water in Almatti dam on the submergence / flooding along river Krishna in the State of Maharashtra By Prof. R. J. Gadre and S. Y. Shukla (November 2007):

I. Scope: The prime objective of this study was to investigate the effect of storage of water in Almatti dam on the submergence / flooding along river Krishna in the State of Maharashtra.

II. Methodology used: Mathematical model “CHARIMA” developed by Lowa Institute of Hydraulic Research (IIHR) in 1988 and capable of simulating unsteady flow in the network of river channel comprising of junctions, bifurcations, loops etc. was used. However, effect of bends could not be accounted in this model. Total 48 cross sections of the river Krishna were modeled. Daily/hourly hydrographs for the river Krishna and various tributaries were used as upstream boundary conditions. The Almatti reservoir water levels at the dam site was used as downstream boundary condition. Since the hourly discharge data at gauging station at Karad and gauging stations on tributaries was not available, attempt was made to generate hourly discharges from the daily discharge data available. Average value of manning’s roughness of 0.0265 was used.

III. Findings:

I. There is practically no effect of Almatti operations and Hippargi barrage on flood levels at Sangli during the 2005 flood event.

II. Flood discharge in Warna affect flood levels at Sangli to some extent. If the Warna discharge is controlled to 20 % that of the actual discharge, predicted water levels in Sangli was reduced by 0.7 m.

III. Effect of discharge of Panchganga on Sangli flood level is marginal. If flood discharge is controlled to 20 % that of the actual maximum observed discharge, the reduction in Sangli floods level is only 0.08 m.

IV. Observations / Inputs for future studies:

I. The Sangli flood can be moderated to some extent by staggering the peak discharges in river Krishna and Warna.

II. The observed water level at Sangli Bridge is deceptive. At the bridge location water level dips by about 0.3m to 0.5m due to effect of constriction of natural cross section at bridge location.

5.1.6 Mission for Geo-Spatial Applications by DoST (2011):

I. **Scope:** Considering the experience of the mission, in flood modeling on the rivers Sabarmati and Tapi, the task of analyzing the impact of raising the height of Almatti dam from 519.6 meters to 524.256 meters was entrusted to this mission.

II. **Methodology Used:** The ground surveyed cross sections, 40 in number, were found inadequate, hence sections were extracted from Digital Elevation Model (DEM) generated and merged with the bathymetric data. Total 140 sections were digitalized in 270 km length of the river Krishna; 16 cross sections digitalized in 32 km reach of the river Warna; 9 cross sections in 19 km reach of the river Panchganga; 10 cross-sections in 25 km length of the river Dudhaganga and 18 cross sections in 28 km length of the river Ghataprabha.

HEC-RAS, the computer program, developed by US Army Crops especially for analysis of backwater effect of highway bridges and culverts, was used. HEC-RAS is designed to perform 1-dimensional hydraulic calculations. The basic computational procedure is based on solutions of energy equation. Energy losses are evaluated by friction (manning's equation) and contraction /expansion coefficients. The momentum equation is utilized in situations where the water surface profile is rapidly varying. These situations include hydraulic jumps, hydraulics of bridges and evaluating stream junctions.

Cross section specific, values of Manning's "n" as provided in Rama Prasad's report were used as these values were calculated on the basis of relevant physical parameters. These values vary from 0.0239 to 0.0283.

III. **Findings of the Study:** The findings of this report based on unsteady flow analysis are:

I. Unsteady state analysis, carried out by setting time step interval for output hydrograph as 2 hours, indicate that water level rise is 40 to 70 cm during flood events of 1994, 1997 and 2005, both in river Krishna and Panchganga, if the Almatti dam height is raised from 519.6 meter to 524.256 meter. The additional submergence would be about 11.7 %., 11.5 % and 6.7 % for the floods of the years 1994, 1997 1nd 2005 respectively.

II. Raising of the Almatti dam, from 519.6 meter to 524.256 meter will cause additional submergence of 118 sqkm in the State of Karnataka.

III. The observation of Tojo Survey Report, (Tojo was tasked by the Tribunal for the specific purpose) regarding sediment deposit is confirmed by this study. As per the Tojo report there has been a rise in bed level due to sedimentation that has occurred between 2000 to 2009.

IV. The scan of satellite imagery from Almatti dam along river Krishna up to Sangli shows that sediment deposits appear to take place at locations:

i. Where the river has sharp bends and meanders;
ii. Where the river is constricted by any structured bridges etc.;

iii. Deposition takes place mostly at confluence points; there has been a shift of confluence points of the tributaries due to sediment deposits. Model output also indicates deposition, mostly at confluence points.

V. The predicted sediment profile after 30 years, with FRL of Almatti dam at 524.256, indicates substantial deposition, mainly at Warna, Panchganga and Dudhaganga confluences. Some erosion is also observed near Sangli. The sediment deposit of 2.41 m, 1.74 m and 1.32 m is observed at confluence of Warna, Panchganga and Dhudhaganga respectively. These, deposits are significant to cause changes in the river regime and increase in the future flood events.

IV. Inferences and observations for further studies:

I. The changes in river bed levels, particularly at confluence points need to be studied from satellite imaginaries and need to be modeled after ground validation at representative locations.
II. More number of cross sections needs to be modeled.
III. Maximum possible structures on river need to be modeled.
IV. Narrow interval of flood hydrograph (1 or 2 hours) needs to be used.

5.1.7 Backwater study for Almatti dam & Hippargi barrage on river Krishna (For event of year 2019):

Study Report by Central Water Commission, New Delhi
(Effect of Almatti backwater in Maharashtra – by Er. Nityanand Rai, CE)

I. Scope of the study: Scope was limited to study backwater effect of Almatti dam considering flood parameters of the year 2019.

II. Methodology used: This study has been carried out by using MIKE-11 mathematical model. Total 30 number of surveyed river cross sections has been used to represent the river reach of about 275.64 km length from Sangli to Almatti. Rating curves (stage-discharge relationship) of Sangli, Mhaisal and Rajapur weirs based on the observed stages and discharge parameters during the 2019 flood event and also the rating curves of Hippargi barrage and Almatti dam are modeled. Results with different scenarios for Manning’s “n” of 0.025, 0.028 and 0.030 are evaluated. Almatti dam flood disposal level of 518.60 m corresponding to discharge of 16,143 cumecs in flood event of year 2019 is considered.

III. Findings of the study: The findings of the study are:

I. Backwater effect of Almatti dam get diminished just downstream of Hippargi barrage.
II. Backwater effect of Hippargi barrage extends up to 40 km downstream of Rajapur K.T. weir.
III. The afflux at Hippargi barrage is 57cm.
IV. Afflux at Sangli, Mhaisal and Rajapur K.T. weirs are 1.31m, 2.87 m and 2.01 m respectively. (For Manning’s “n” 0.030 m).

V. The higher values of observed afflux values may be due to congestion of flow at Warna- Krishna confluence and Panchganga-Krishna confluence.

IV. Observations and points for further studies:

Afflux at Sangli, Mhaisal and Rajapur K.T. weirs when calculated in isolation with methodology adopted for calculating afflux due to submersible bridge gives afflux value of 0.087 m and 0.037 m at Sangli and Rajapur K.T. weirs respectively. There is a large variation in afflux values calculated in isolation by classical empirical formula and that simulated by mathematical model. This may be due to combined effect of narrow width of river course at the K.T. weir locations, afflux due to construction of K.T. weirs structure and the congestion of flow due to confluences. Also, there exist approximations in representation of ground features in mathematical model due to limitation of data availability and modeling approximations in solving the flow equations. Therefore, it is prudent to consider this mathematical model results in relative terms, rather than absolute terms. It is necessary to study afflux phenomenon of K.T. Weirs coupled with confluence effect on physical models. Based on the results of such physical model study, structural modifications may be considered if found necessary,

5.2 Necessity for further studies:

Overview of the studies done so far, reveals that, the focus of the earlier studies was on ascertaining the adverse impact of the backwater of Almatti reservoir on the territory of the Maharashtra if any. Earlier studies were not focused on investigation of probable reasons behind the flooding in the territory of the Maharashtra. Thus, fresh hydrodynamic simulation studies are necessary for addressing the issues included in the TOR. Further, it is felt that there is a scope for refinement in mathematical river flow analysis, by addressing the following issues.

- Most of the studies (Except Garde) have considered steady state flow condition, in which temporal and spatial variation in the river flow, existing during flood event are not considered.
- The models used so far are straight line without curves and meanders.
- Except Garde, all have studied only Krishna River and not the tributaries. Confluence effect need to be studied as the inundation at confluence locations is more severe.

- Number of cross sections considered in earlier studies is inadequate (29 to 74). More number of cross sections will represent the river geometry in a better way and will give results close to the actual scenario.
- Maximum possible structures (K.T. weirs, bridges and barrages) on river need to be modeled.
- Narrow interval of flood hydrograph (1 or 2 hours) will give better results.
- Sediment deposition already occurred on river beds and particularly at confluence points needs to be ascertained and considered in the modeling after validation at few locations.
- No models are validated for observed flood levels (except Garde).
- In Garde's conclusion of observed levels, at Arjunwad and Kurundwad, which are more than simulated values, needs to be addressed.
- Velocity distribution within river reach, between reaches, across cross sections need to be studied in details.

5.3 Hydrodynamic study for Krishna basin:

5.3.1 Methodology:

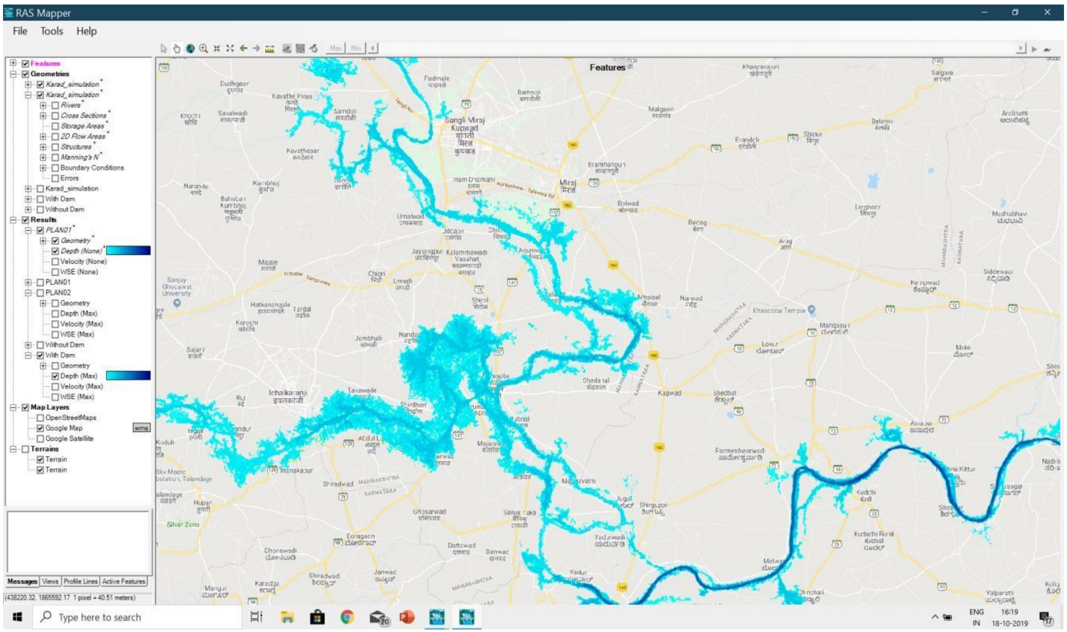
One dimensional unsteady flow analysis is done using Version 5.06 of the HEC-RAS software developed by the Hydrologic Engineering Centre of the US Army Corps of Engineers. This software has ability to carry out:

- Steady flow water surface profile computations.
- One- and two-dimensional unsteady flow simulations
- Movable boundary sediment transport computations;
- Water temperature and constituent transport modeling.

The software has capabilities to solve 2D full Saint Venant shallow water equations (with optional momentum additions for turbulence and Coriolis effects) or the 2D Diffusion Wave equations, as chosen by the user. The 2D unsteady flow equations solver uses an Implicit Finite Volume Algorithm, allowing for larger computational time steps with improved stability and robustness in handling subcritical, supercritical and mixed flow regimes. The 1D and 2D solution algorithms are coupled through time steps. Each cell and cell face are defined as table to have properties like elevation-volume, elevation- area, elevation wetted perimeter and roughness based on the resolution of the terrain model which is much smaller than the grid size of the mesh used for 2D computation. This allows much faster computation without losing details. It also has detailed flood mapping and flood animation capabilities. This software has following facilities:

- Modelling with geo-referenced geometry of the channel along with bends and meanders. RAS mapper can be used to represent the river geometry
- Modelling of the tributaries.
- Modeling inline structures.
- Superimposition with Google maps,
- To animate water surface profiles in various reaches at various time instances and over the cross sections. It makes easy to understand and observe.
- Animation of flood inundation with respect to time. If the geometry is geo-referenced, the Ras mapper (in built in HEC RAS) has facility to superimpose flood inundation on Google maps and animate unsteady flow results. Typical screen shot of inundation is shown in Figure 5.1.
- Graphical representation of the velocity distribution along the reach of the river and also across the cross section. In the cross section, vertical as well as horizontal velocity distribution can be presented in color contours.

Figure 5.1: Typical Flood Inundation mapping near Sangli on Google map



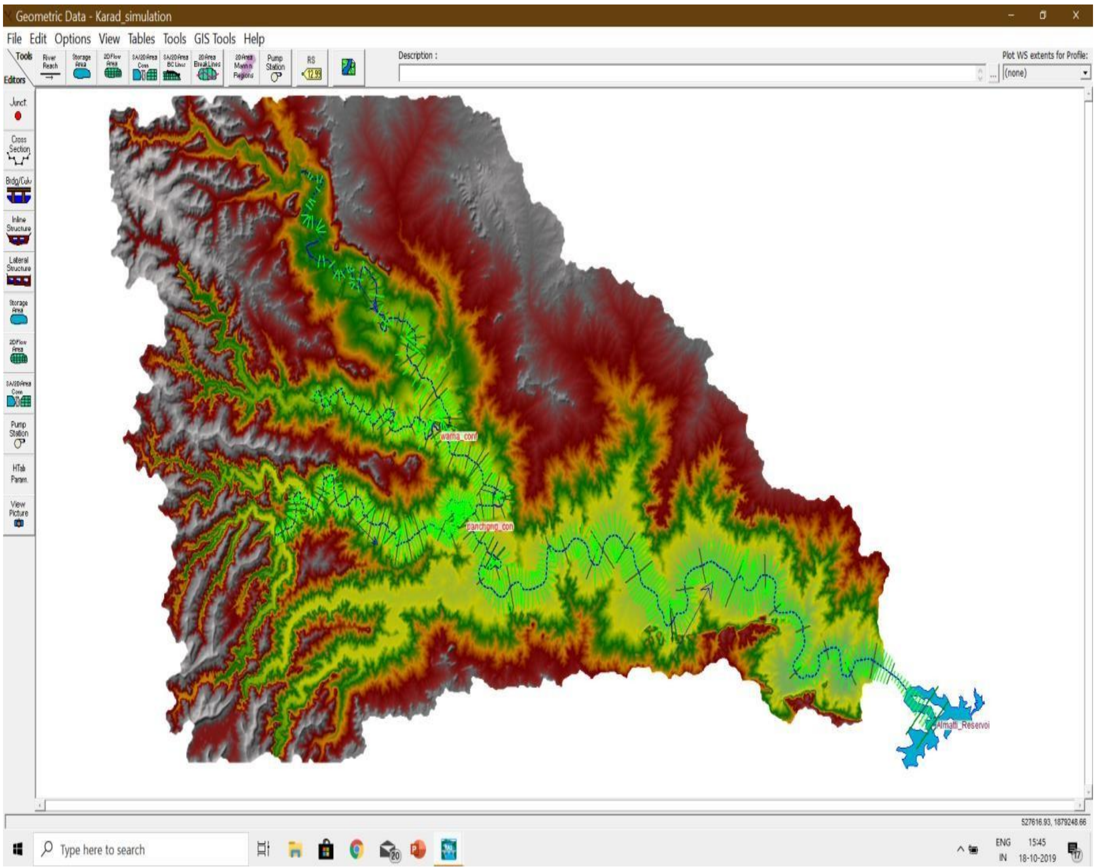
5.3.2 Geometric data of River system:

The river system comprising of River Krishna from Karad to Almatti along with its three tributaries Yerala, Warna and Panchganga is considered for modeling. The river system is developed on reach by reach basis. The river system is geo-referenced in order to have inundation mapping. Total length of the river system considered is 544.59 km. Meanders and curves are also modeled. The modeled geometry of the river system with terrain background is as shown in Figure 5.2. Following river reaches are modeled with adequate number of cross sections.

Table 5.1: Reach wise cross sections modeled

River	Reach	Length (km)	Surveyed Cross Sections	Interpolated Cross Sections	Total Cross Sections	Year of Survey
Krishna	Karad Bridge to Yerala Confluence at Brahmnal	95	48	44	92	2012
Krishna	Brahmnal – Warna Confluence at Haripur	12.6	18	8	26	2012
Krishna	Haripur - Rajapur	48	19	27	46	2012
Krishna	Rajapur-Almatti	211	31	189	220	2008
Yerala		32.9	7	63	70	2012
Warna	National Highway – Confluence at Haripur	58.5	20	36	56	2012
Panchg-anga	Kolhapur-Confluence at Kurundwad	93.7	34	101	135	2012

Figure 5.2: River system



5.3.3 River Bed Gradient:

Table No.5.2: Natural River bed gradients

River	Reach	Length	From		From		Bed Gradient	01:X
			Station	RL	Station	RL		
Krishna	Karad to Sangli	102.1 km	1.36 km U/S of Karad bridge	548.65	Irwin Bridge	525.90	0.00022282	4487
	Sangli to Rajapur	53.40 km	Irwin Bridge	525.90	Rajapur K. T weir	518.2	0.00014419	6935
	Rajapur Hippargi	91.50 km	Rajapur	518.20	Hippargi Barrage	509.00	0.00010055	9945
	Hippargi Alamtti	119km	Hippargi Barrage	509.00	Almatti Dam	495.50	0.000113368	12185
Warna	NH to Confluence	58.506 km	Highway Crossing	536.50	Confluence	525.80	0.00018289	5467
Panchganga	Kolhapur - Confluence	93.478 km	Kolhapur U/S	532.44 km	Confluence	517.70	0.00015768	6341

Note : Due to sills of Hippargi and Almatti , the effective bed gradient between Rajapur to Hippargi and between Hippargi to Almatti reduce to about 1:16000.

5.3.4 Manning's roughness coefficient "n":

Manning's roughness coefficient modeled for main channel portion is **0.031**.

For overbank portions higher value of **0.07** is adopted duly considering the standing crops (prominently sugarcane).

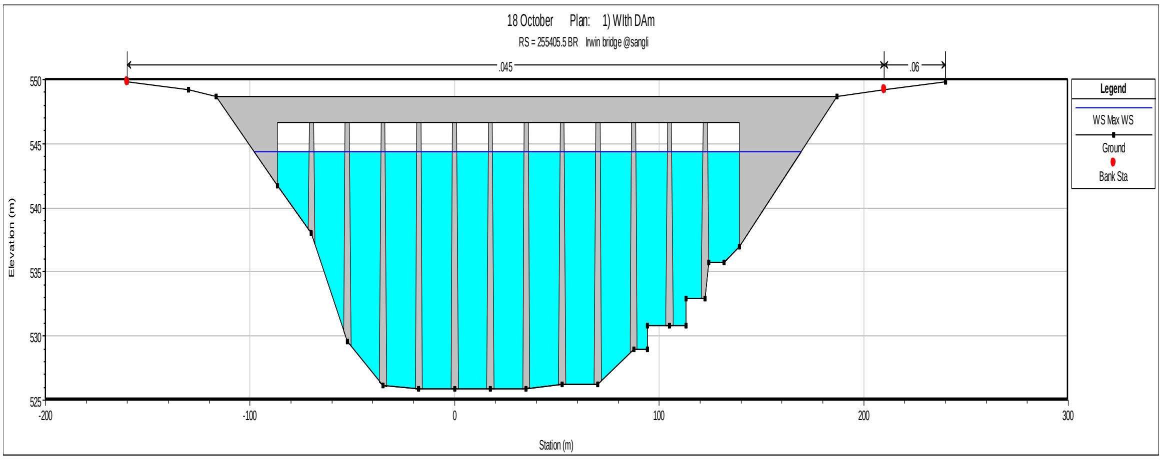
5.3.5 Hydraulic Structures:

Following inline hydraulic structures have been modeled. Typical sections are shown as Graph 5.3, 5.4, and 5.5.

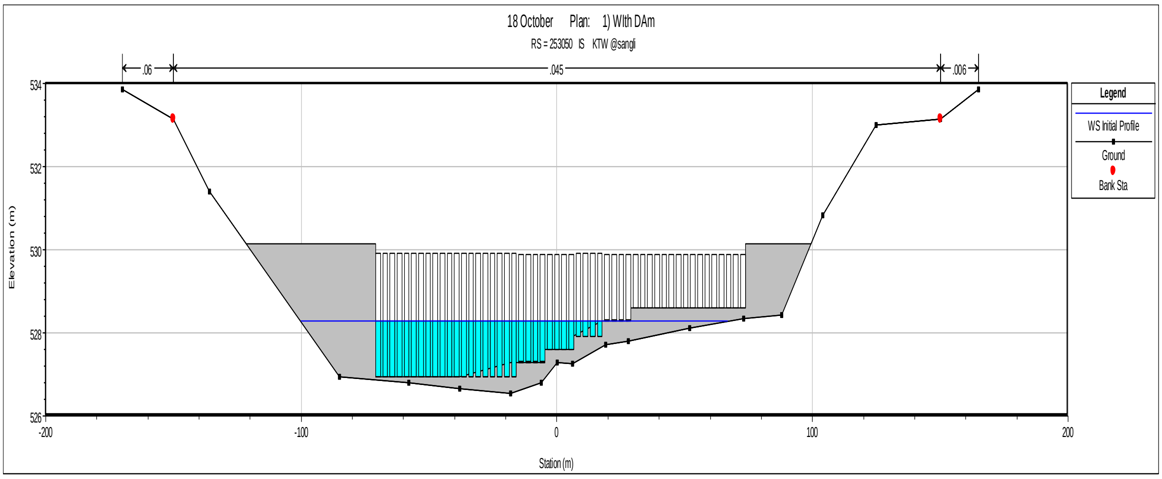
- Bridges: Karad, Sangli Bye pass, and Sangli (Irwin).
- Barrages: Tembhu, Takari and Hippargi.
- Kolhapur Type Weirs (submersible): Mhaisal, Sangli and Rajapur.

During the flood, needles of the K.T. weir are out of span hence the hydraulic behavior of the structure is similar to that of the bridge.

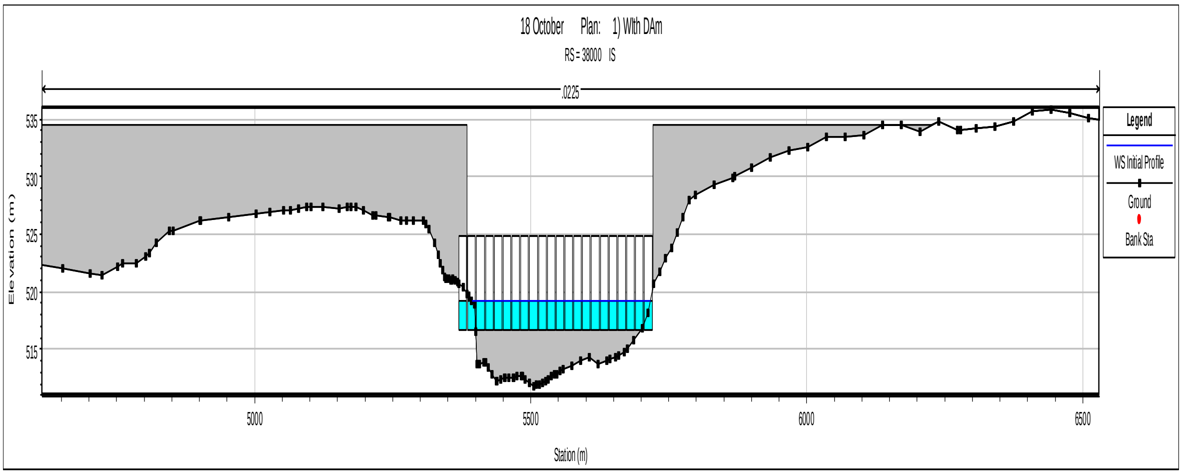
Graph 5.1: Irwin Bridge at Sangli



Graph 5.2: K.T. Weir at Sangli (Submersible)



Graph 5.3: Hippargi Barrage



5.3.6 Unsteady flow analysis:

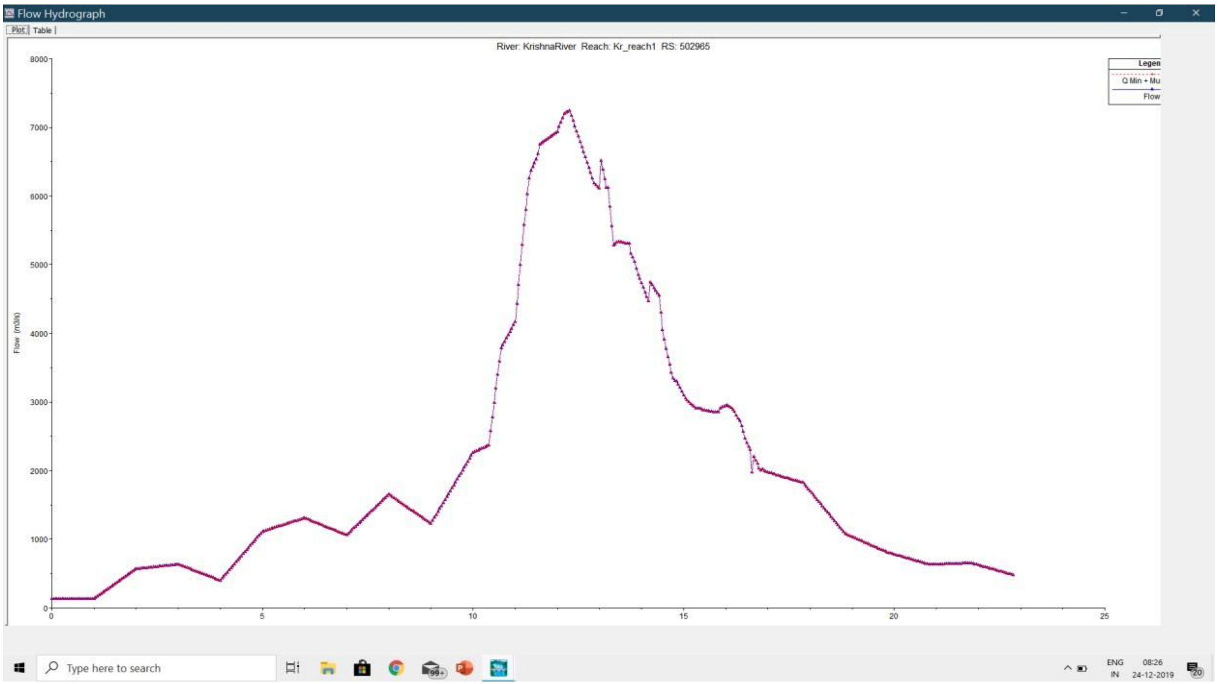
As the discharge and velocities in the river system are changing with respect to both time and space, unsteady flow option is chosen for the analysis. Unsteady flow analysis is done for period of 25th July, 2019, 8.0 am too 17th August, 2019, 8.0 am.

5.3.7 Boundary conditions for the analysis:

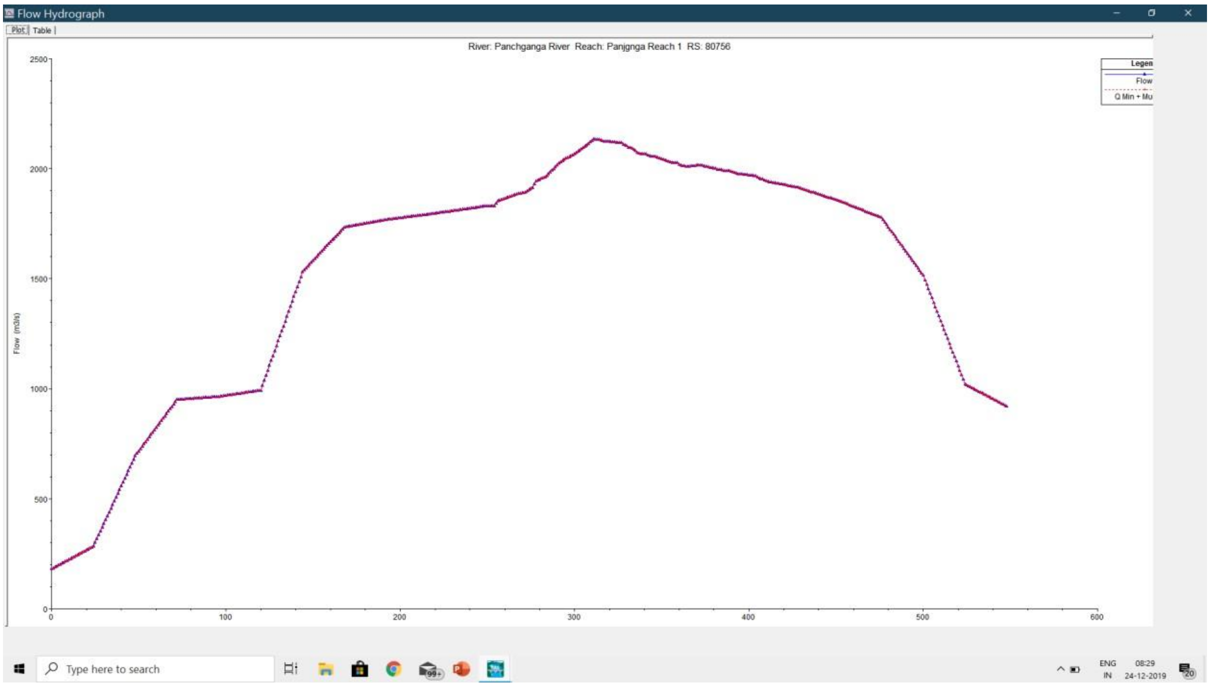
Analysis has been done with following boundary conditions.

- Upstream boundaries are defined with flood hydrograph of river Krishna at Karad, river Warna at national highway crossing, river Panchganga at Kolhapur for a period of 25th July to 17th August, 2019.
- Downstream boundary condition is defined by flood hydrograph of river Krishna at Almatti and observed levels at Almatti. The hydrographs considered are as below.

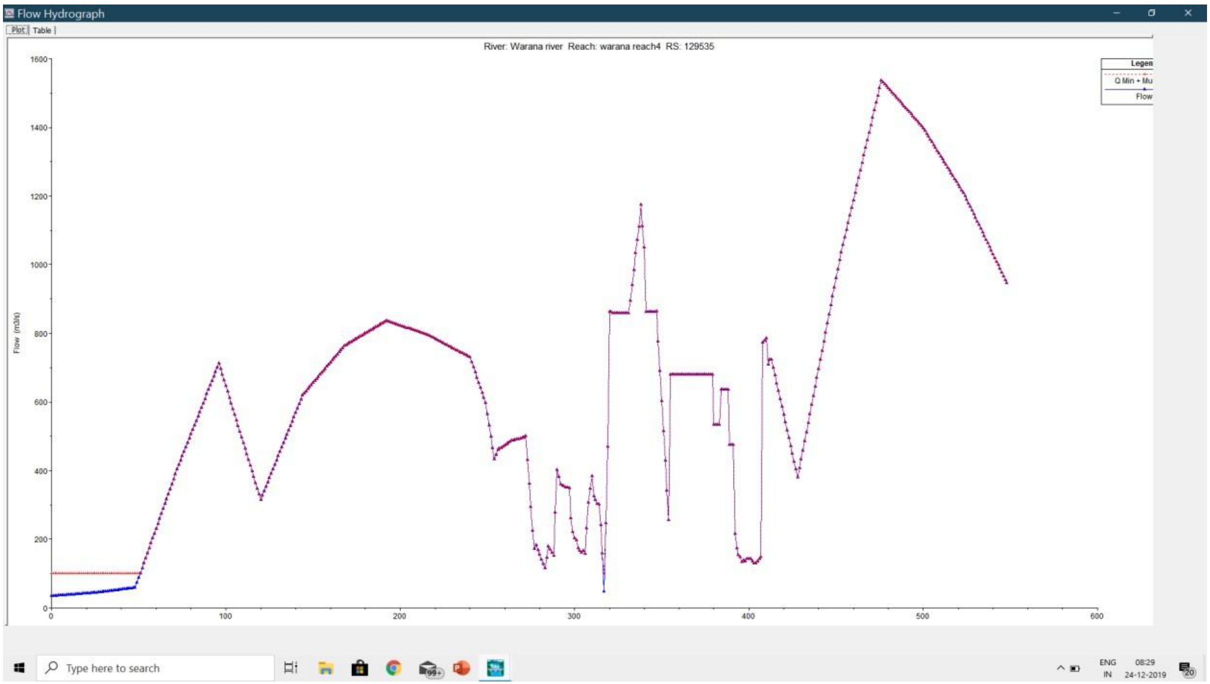
Graph 5.4: Flow Hydrograph of river Krishna at Karad



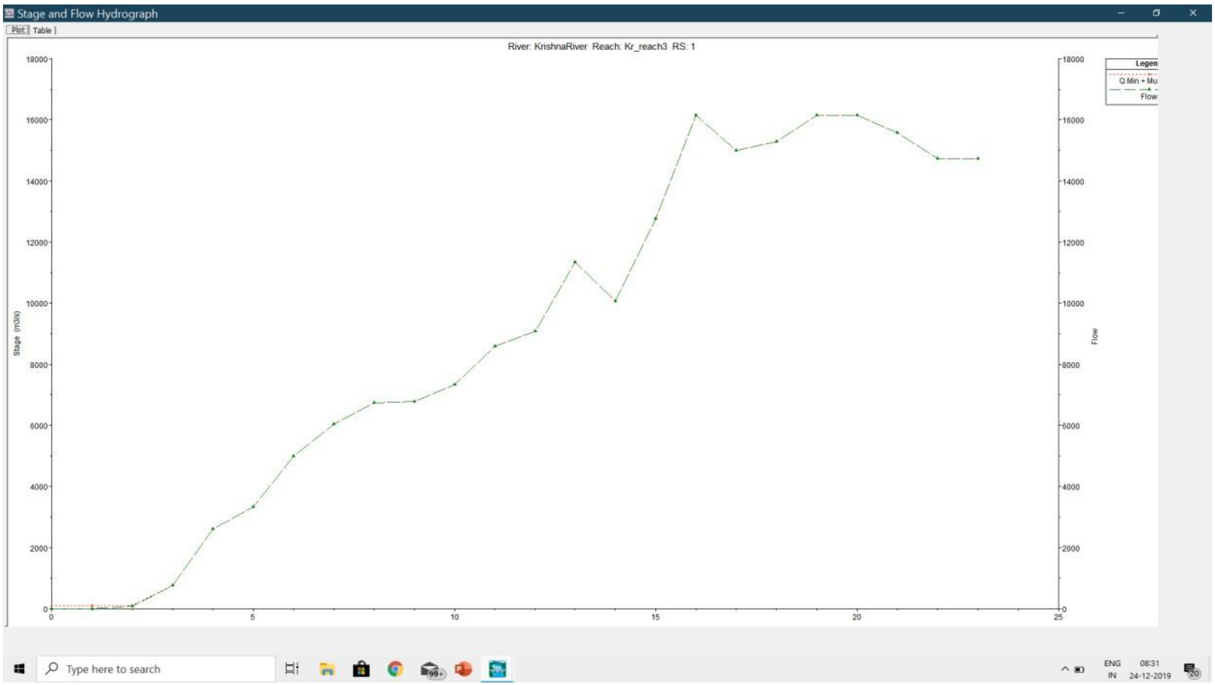
Graph 5.5: Flow Hydrograph of river Panchganga at Rajaram Bridge



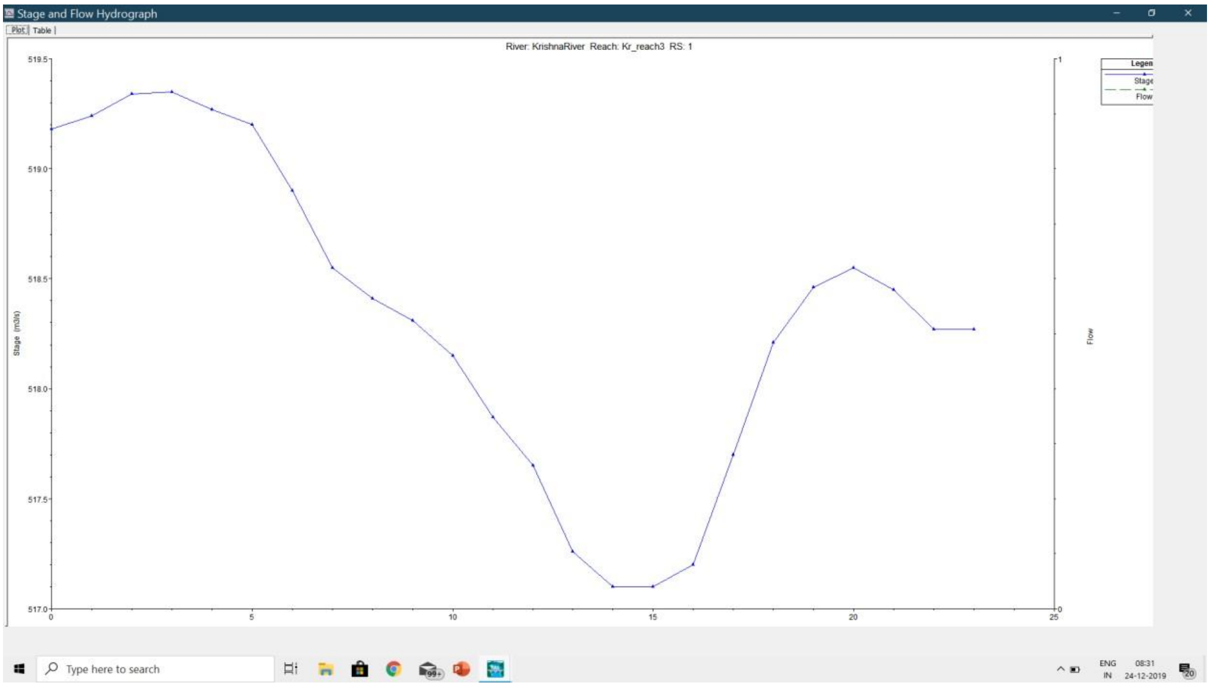
Graph 5.6: Flow Hydrograph of river Warna a t NH4 crossing



Graph 5.7: Outflow Hydrograph of river Krishna at Almatti



Graph 5.8: Observed Stages of river Krishna at Almatti from 25th July to 17th August, 2019.



5.3.8 Various scenarios studied

i. Steady state analysis to simulate backwater profile of Almatti dam for FRL at 524.25 m and PMF discharge of 31,000 cumecs at Almatti and 24, 609 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs.

ii. Steady state analysis to simulate backwater profile of Almatti dam for FRL at 519.60 m and PMF discharge of 31,000 cumecs at Almatti and 24,609 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs.

iii. Unsteady state analysis of flood event from 25th July 2019, 8 am to 17th august 2019, 8 a.m., (2019 flood event) with and without Almatti and Hippargi dams.

iv. Unsteady state analysis of 2019 flood event with Almatti reservoir (FRL 519.6 m) and Hippargi reservoir by controlling the discharge of river Warna (75 %, 50 %, 25 %).

v. Unsteady state analysis of 2019 flood event with Almatti reservoir (FRL 519.6 m) and Hippargi reservoir by controlling the discharge of river Panchganga (75 %, 50 %, 25 %).

vi. Unsteady state analysis of 2019 flood event with Almatti reservoir (FRL 519.6 m) and Hippargi reservoir by controlling the discharge of river Koyna (75 %, 50 %, 25 %).

5.3.9 Results

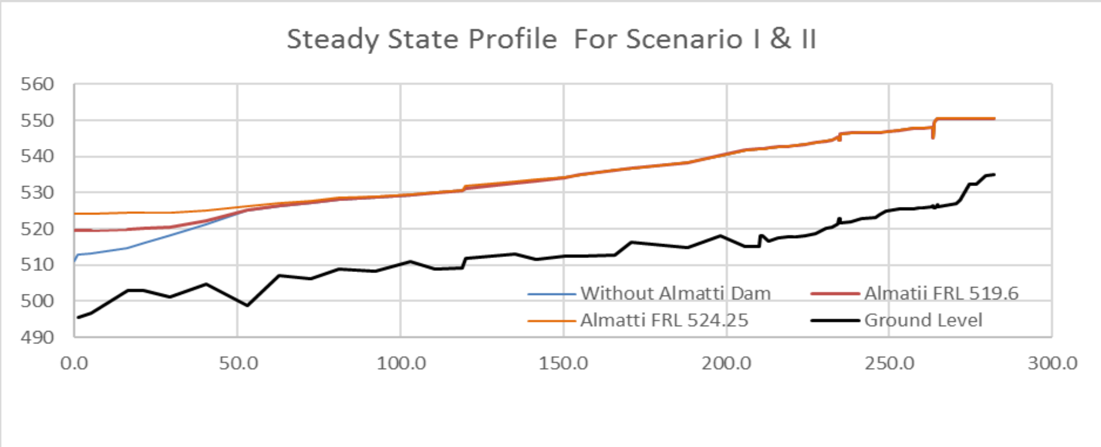
The results of the mathematical model studies for various scenarios are as under:

Scenario i: In steady state analysis, the backwater profile of Almatti dam, for FRL at 524.25 m and PMF discharge of 31,000 cumecs at Almatti and 24, 609 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs and siltation occurred up to the year 2011, merges with water profile in natural condition of the river Krishna at km 206 from Almatti dam i.e. in the territory of the State of Karnataka. (As per CWC report (2006) this location was at 221 km). These, results are consistent with the earlier studies.

Scenario ii: In steady state analysis, the backwater profile of Almatti dam for FRL at 519.6 m and PMF discharge of 31,000 cumecs at Almatti and 24, 609 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs and siltation occurred up to the year 2011, merges with water profile in natural condition of the river Krishna at 135 km from Almatti dam i.e. in the territory of the State of Karnataka. These, results are also consistent with the earlier studies.

Simulated elevations of water surface profile for Scenario i and Scenario ii are tabulated in Table No. 5.8 (Para. 5.6), and graphically presented in Graph 5.9.

Graph No. 5.9: Steady State Water Surface Profile For no dam condition, Almatti dam with FRL at 519.6m & Almatti dam with FRL at 524.25 m, corresponding to PMF at Almatti



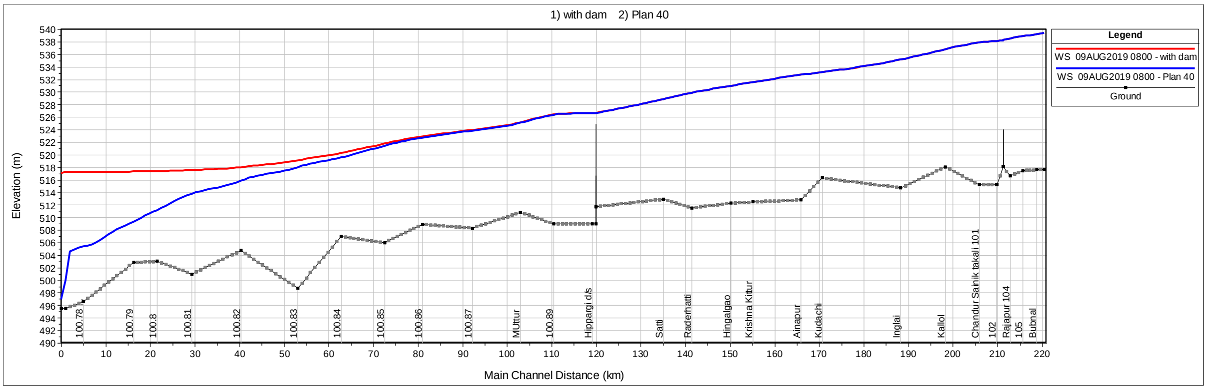
Near Sangli city, there exist three structures on river Krishna in a reach of 1.6 km, (from chainage 264.9 Km to 263.30 Km) namely Sangli bypass bridge, Irwin bridge and Sangli K.T. weir. Simulated water surface profile, for steady state condition indicates elevation difference of 2.51 meters (550.54 m – 548.03 m) in simulated water surface profile, in this reach. Abrupt changes in water surface elevations are also seen in this reach. This may be due to combined effect of contraction in waterway at the locations of structures, confluence effect of the tributaries which are meeting to the river Krishna on the downstream side of this reach and existence of two humps in the river bed in this reach (hump of 0.4 m between chainage 264.9 km to 264.6 km and hump of 0.6 m between chainage 263.5 km to 263.4 km). The ground level data indicating humps need to be verified on site for its correctness.

Scenario III: In unsteady state analysis of the flood event, on 9th August, 2019 at 4.00 am (Time instant at which simulated water level at Irwin bridge is maximum) with and without Almatti and Hippargi dams and siltation occurred up to the year 2011, indicates that backwater profile of Almatti dam with FRL at 519.6 m , considering both Almatti and Hippargi reservoirs , merges with water surface profile in natural condition of the river Krishna at 119 km from Almatti dam i.e. just upstream of Hippargi dam in the territory of the State of Karnataka. Further, the backwater profile of the Hippargi Barrage merges with the water surface profile in natural condition at 69 Km upstream of Hippargi Barrage.

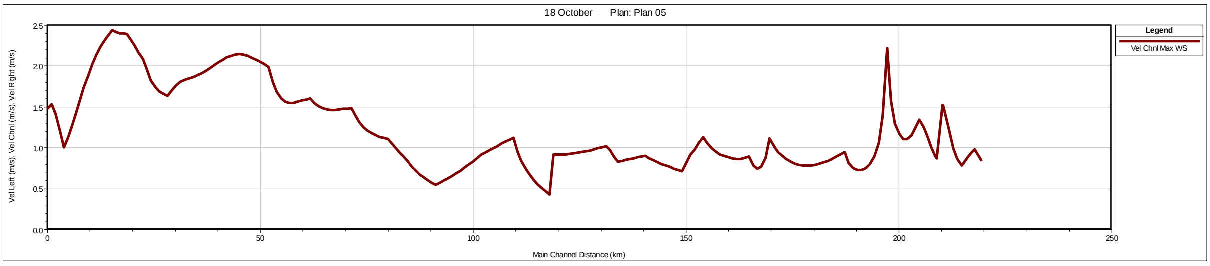
This scenario indicated simulated water surface elevation, upstream of Hippargi barrage, as 526.53m. However, level actually observed at this location is 527.33 m. Hence, one more run was taken by giving actual stages at Hippargi as internal boundary condition. It is observed that this effect extends up to 70 km up stream of Hippargi i.e. 22 km downstream of the State boundary.

Elevations of simulated water profiles for scenario iii are tabulated in Table No. 5.9 (Para. 5.6) and graphically presented in Graph 5.10 below.

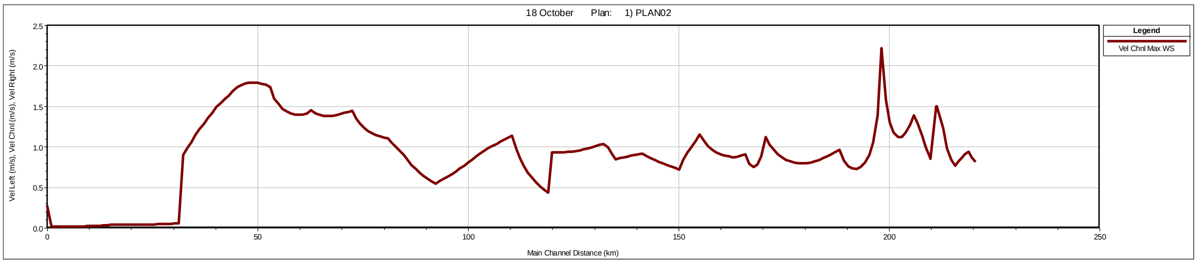
Graph No. 5.10: Simulated Water Surface Profiles for Flood Event of 9th, 2019, 4 am, with Almatti FRL 519.6 m and without Dam



Graph 5.11: Spatial Variation in Channel Velocities from Almatti to Sangli without Almatti Dam (Scenario iii)



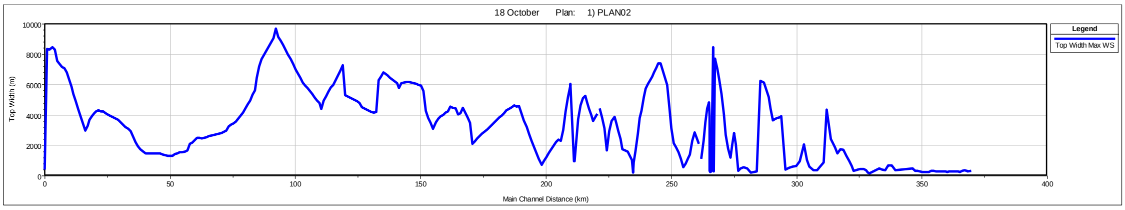
Graph 5.12: Spatial Variation in Channel Velocities from Almatti to Sangli with Almatti Dam (Scenario iii)



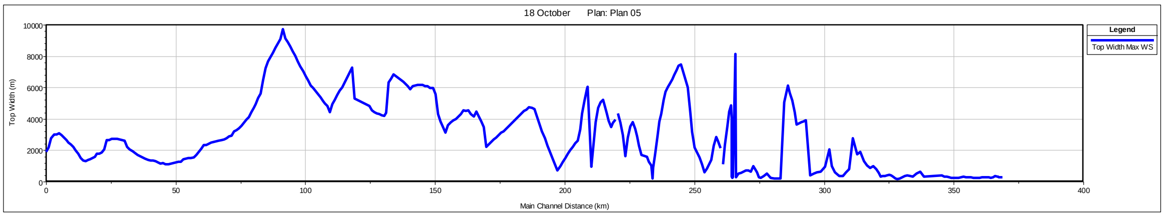
Graph 5.11, indicates that in the initial reach of 35 km from Almatti dam, the velocities are very less. This initial reach is the water pool behind the dam. However, after 50 km the shape of the graph is similar to that of velocity plot in Graph 5.12 i.e. without Almatti dam for natural condition. Further, the velocity plots after 200 km are almost identical.

Spatial Water Spread Area Variation (Scenario iii)

Graph 5.13: Water spread area with Almatti dam on 9 August 2019



Graph 5.14: Water spread area without Almatti dam on 9 august 2019



The shapes of the water spread area plots are exactly opposite to that of the velocity plots. In Graph 5.13, near the dam site water spread is wide. After initial reach of 35 km from Almatti dam, the water spread graphs for both the conditions, (with and without Almatti) is similar. The spread also matches with observed floods at Rajapur, Sangli and Bramhanal.

Thus, the results of the mathematical model study, do not indicate any adverse impact of Almatti or Hippargi reservoirs of the Karnataka and its operations, on the flooding, in the territory of the Maharashtra, during the Flood Event 2019.

Near Sangli city, there exist three structures on river Krishna in a reach of 1.6 km, (from chainage 264.9 Km to 263.30 Km) namely Sangli bypass bridge, Irwin bridge and Sangli K.T. weir. Simulated water surface profile for unsteady state condition indicates elevation difference of 0.70 meters (544.5 m – 543.80 m). Separate run of scenario iii, taken by deleting the Sangli K.T. weir from the model, indicated reduction in water level by 0.34 m at K.T. weir location and Irwin bridge location. Similar, run taken, by deleting Mhaisal K.T. weir from the model, showed level drop of only 0.02 m at Mhaisal K.T. weir location.

During the site visit of the committee it was reported by the in-charge of the river gauging station at Kurundwad that during flood velocity of flow was 1.2 m /sec. simulated model gives velocity of 1.17 m/sec at this location on 9th August, 2019.

Table 5.3: Comparison of Observed and Simulated Water Levels

Location	Observed level on 9th August	Simulated level on 9th August
Irwin Bridge	544.55	544.34
Miraj Arjunwad Bridge	544.175	542.92
Kurundwad CWC RG Station	540.955	539.34
Rajapur Weir	537.97	538.18

The observed water levels at Irwin Bridge and Rajapur KT Weir fairly tallies with that of simulated water levels where as the observed water levels at Miraj Arjunwad Bridge and Kurundwad RG station are higher than that of simulated values. This difference was existed earlier in Prof. Gadre's study also. At Arjunwad Bridge, the difference may be due to naturally contracted river section and obstruction of the structure. At Kurundwad, difference may be due to siltation occurred, which was predicted in earlier studies and due to confluence effect. The reasons need to be ascertained with the detail study.

Scenario IV: Unsteady state analysis of 2019 flood event, with Almatti dam, by controlling the discharge of river Warna (75 %, 50 %, 25 %) indicates that the water level of Irwin bridge would have been 0.29 m less if the flow hydrograph of river Warna (Graph 5.6) is reduced to 25% of the discharges prevailing during the flood event. The corresponding reduction in water level at confluence point would have been 0.34 meter. Simulated water profile elevations are tabulated in Table No. 5.10 (Para. 5.6).

Table No. 5.4: Simulated elevations of water surface at Irwin Bridge with controlling the discharge of river Warna

Discharge in River Warna	Simulated elevations of water surface at Irwin Bridge (m)
As per flood hydrograph in Graph 5.6 above	544.34
Restricted to 75 %	544.24
Restricted to 50%	544.15
Restricted to 25 %	544.05

Scenario V: Unsteady state analysis of 2019 flood event, with Almatti dam and Hippargi Barrage, by controlling the discharge of river Panchganga, indicates that the level of Irwin bridge would have been 0.14 m less if the discharge values in flood hydrograph in Graph 5.5 above are restricted to 25%. However, the corresponding reduction in water level at Panchganga confluence point would be about 1.1 meter. Simulated elevations of water surface profile are tabulated in Table No. 5.11 (Para. 5.6).

Scenario VI: Unsteady State Analysis, of 2019 Flood Event, with Almatti dam and Hippargi Barrage, by controlling the discharge of river Koyna at Karad (by controlling releases from Koyna dam) indicates that water level in Sangli at Irwin bridge is significantly influenced by the releases from Koyna dam. The elevations of water surface profiles for this Scenario are tabulated in Table No.

5.12 (Para. 5.6). The extract of elevations of simulated water surface profiles in different scenarios, at Irwin Bridge is as under.

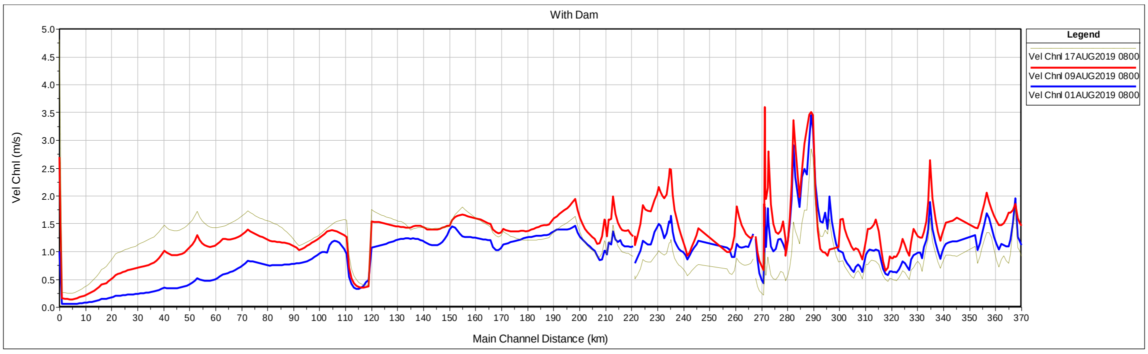
Table No.5.5: Simulated elevations of water surface at Irwin Bridge with controlling the discharge of river Korana

Discharge in River Korana at Karad (cumecs)	Simulated elevation of water profile at Irwin bridge. (m)
As per flood hydrograph in Graph 5.4 above	544. 34
Restricted to 75%	543.
Restricted to 50 %	541.
Restricted to 33 %	540.

5.3.10 Influence of tributaries on river Krishna

Channel velocity variation of river Krishna from Almatti reservoir to Karad is presented in Graph 5.15 below. Velocity plot indicates reduction in velocity at confluence points due to formation of stagnation zone. (Yerala confluence 271.6 km, Warna confluence at chainage 259 km, Panchganga confluence at chainage 220.5 km). This reduction in velocity is accompanied by consequential increase in flow area i.e. water spreading on the banks.

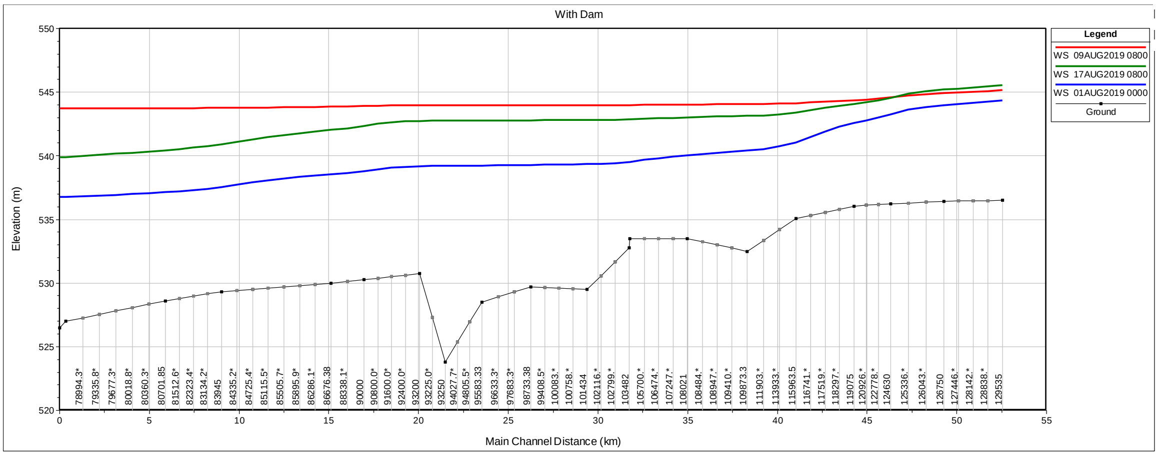
Graph 5.15: Velocity plot of channel flow along river Krishna



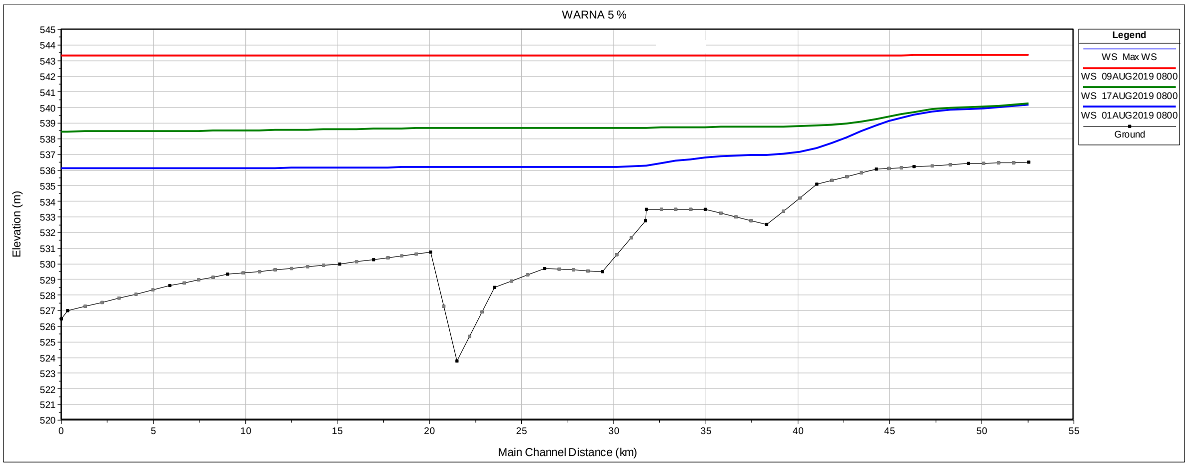
5.3.11. Influence of flow in river Krishna on the river Warna:

During the flood it is observed that the flow in the river Warna is predominantly influenced by the flow in the river Krishna**.** The Krishna – Warna confluence effect can be demonstrated with the help of HAC-RAS outputs (Graph 5.16 to Graph 5. 21)

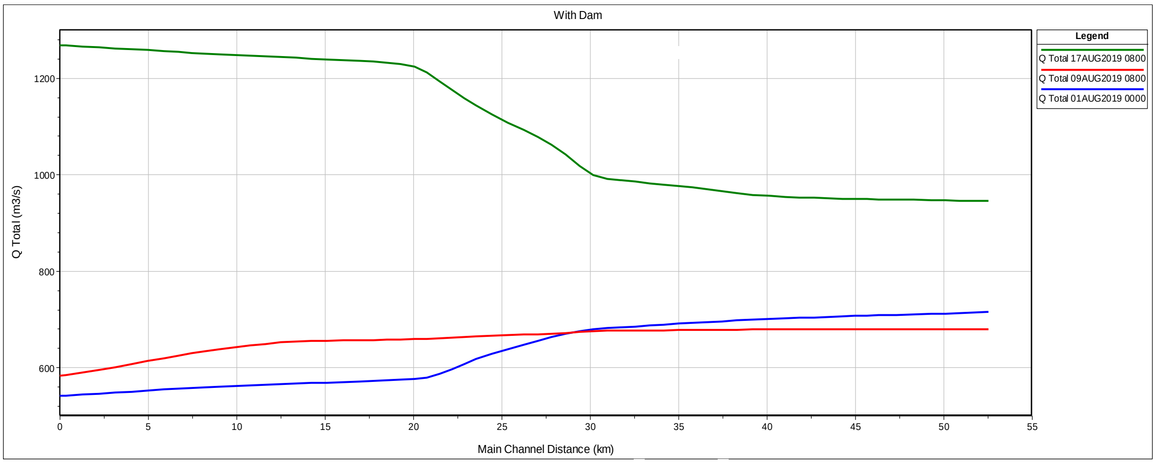
Graph 5.16: Water Surface Profile in river Warna on 1 August, 9 August and 17 August (Ch "0" is confluence point)



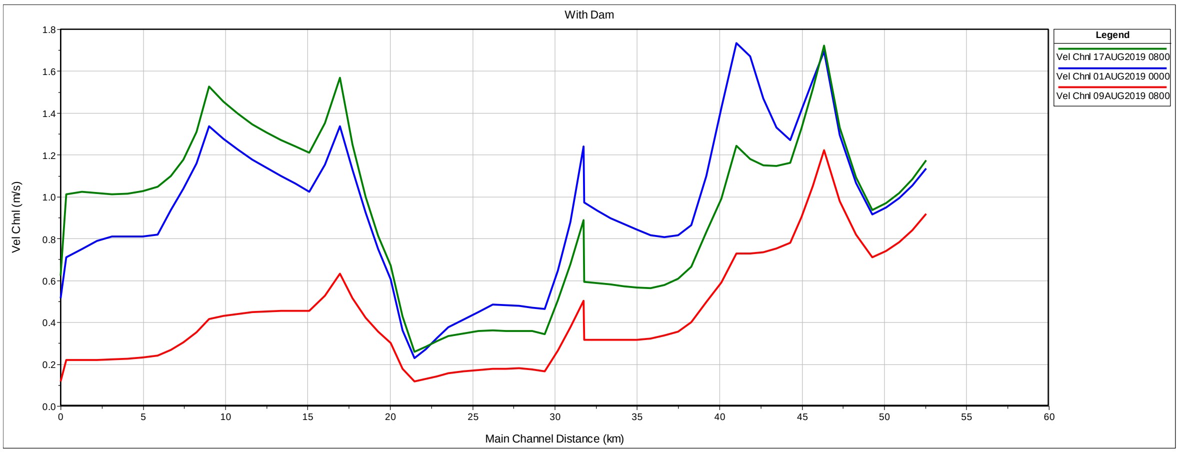
Graph 5.17: Water Surface Profile in river Warna on 1 August, 9 August and 17 August, when Warna flow is restricted to 5%. (Ch "0" is confluence point)



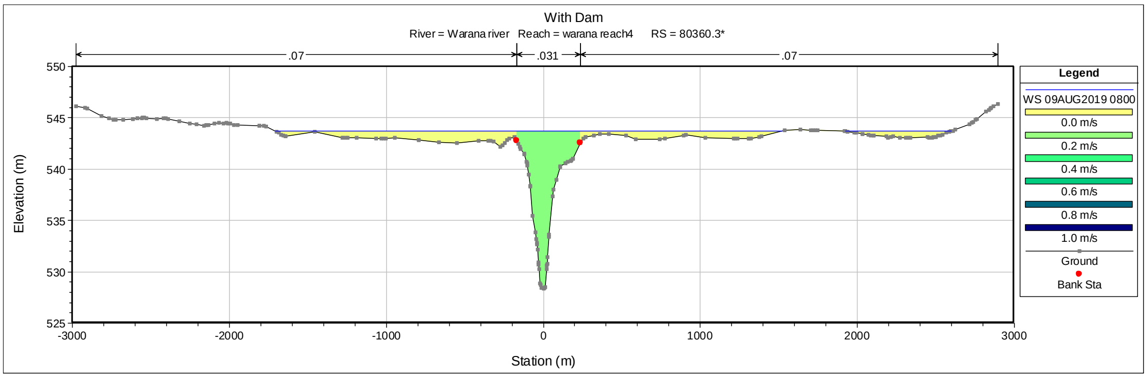
Graph 5.18: Discharges in river Warna on 1 August, 9 August and 17 August (Ch "0" is confluence point)



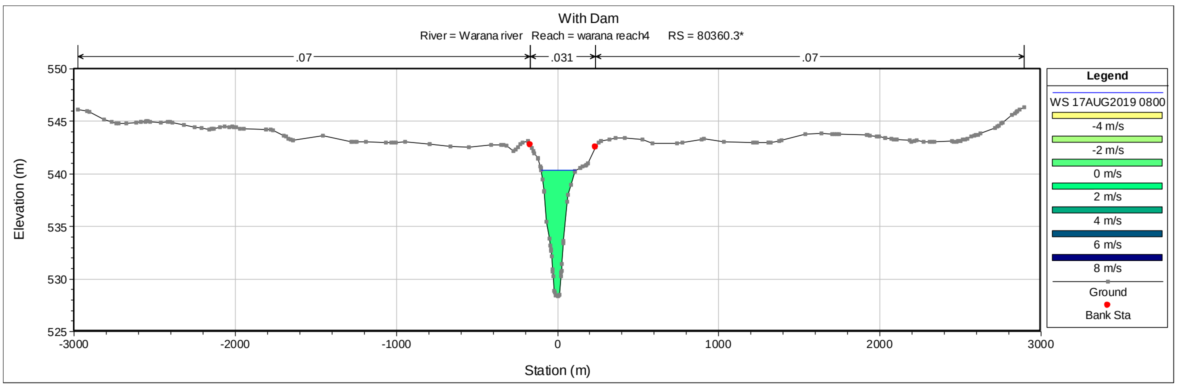
Graph 5.19: Velocities along river along Warna on 1 August, 9 August and 17 August (Ch "0" is confluence point)



Graph 5.20: Velocity distribution along cross section of river Warna at chainage 5 km upstream of confluence point on August 9, 2019. (Peak Flood position)



Graph 5.21: Velocity distribution along cross section at chainage 5 km upstream of Confluence in river Warna on August 17, 2019 (Receding Flood position)



Plot of water surface profiles on various dates in river Warna (Graph 5.16) indicates that there exists backwater effect in river Warna up to about 40 to 50 km. This indicates that during the flood, water levels in river Warna are controlled by water levels existing in river Krishna. Confluence effect is studied by reducing the discharge of the river Warna. It is observed that the simulated water levels (Graph 5.16) in river Warna are not reduced substantially even for no-flow condition in river Warna. (Warna discharge restricted to 5 %)

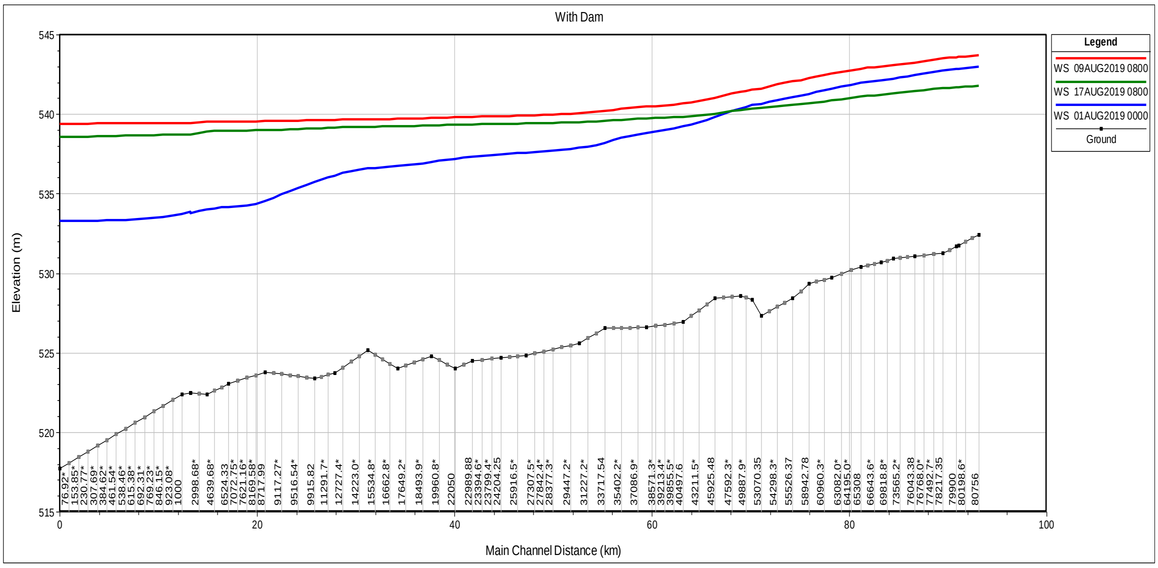
This phenomenon is also confirmed by the velocity plots in Graph 5.17. The velocity of flow near confluence point at the time of peak flood (9^{th} August) was less as compared to that on other dates (1 August and 17 August). The velocity is generated in river Warna only when the water level in river Krishna is reduced (17^{th} August profile).

The confluence effect can be better explained by correlating the discharge variations in Graph 5.19 and velocity distribution along the cross section in Graph 5.20 and 5.21. It is observed that on August 9, 2019 (during peak flood) the flow in river Warna, instead of going into river Krishna is spread along the banks of river Warna. The water spread on left bank is about 1.7 km and on right bank it is about 2.7 km. Velocity in main channel during peak flood on August 9, 2019 is about 0.2 meters/sec and on banks it is almost zero. As we go towards confluence the discharge in the channel is reducing due to its lateral spreading. On the contrary, on August 17, 2019, it is seen that after water level in river Krishna is reduced, the velocity is generated in river Warna. Velocity in main channel is slightly more than 1 meter/sec. The flow spread on banks is returned to main channel (Graph 5.21). As an effect the gross discharge in river Warna is increasing in river Warna as we go towards confluence point (Graph 5.18).

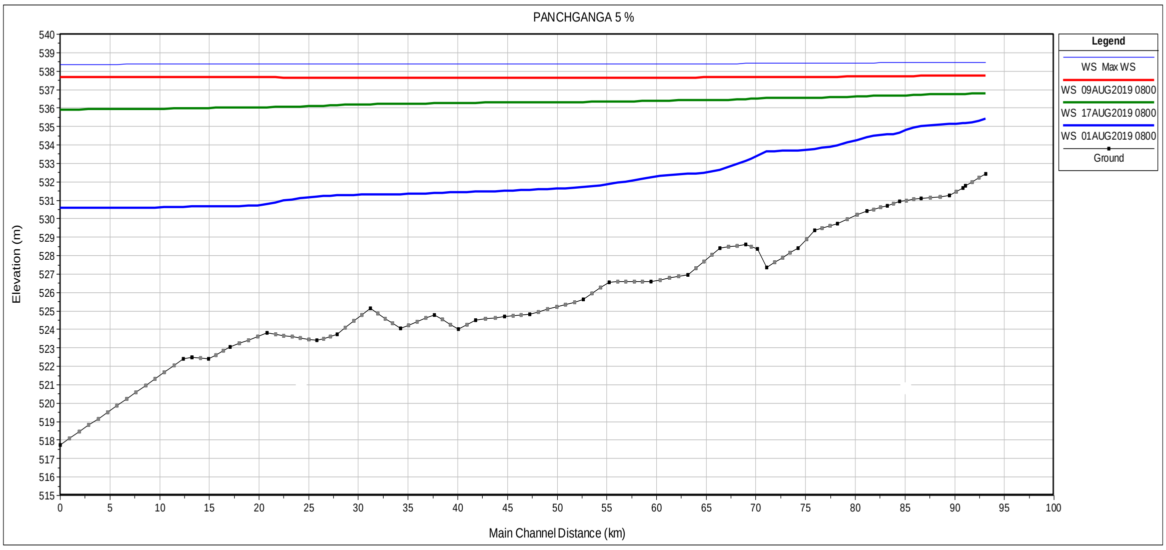
5.3.12 Influence of flow in river Krishna on the river Panchganga:

During the flood, behavior of river Panchganga is very similar to that of river Warna. It is observed that the flow in the river Panchganga is also predominantly influenced by the flow in the river Krishna. The Krishna – Panchganga confluence effect can be demonstrated with the help of HAC-RAS outputs (Graph 5.22 to Graph 5.26).

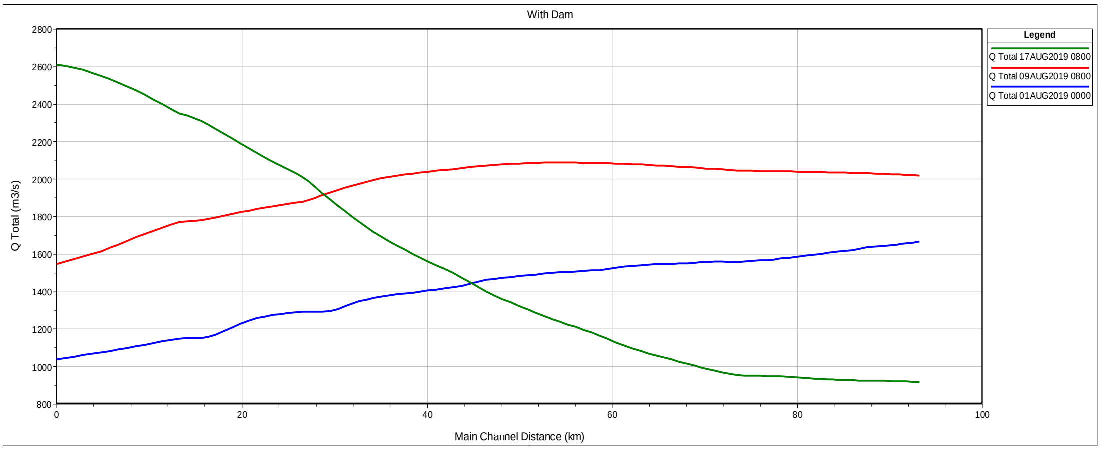
Graph 5.22: Water Surface Profile in river Panchganga on 1 August, 9 August and 17 August (Ch "0" is confluence point)



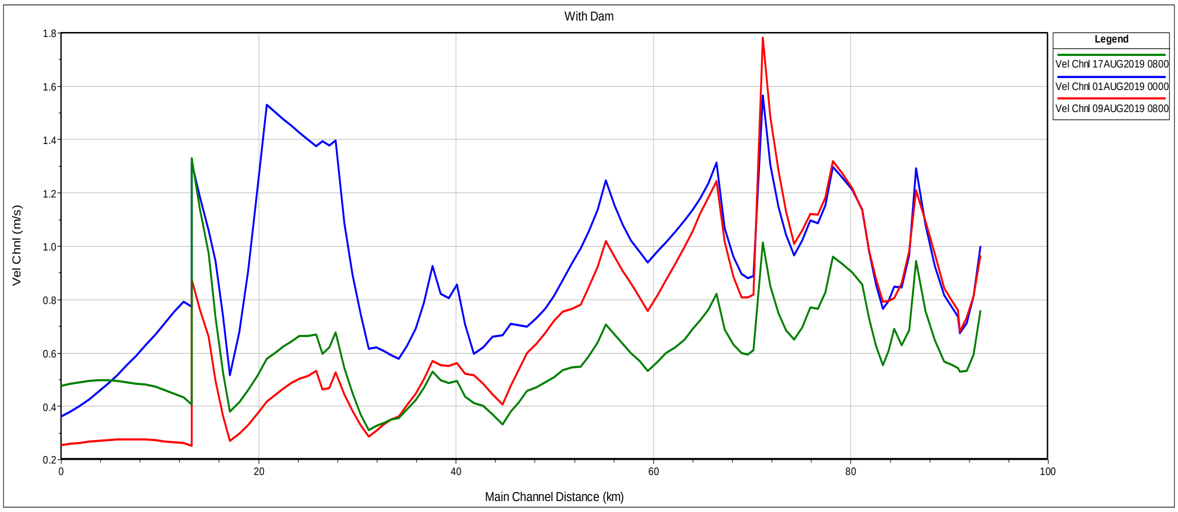
Graph 5.23: Water Surface Profile in river Panchganga on 1 August, 9 August and 17 August, when Panchganga flow is restricted to 5% (Ch "0" is confluence point)



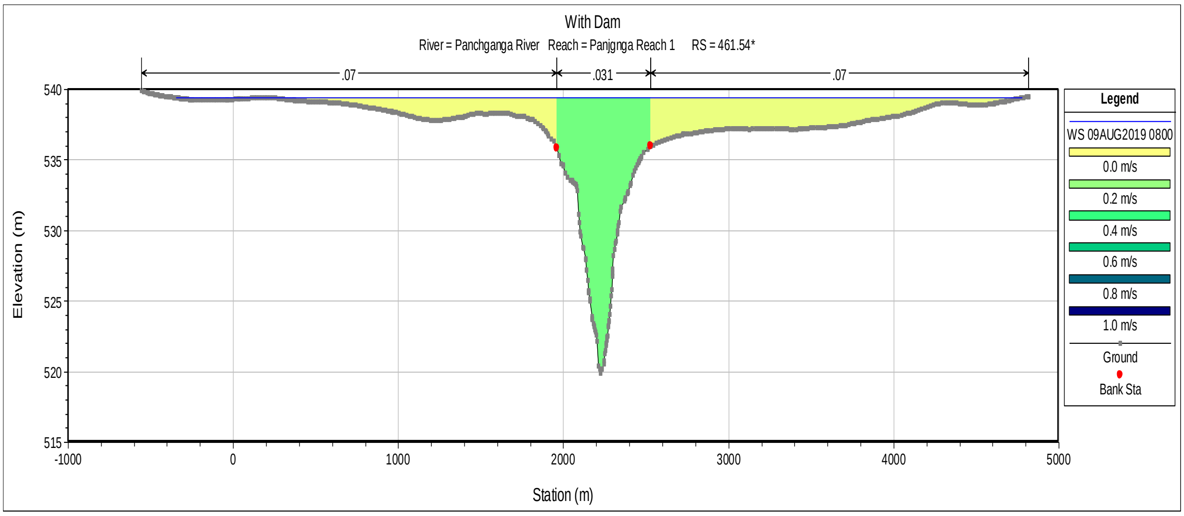
Graph 5.24: Discharges in river Panchganga on 1 August, 9 August and 17 August (Ch "0"is confluence point)



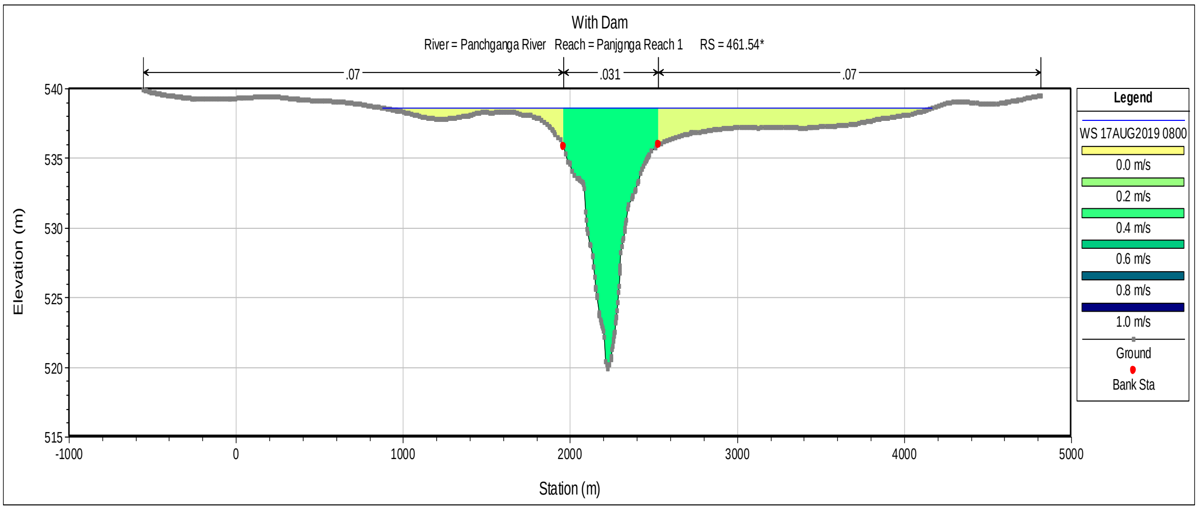
Graph 5.25: Velocities along river along Panchganga on 1 August, 9 August and 17 August (Ch "0" is confluence point)



Graph 5.26: Velocity distribution and Water spread at 5km upstream of confluence in river Panchganga on 9th August



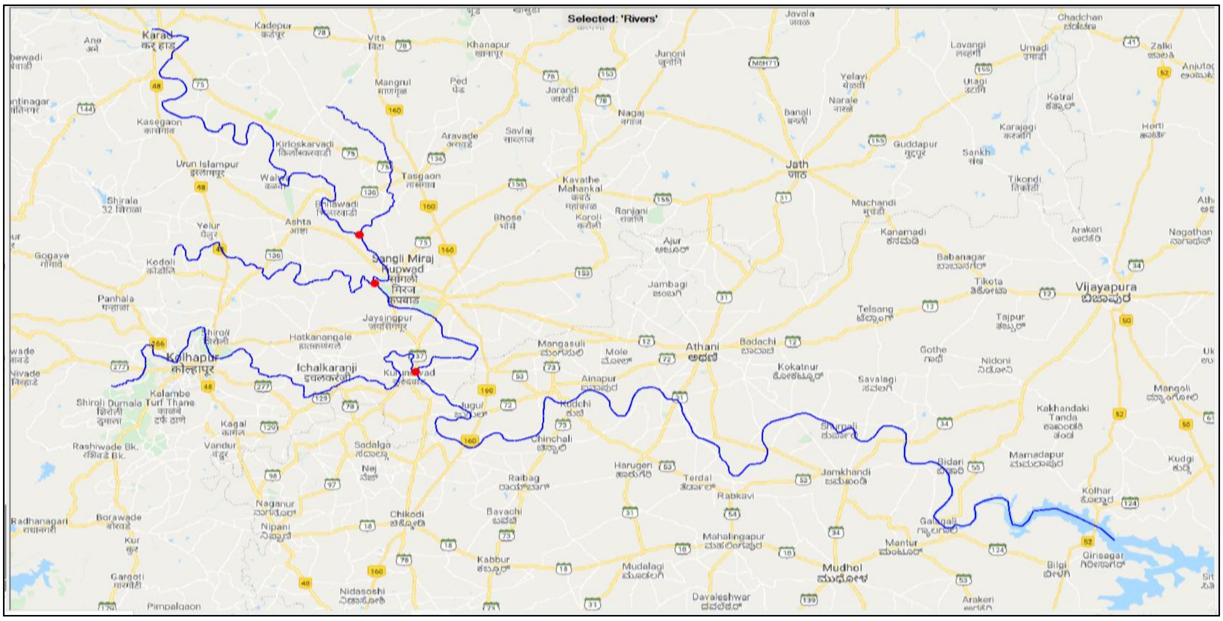
Graph 5.27: Velocity distribution and Water spread in river Panchganga at chainage 5km upstream of confluence on 17th August



5.3.13 Straightening of Meanders of the Rivers

The River Krishna has so many curves and meanders on it. The major considerable meanders are 17, as shown in figure below.

Figure 5.3: Krishna River Meanders

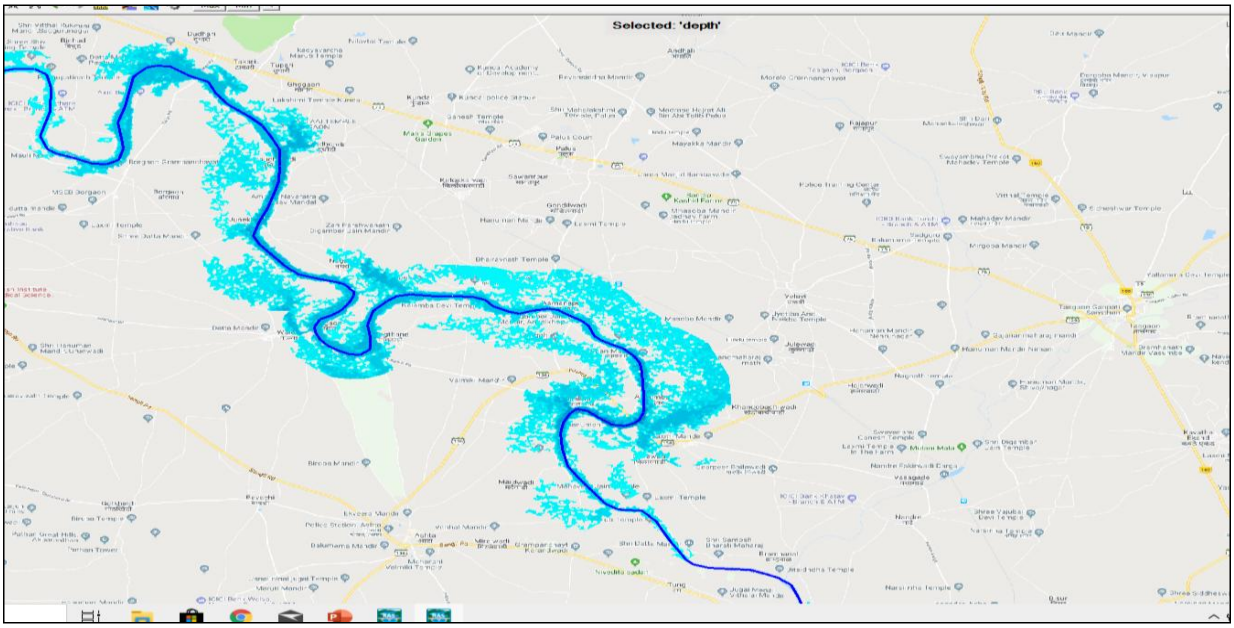


5.3.13.1 Major Meanders on Krishna River

Karad Sangli reach

It can seen from inundation map, at the meanders on inner as well outer sides of meanders have more inundation areas as compared to straight reaches of the river few meanders on Krishna river are shown below Shirgao Nagthane

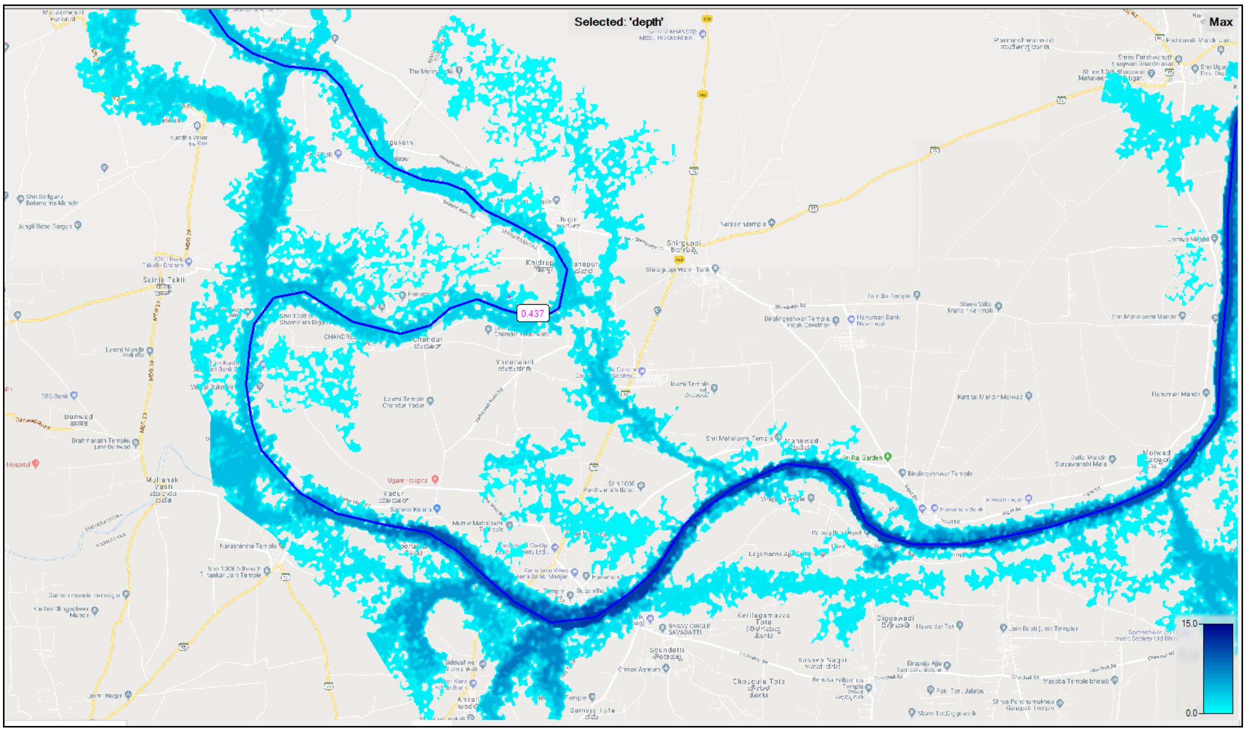
Figure 5.4: Karad Sangli reach



Sangli Rajapur Reach

Khidrapur Meander

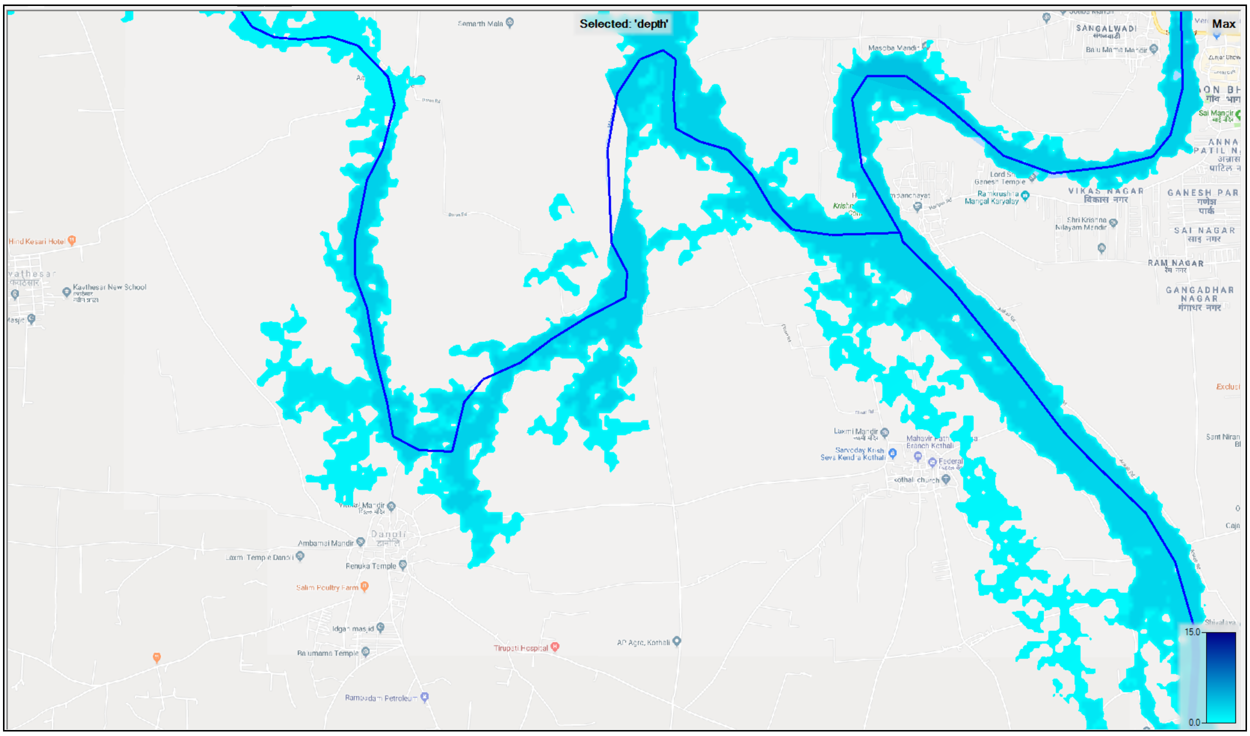
Figure 5.5: Khidrapur Meander



Meanders on Warna River

Near Danoli

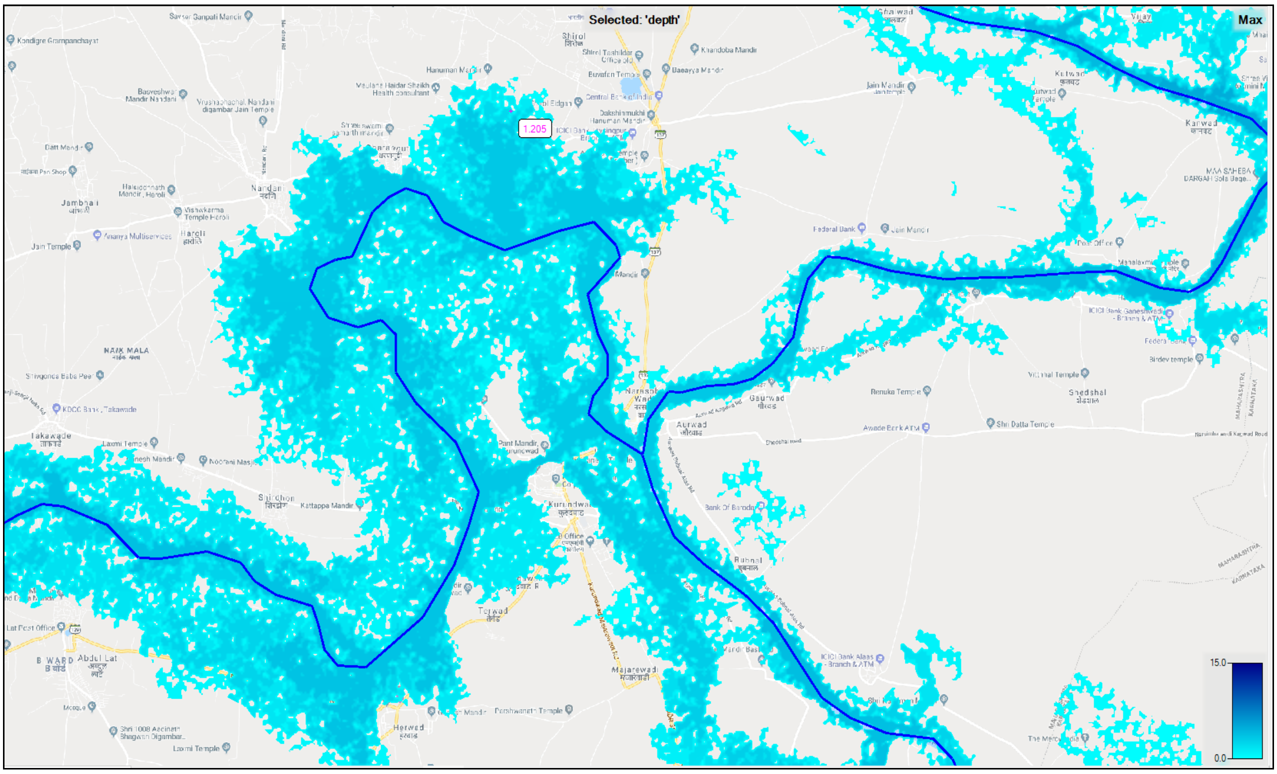
Figure 5.6 Meanders on Warna River



Panchganga River

Near Kurundwad

Figure 5.7 Meanders on Panchganga River



5.3.13.2 Straightening of Meander near Ghalwad to Shirati

Figure 5.8 Straightening of Meander near Ghalwad to Shirati

Considered meander is just Upstream of Panchganga confluence passing through Ghalwad, Dhavali, Kutwad, Mhaisal, Kanwad, Hasur, Kawathe guland, Shirati. Natural river has length of 14.93 km. This is shown in yellow colour. If this meander is straightened, it can be connected straight from Ghalwad to Shirati, the length reduces to 4.19 km (reduction by 10.74 km).

Actual River with Meander

Meander cut and straightened

5.3.13.3 Meandering Effect: Rajapur- Ingali Meander

Meander on border of Maharashtra Karnataka from Rajapur to Ingali is selected for studying the behavior of the meander. This meander is roughly circular in shape having diameter around 7.5 km. The meander is shown in Graph 5.30 below. Output of the analysis is reproduced in Table 5.6 below. Output indicates maximum, minimum and average channel velocities as 2.24, 0.97 and 1.4 m/s respectively. These values are inconsonance with velocities on upstream and downstream reaches. Thus, meander at this location has behavior, similar to normal river channel. The effect of straightening the meander needs to be further studied both on mathematical as well as physical model. Artificial straitening of the meander will increase in bed gradient, which may result into increase in velocity and consequential release of flood congestion.

Figure 5.9: Rajapur Ingali Meander

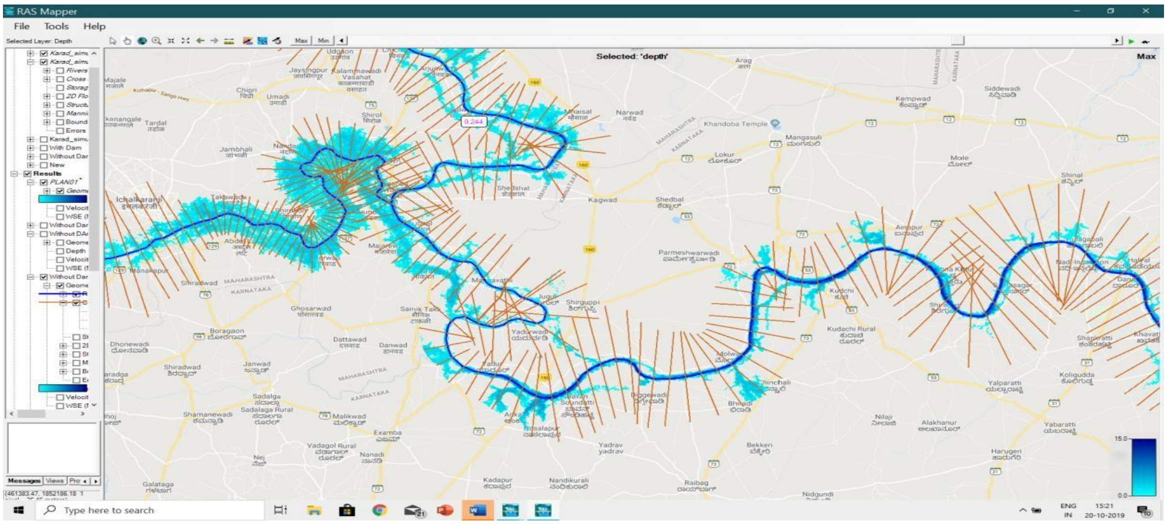


Table No. 5.6: Output Table at Rajapur-Ingali Meander

HEC-RAS Plan: 19 Meander Effect Locations: User Defined

River	Reach	River Sta	Profile	Top Width	Area Channel	Area Left	Area Right	Area	Q left	Q Channel	Q Right	Q Total	Vel Left	Vel Chnl	Vel Right	Vel Total
				(m)	(m2)	(m2)	(m2)	(m2)	(m3/s)	(m3/s)	(m3/s)	(m3/s)	(m/s)	(m/s)	(m/s)	(m/s)
KrishnaRNe	Kr_reach;	108650	Max WS	658.46	4776.6			4776.62		8843.32		8843.32		1.85		1.85
KrishnaRNe	Kr_reach;	107058.*	Max WS	735.21	5736			5735.98		8864.84		8864.84		1.55		1.55
KrishnaRNe	Kr_reach;	105466.*	Max WS	846.82	5872.9	91.67		5964.56	13.84	8872.52		8886.36	0.15	1.51		1.49
KrishnaRNe	Kr_reach;	103874	Max WS	1535.7	5017.2	599.17	6.72	5623.09	120.69	8786.76	0.41	8907.86	0.2	1.75	0.06	1.58
KrishnaRNe	Kr_reach;	93074	Max WS	1533	5454.9	159.28	6.61	5620.75	24.14	8893.76	0.55	8918.45	0.15	1.63	0.08	1.59
KrishnaRNe	Kr_reach;	91872.5*	Max WS	1535.05	5823.7	127.93	9.46	5961.09	16.98	8945.85	0.78	8963.61	0.13	1.54	0.08	1.5
KrishnaRNe	Kr_reach;	90671.0*	Max WS	1547.26	6005	98.81	13.77	6117.56	11.98	8995.64	1.18	9008.8	0.12	1.5	0.09	1.47
KrishnaRNe	Kr_reach;	89469.5*	Max WS	1576.17	5989.7	68.43	19.68	6077.76	7.68	9031.3	1.71	9040.68	0.11	1.51	0.09	1.49
KrishnaRNe	Kr_reach;	88268	Max WS	1588.56	5765.7	36.22	27.11	5829.07	3.66	9080.77	2.56	9086.99	0.1	1.57	0.09	1.56
KrishnaRNe	Kr_reach;	86068.7*	Max WS	2198.55	5894.5	53.08	398.46	6346.03	6.16	9038.88	89.16	9134.2	0.12	1.53	0.22	1.44
KrishnaRNe	Kr_reach;	83869.5*	Max WS	2141.71	5915.6	74.51	916.5	6906.6	9.95	8853.27	318.19	9181.41	0.13	1.5	0.35	1.33
KrishnaRNe	Kr_reach;	81670.2*	Max WS	1954.99	5817.2	97.12	1314.4	7228.66	14.65	8637.44	576.58	9228.67	0.15	1.48	0.44	1.28
KrishnaRNe	Kr_reach;	79471.0*	Max WS	1729.97	5592.8	118.33	1550.7	7261.87	19.87	8351.98	796.99	9168.83	0.17	1.49	0.51	1.26
KrishnaRNe	Kr_reach;	77271.7*	Max WS	1486.67	5237.3	135.75	1609.3	6982.31	25.76	8215.09	966.81	9207.66	0.19	1.57	0.6	1.32
KrishnaRNe	Kr_reach;	75072.5*	Max WS	1233.85	4746.6	146.65	1484.6	6377.86	31.82	8157.11	1058.56	9247.48	0.22	1.72	0.71	1.45
KrishnaRNe	Kr_reach;	72873.2*	Max WS	974.74	4116.5	146.43	1174.2	5437.2	36.98	8216.01	1034.87	9287.86	0.25	2	0.88	1.71

HEC-RAS Plan: 19 Meander Effect Locations: User Defined

River	Reach	River Sta	Profile	Top Width	Area Channel	Area Left	Area Right	Area	Q left	Q Channel	Q Right	Q Total	Vel Left	Vel Chnl	Vel Right	Vel Total
				(m)	(m2)	(m2)	(m2)	(m2)	(m3/s)	(m3/s)	(m3/s)	(m3/s)	(m/s)	(m/s)	(m/s)	(m/s)
KrishnaRNe	Kr_reach;	70674	Max WS	704.9	3337.3	122.06	683.89	4143.25	36.11	8456.28	802.92	9295.31	0.3	2.53	1.17	2.24
KrishnaRNe	Kr_reach;	69674.0*	Max WS	1012.03	3527.1	161.72	1666.6	5355.36	47.37	7687.55	1605.65	9340.57	0.29	2.18	0.96	1.74
KrishnaRNe	Kr_reach;	68674.0*	Max WS	1320.06	3702.6	200.04	2504.4	6407.08	58.32	7276.83	2068.87	9404.03	0.29	1.97	0.83	1.47
KrishnaRNe	Kr_reach;	67674.0*	Max WS	1627.44	3868.2	240.43	3197.1	7305.69	70.92	7054.76	2325.76	9451.44	0.29	1.82	0.73	1.29
KrishnaRNe	Kr_reach;	66674.0*	Max WS	1935.65	4024.4	283.34	3744.9	8052.6	85.68	6978.81	2451.21	9515.7	0.3	1.73	0.65	1.18
KrishnaRNe	Kr_reach;	65674.0*	Max WS	2346.11	4171.6	330.74	4155.5	8657.87	101.38	7000.44	2478.08	9579.9	0.31	1.68	0.6	1.11
KrishnaRNe	Kr_reach;	64674.0*	Max WS	2784.47	4308.9	404.98	4436.3	9150.17	113.75	7092.35	2422.98	9629.07	0.28	1.65	0.55	1.05
KrishnaRNe	Kr_reach;	63674.0*	Max WS	3207.75	4437.8	507.68	4594.1	9539.55	180.68	7216.6	2296.67	9693.96	0.36	1.63	0.5	1.02
KrishnaRNe	Kr_reach;	62674.0*	Max WS	3793.21	4555 .66	643	4625.3	9823.93	234.18	7390.79	2119.7	9744.66	0.36	1.62	0.46	0.99
KrishnaRNe	Kr_reach;	61674.0*	Max WS	4315.37	4664.6	811.55	4552.2	10028.3	309.26	7597.67	1903.25	9810.19	0.38	1.63	0.42	0.98
KrishnaRNe	Kr_reach;	60674	Max WS	4665.42	4765.6	1021.9	4393.4	10181	402.97	7670.48	1789.02	9862.47	0.39	1.61	0.41	0.97

5.3.13.4 Methodology:

With same geometry and all other conditions the unsteady flow analysis is done with giving straight line geometry with deleting the cross sections in between. The results with and without meander are tabulated below and the reduction in water level is shown.

Table No. 5.7 The results with and without meander

Reach	River Sta	Chainage (m)	from Almatti Chainage (km)	Location	W.S. Elevation in (m)		Reduction in level
					Without Meander	Natural	
Kr_reach1	273964.6	20669.1	279.7		548.85	549.02	0.17
Kr_reach1	271817.4	17780.5	276.8		547.38	547.53	0.15
Kr_reach1	270000	15540.8	274.6		545.51	545.8	0.29
Kr_reach1	258741	12525.5	271.6		545.28	545.6	0.32
Kr_reach1	258740.5	12515.5	271.6		545.26	545.58	0.32
Kr_reach1	258740	11620.25	270.7		545.07	545.42	0.35
Kr_reach1	255584	5864.78	264.9		544.05	544.5	0.45
Kr_reach1	255575	Bridge	264.9	Sangli Bye pass			
Kr_reach1	255565	5834.18	264.9		543.97	544.43	0.46
Kr_reach1	255416	5599.48	264.6		543.99	544.49	0.5
Kr_reach1	255406	4874.98	263.9		543.85	544.34	0.49
Kr_reach1	255405.5		263.9	Irwin Bridge			0
Kr_reach1	255405	4854.98	263.9		543.79	544.28	0.49
Kr_reach1	255400	4437.98	263.5		543.33	543.85	0.52
Kr_reach1	253084	4322.38	263.4		543.71	544.2	0.49
Kr_reach1	253050		263.4	Sangli KT KT Weir			
Kr_reach1	251450	4240.78	263.3		543.27	543.8	0.53
Kr_reach1	247066.6	4120.78	263.2		543.35	543.87	0.52
Kr_reach1	245320	968.88	260.0		543.11	543.67	0.56
Kr_reach2	259181	38572.49	259.0	Warna Confluence	543.11	543.67	0.56
Kr_reach2	258181	36827.49	257.3		542.89	543.5	0.61
Kr_reach2	254181	32444.49	252.9		542.34	543.05	0.71
Kr_reach2	242937	28478.49	248.9		542.13	542.92	0.79
Kr_reach2	237688		247.0	Miraj Arjunwad Bridge			0
Kr_reach2	237687	25154.49	245.6		540.97	542.26	1.29
Kr_reach2	229887	20906.5	241.4			542.09	CS removed
Kr_reach2	218942	17586.5	238.1			541.88	CS removed
Kr_reach2	206189	14451.5	234.9			541.28	CS removed
Kr_reach2	205000	14266.5	234.7			541.24	CS removed
Kr_reach2	204500		234.7	Mhaisal KT Weir			CS removed
Kr_reach2	204000	14226.5	234.7			541.22	CS removed
Kr_reach2	196439	13886.5	234.4			541.18	CS removed
Kr_reach2	186689	12069.5	232.5			540.95	CS removed
Kr_reach2	178189	9935.5	230.4			540.56	CS removed
Kr_reach2	166939	6771.5	227.2		539.73	540.14	0.41

5.3.13.5 Results and discussions

It can seen from above table that at the reduction in water level is 0.41 m at d/s end of meander and 1.21 m at u/s end. This effect is continued on u/s near Irwin bridge Sangli reduction in water level is 0.49 m. The effect is upto 16 km on u/s of Sangli. This reduction is obvious because, the length with same ground level is reduced from 14.93 km to 4.19 km. The bed gradient increases 3.54 time.

5.3.13.6 Recommendations on Meanders

There is significant decrease in water level by removing meander and straightening. However it can not be general conclusion, but individual meander has to be studied separately. The practicality of the removing meanders is also to be studied.

5.3.14 Flow through and outside river channels

Natural top width of River Krishna varies from 267 m to 1021 m. However, during the Flood Event 2019, it was observed that the flood was spread outside the normal river channel. The width of flooded area was about 2.5 km to 7.9 kms. Maximum flow width was 7.98 km, in river stretch between Sangli Bye pass Bridge and Irwin Bridge. Flow width downstream of Miraj – Arjunwad bridge was 5.7 Km. Therefore, attempt is made to analysis the flow pattern in main river channel and on overbanks. It is observed that although the water spread on overbanks was 7 to 10 times that of river channel, the discharge flowing overbanks was less than 5 %. Almost, 95 % discharge was flowing through main channel. Velocity of flow on the overbanks was very merger (about 0.05 to 0.1 m /sec). In some reaches there is no connectivity for flow on overbanks in forward (downstream) direction. In such areas, water doesn't flow in downstream direction but spreads on banks and comes back to channel in receding period.

5.3.15 Channel storages

In HEC-RAS software, there is facility to compute channel storages. Although, the water is continuously flowing, the storage can be computed at any particular locations at any particular instant of time. Channel storages, in various reaches on various dates are extracted and tabulated below in table 5.8 below.

Table 5.8: Channel Storages on Various Dates

River Reach		28-Jul	05-Aug	09-Aug	12-Aug	15- Aug
Hippargi Rajapur		229645.91	918299.38	1344906.5	2335268.25	3970240
Rajapur Sangli		31084.86	97573.52	180425.94	311878.44	334511
Sangli Karad		77216.2	260071.89	479067.41	446946.97	246563

River Reach		28-Jul	05-Aug	09-Aug	12-Aug	15- Aug
Warna		23539.43	64235.05	178175.27	271086.06	330414
Panchganga		83126.79	327564.59	523113.72	851507.44	872739
Total Channel Storage in Maharashtra river Portion	In TCM	444613.19	1667744.43	2705688.84	4216687.16	5754467
	In MCM	444.61	1667.74443	2705.68884	4216.68716	5754.67
	In TMC	**15.70**	**58.89**	**95.54**	**148.89**	**162.97**

This information, if timely shared with the state of Karnataka, will help them to take timely decisions regarding releases from Almatti reservoir.

5.4 Limitations :

a) The accuracy of results of any mathematical model study depends on data accuracy and approximations in solving the governing flow equations. The basic computational procedure is based on solutions of energy equation. Energy losses are evaluated by friction (manning's equation) and contraction /expansion coefficients. Modeling more and more structures with realistic geometry will give more accurate results. Hence, it is more prudent to understand the model results in relative terms rather than absolute terms.

b) The effect of Kudchi and Darur bridge existing in Karnataka territory upstream of the Hippargi Barrage which could not be studied at this stage due to non-availability of data. The effect of these, along with many such structures built across the Krishna river need to be studied further, to ascertain its afflux in the State of Maharashtra.

c) The river cross sections modeled are on the basis of survey done in the year 2012. . Sediment analysis done by Prof. Gadre (2007) predicted rise in bed levels after 10 years at confluences of Panchganga and Warna are 1.15 m and 0.87 m respectively.

d) Contribution of the free catchment between the two River Gauging stations is not generated in the model from the rainfall data but uniformly distributed as lateral flow, on the assumption that the runoff contribution is uniform between the two stream flow gauging locations.

e) During the Flood Event of 2019, the water levels, at many locations, crossed the previously recorded highest levels. The levels recorded at Sangli, Ankali, Rajapur and Rajaram R-G stations were above the range of their respective S-D curves. Hence, the discharge observations recorded at these R-G stations are based on extrapolated S-D curves. Further, gauges at Rajaram Bridge on river Panchganga were severely affected due to backwater of River Krishna. This, has influenced to some extent the hydrographs at these R-G stations and hence the results.

f) In present model, hydraulic structures, mentioned in para 5.3.5 above, in the Maharashtra reach, along with Almatti dam and Hippargi Barrage in the Karnataka reach are modeled. Similarly, Warna and Panchganga rivers along with their confluences are modeled. Other structures and tributaries could not be modeled due to non-availability of the data.

g) Absolute elevations of water surface profiles at Irwin Bridge and at Rajapur fairly tally with observed values at respective locations. Further, as observed stages (water surface elevations) at Hippargi Barrage are modeled as an internal boundary condition, in the hydrodynamic analysis, the conclusions drawn in respect of backwater profiles of Almatti and Hippargi are fairly reliable.

h) The reasons for variation in the simulated values and observed values of water surface elevations at few locations such as Mhaisal K.T. weir, Arjunwad and Kurundwad need to be further analyzed. This difference also existed earlier in Prof. Gadre's study. The difference may be due to combined effect of local contraction / obstructions to the flow, afflux due to structures, confluence effect and also due to reasons mentioned at sub-para "d" above.

i) This model study has given relative water surface profiles in river Krishna with and without Almatti and Hippargi reservoirs and also thrown light on confluence phenomenon. However, this model needs to be further refined, by using more accurate data, as many structures as possible, existing on River Krishna and its tributaries, both in Maharashtra and Karnataka territory, all the tributaries along with their confluences with River Krishna. Further, all the structures and particularly river gauging stations need to be connected to G.T.S bench marks for confirming its levels. The S-D curves at various R-G stations need to be calibrated. So also, the river cross sections need to be verified for siltation occurred since 2012. Such a refined model will help in integrated reservoir operations and also for designing the remedial measures.

5.5 Conclusions:

The important conclusions drawn from hydrodynamic simulation study are summarized as under.

5.5.1 Reasons for flooding:

a) Simultaneous occurrence of unprecedented heavy precipitation in the catchments of river Krishna and its tributaries.

b) Tribunal Constraints on utilization of available waters in Krishna basin in Maharashtra.

c) Inadequate discharge carrying capacity of river Krishna to accommodate the releases of Koyna dam, the contribution of tributaries and the runoff of the free catchment.

d) The river Krishna flows, a near plain land, between Sangli and the State border. It is a general phenomenon that the river takes meandering course, while traversing on a plain land. Thus, River Krishna has so many curves and meanders. The velocity of the river is comparatively less while traversing curves and meanders, causing thereby more inundation on inner as well as outer sides of the curves and meanders as compared to straight reaches of the river. The Sangli city is on one of the curve of River Krishna and Kolhapur city is on the curve of river Panchganga

e) Flow stagnation in River Krishna from Sangli city to the State border, due to confluence effect. There exists series of confluences, Yerala-Krishna, Warna-Krishna, Panchganga-Krishna and Dudhganga-Krishna within a reach of about 50 to 55 km length. At confluence points due to formation of stagnation zone velocity is reduced.

f) Generation of backwater effect in the tributaries and nallas meeting the River Krishna. Due to comparatively higher discharge in the River Krishna, the backwater effect is generated in the tributaries. The flooding in tributaries viz. Yerala, Warna, Panchganga and other nallas like Bhilwadi, Nagthane was primarily due to backwater effect of the river Krishna. The situation was further aggravated as these tributaries could not drain out their own discharge, till the flood in the river Krishna was receded. Water was spread on the side banks of the tributaries due to ponding effect. The river Krishna was flooded for a long duration from 5th August up to 13th of August. Consequently, the backwater spread in the tributaries and on floodplains could not return back to the river course. Thus, the floodplains were under water for a prolonged time.

g) Reduced discharge carrying capacity of river Warna and Panchganga due to siltation, vegetation growth and encroachments further delayed the reception of floods.

h) The lateral slope of the flood plains is very gentle. The flood plains are almost flat. This has resulted into spreading of flood on larger area on both the banks of the river.

5.5.2 Effect of Almatti dam and Hippargi Barrage:

Although this mathematical model study, has certain limitations, this study indicates that Almatti and Hippargi reservoirs in Karnataka and its flood operations, during the Flood Event of 2019, has not adversely affected the flood situation in the State of Maharashtra.

N. N. Rai, Director, Hydrology (South) Central Water Commission, New Delhi, and the Member of this Committee, with independent analysis, has also drawn similar conclusions.

The crest level and FRL of Hippargi Barrage are at 516.640 m and 524.870 m respectively whereas the ground level at Rajapur Barrage (State Boundary) is 518.20 m. Thus, when the gates of Hippargi Barrage are in place, (in normal condition) there is a submergence of about 6.67 m at Rajapur. However, during flood as the gates are raised, this scenario is not governing.

Unsteady state analysis of the Flood Event 2019, indicates that backwater profile of Almatti dam, considering both Almatti and Hippargi reservoirs, merges with water profile in natural condition of the river Krishna (no Almatti and Hippargi condition) at 119 km from Almatti dam i.e. just downstream of Hippargi dam in the territory of the State of

Karnataka. Further, the effect of Hippargi is up to 80 km from Hippargi i.e. 22 Km downstream of Rajapur, in the territory of the Karnataka.

Various assumptions / approximations in the input data, may alter the absolute water surface profiles to some extent, for both, with Almatti and Hippargi in reservoirs place and in natural condition (No Dam condition).

Dr. Subimal Ghosh, Professor, Department of Civil Engineering, Indian Institute of Technology, Bombay, after scrutiny of this study, has opinioned that, he agrees with the conclusions drawn, as they are based on technically correct and scientifically rigorous study.

5.5.3 Suggestions for reducing severity of flood situation:

a) Restoration of cross sections of river Krishna including tributaries, nallas and any local drains, in the flood prone area, to its natural state (natural width and depth) by removing dumped debris, siltation occurred, vegetation developed and manmade encroachments. Further, the Local Authorities, as a routine, should clean the drains in its jurisdiction, every year before monsoon. River Krishna is susceptible to siltation below the confluence of river Warna, due to its flatter bed slopes.

b) Raising the low riverbanks to restrict inundation, coupled with pumping facility to prevent interior flooding problems. Hydrodynamic mathematical model already developed is capable of identifying such potential reaches and also suggest bank raising requirement. The decision in this regard can be taken on inspection of banks from stretch to stretch

c) Converting the spillway of Radhanagari dam to gated spillway with flood cushion.

d) City specific detailed studies for ascertaining the location specific reasons of inundation (on the lines of study done by ISMR for Chennai city), and designing site specific mitigating measures.

e) Reliving the flow congestion particularly below Warna and Panchganga confluences by straightening the meanders. The decision in this regard needs to be taken after physical model studies

f) Planning and managing all the reservoirs in the K1 sub-basin in integrated manner with well-designed Decision Support System (DSS). The reservoir releases of Koyna and that of Warna and dams in Panchganga valley must be staggered.

g) Integrated operation of reservoirs in the State of Karnataka by sharing real time data regarding rainfall, reservoir storages, river gaging and channel storages (as computed from HEC-RAS) with the State of Karnataka.

h) Creating temporary flood cushion, in the dams, by ‘Pre Releases’, before the flood actually strikes the reservoir. This is possible without materially sacrificing the conservation storage, by ensuring proper coordination with IMD and with appropriate Decision Support System

i) Revising the Reservoir Operation Schedule (ROS) of Koyna dam, giving equal weightage to flood control, considering the fact that releases from Koyna during flood, governs the flood situation,

j) Revising the ROS of Warna dam considering the existing utilization pattern.

k) Using the capability of HEC-RAS, for simulating inundation and deploying automated system to send alert signals to all concerned.

5.6 Simulated elevations of water surface profile for Various Scenario:

The simulated elevations of water surface profiles for various scenarios studied are tabulated in following tables.

Scenario I: Simulated back water surface profile of Almatti dam with FRL at 524.25 m, generated from steady state analysis, PMF discharge of 31,000 cumecs at Almatti and 24,600 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs.

Scenario II: Simulated back water surface profile of Almatti dam with FRL at 519.60 m, generated from steady state analysis, PMF discharge of 31,000 cumecs at Almatti and 24,600 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs. Table 5.9 shows the details of above scenario.

Table no: 5.9

Chainage (km)	Ground Level	Location	W.S. Elevation in (m)		
			Without Almatti Dam	Almatt FRL 519.60	Almatti FRL 524.24
367.4	548.7		551.47	551.47	551.47
366.0	549.0		551.24	551.24	551.24
366.0		**Karad Bridge**			
365.9	548.3		551.23	551.23	551.23
365.8	548.5		551.21	551.21	551.21
363.2	547.8		550.93	550.93	550.93
360.8	547.6		550.77	550.78	550.78
359.8	547.5		550.69	550.7	550.7
358.0	546.0		550.63	550.64	550.64
357.4	545.3		550.62	550.63	550.63
353.6	544.5		550.6	550.61	550.61
348.8	543.1		550.57	550.58	550.58
345.3	543.1		550.56	550.57	550.57
344.3	542.8		550.56	550.57	550.57
337.3	542.5		550.55	550.56	550.56
333.2	542.1		550.55	550.56	550.56
331.0	541.6		550.55	550.56	550.56
327.1	540.5		550.55	550.55	550.55
324.0	538.3		550.55	550.55	550.55
320.7	537.9		550.54	550.55	550.55
319.1	537.5		550.54	550.55	550.55
316.6	537.0		550.54	550.55	550.55

Chainage (km)	Ground Level	Location	W.S. Elevation in (m)		
			Without Almatti Dam	Almatt FRL 519.60	Almatti FRL 524.24
314.2	536.4		550.54	550.55	550.55
313.7	536.2		550.54	550.55	550.55
312.8	536.2		550.54	550.55	550.55
311.6	536.5		550.54	550.55	550.55
310.0	536.7		550.54	550.55	550.55
307.3	536.6		550.54	550.55	550.55
306.2	536.5		550.54	550.55	550.55
303.1	536.3		550.54	550.55	550.55
300.9	536.2		550.54	550.55	550.55
297.7	536.1		550.54	550.55	550.55
293.4	535.9		550.54	550.55	550.55
291.9	535.8		550.54	550.55	550.55
288.4	535.5		550.54	550.55	550.55
286.7	535.3		550.54	550.55	550.55
284.8	535.2		550.54	550.55	550.55
282.1	535.0		550.54	550.55	550.55
279.7	534.8		550.54	550.54	550.55
276.8	532.3		550.54	550.54	550.54
274.6	532.3		550.54	550.54	550.54
271.6	528.0	**Yerala Confluence**	550.54	550.54	550.54
271.6	528.0		550.54	550.54	550.54
270.7	526.9		550.54	550.54	550.54
264.9	525.9		550.54	550.54	550.54
264.9		**Sangli Bye pass**			
264.9	525.9		550.54	550.54	550.54
264.6	526.5		**550.52**	**550.52**	**550.52**
263.9	525.9		549.72	549.73	549.73
263.9		**Irwin Bridge**			
263.9	525.9		549.24	549.25	549.25
263.5	525.9		545.1	545.13	545.13
263.4	526.3		547.97	547.99	547.99
263.4		**Sangli K. T. Weir**			
263.3	526.3		548.02	548.03	548.03
263.2	526.0		548.02	548.03	548.03
260.0	525.9		547.87	547.89	547.89
259.0	525.8	**Warna Confluence**	547.84	547.86	547.86
257.3	525.6		547.76	547.78	547.78
252.9	525.5		547.25	547.28	547.28
248.9	524.8		546.7	546.74	546.74
247.0		**Miraj Arjunwad**			
245.6	523.2		546.58	546.62	546.62
241.4	522.7		546.52	546.57	546.57
238.1	522.0		546.43	546.48	546.48
234.9	521.6		546.19	546.25	546.25
234.7	522.9		544.66	544.62	544.62
234.7		**Mhaisal KT Weir**			

Chainage (km)	Ground Level	Location	W.S. Elevation in (m)		
			Without Almatti Dam	Almatt FRL 519.60	Almatti FRL 524.24
234.7	522.9		544.52	544.46	544.46
234.4	521.4		545.34	545.34	545.34
232.5	520.5		544.62	544.63	544.63
230.4	520.0		544.29	544.3	544.3
227.2	518.7		543.75	543.76	543.76
224.2	518.0		543.32	543.34	543.34
221.3	517.7		543.02	543.05	543.05
221.3	517.7		543.02	543.05	543.05
220.5	517.7	**Panchanga Confluence**	542.95	542.98	542.98
218.8	517.6		542.79	542.82	542.82
215.6	517.5		542.54	542.58	542.58
212.8	516.7		542.35	542.39	542.4
211.3	518.2		542.06	542.11	542.11
211.0		**Rajapur KT weir**			
210.5	518.2		542.03	542.08	542.08
209.9	515.2		541.94	541.98	541.99
209.7	515.2		541.94	541.99	541.99
205.9	515.2		541.69	**541.74**	**541.74**
198.3	518.1		540.16	540.21	540.22
188.3	514.8		538.32	538.31	538.32
170.7	516.3		536.65	536.65	536.68
165.9	512.8		536.21	536.21	536.24
155.1	512.5		534.89	534.89	534.97
150.1	512.3		534.16	534.16	534.29
141.5	511.5		533.31	533.31	533.54
135.1	512.9		**532.69**	**532.69**	533.04
119.9	511.7		531.14	531.15	531.93
119.0		**Hippargi Barrage**			
119.0	509.0		530.62	530.63	530.7
110.5	509.0		530.08	530.09	530.19
103.0	510.8		529.23	529.26	529.43
92.2	508.4		528.72	528.75	528.99
81.0	508.9		528.09	528.15	528.5
72.5	506.0		527.24	527.32	527.84
62.7	507.0		526.18	526.32	527.08
53.0	498.8		525.04	525.26	526.37
40.2	504.8		521.15	522.09	524.92
29.3	501.0		518.09	520.49	524.47
21.5	503.1		516.2	520.07	524.38
16.2	502.9		514.69	519.82	524.31
5.0	496.6		513.3	519.64	524.26
1.0	495.5		512.99	519.6	524.25
0.0		**Almatti**	511.05	519.6	524.25

Note:

1) It is seen that the backwater profile of Almatti dam for FRL at 524.25 m and PMF discharge of 31,000 cumecs at Almatti and 24, 609 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs and siltation occurred up to the year 2011, merges with water profile in natural condition of the river Krishna at km 206 from Almatti dam i.e. in the territory of the State of Karnataka.

2) It is seen that the backwater profile of Almatti dam for FRL at 519.6 m and PMF discharge of 31,000 cumecs at Almatti and 24,609 cumecs at Hippargi, considering both Almatti and Hippargi reservoirs and siltation occurred up to the year 2011, merges with water profile in natural condition of the river Krishna at 135 km from Almatti dam i.e. in the territory of the State of Karnataka.

Scenario III : Unsteady State Analysis of flood event, on 9TH august at 4 am (time instant at which simulated water level at Irwin is max) with and without Almatti and Hippargi Dam. In Hippargi BC scenario actually observed stages at Hippargi are considered as internal boundary. (See Note 2 below Table 5.10) Table 5.10 shows the details of above scenario.

Table 5.10

Chainage	Location	W.S. Elevation in (m)		
(km)		Without	Almatti	Hippargi BC
		Almatti Dam	FRL 519.6	
367.4		561.93	561.93	561.93
366.0		561.76	561.76	561.76
366.0	Karad Bridge			
365.9		561.75	561.75	561.75
365.8		561.74	561.74	561.74
363.2		561.46	561.46	561.46
360.8		561.15	561.15	561.15
359.8		560.92	560.92	560.92
358.0		560.62	560.62	560.62
357.4		560.52	560.52	560.52
353.6		560.12	560.12	560.12
348.8		559.19	559.19	559.19
345.3		558.71	558.71	558.71
344.3		558.57	558.57	558.57
337.3		557.53	557.53	557.53
333.2		557	557	557
331.0		556.81	556.81	556.81
327.1		555.92	555.92	555.92
324.0		555.76	555.76	555.76
320.7		555.43	555.43	555.43
319.1		555.37	555.37	555.37
316.6		555.21	555.21	555.21
314.2		555.12	555.12	555.12
313.7		555.11	555.11	555.11
312.8		555.07	555.07	555.07
311.6		555.02	555.02	555.02
310.0		554.98	554.98	554.98
307.3		554.68	554.68	554.68
306.2		554.48	554.48	554.48
303.1		554.08	554.08	554.08
300.9		553.94	553.94	553.94
297.7		553.72	553.72	553.72
293.4		553.27	553.27	553.27
291.9		553.15	553.15	553.15
288.4		552.95	552.96	552.96
286.7		552.85	552.85	552.85
284.8		552.69	552.69	552.69
282.1		551.26	551.26	551.26

Chainage (km)	Location	W.S. Elevation in (m)		
		Without Almatti Dam	Almatti FRL 519.6	Hippargi BC
279.7		549.02	549.02	549.02
276.8		547.53	547.53	547.53
274.6		545.79	545.8	545.8
271.6	**Yerala Confluence**	545.6	545.6	545.61
271.6		545.58	545.58	545.59
270.7		545.41	545.42	545.42
264.9		544.5	544.5	544.51
264.9	**Sangli Bye pass**			
264.9		544.42	544.43	544.44
264.6		544.49	544.49	544.5
263.9		544.33	544.34	544.34
263.9	**Irwin Bridge**			
263.9		544.28	544.28	544.29
263.5		543.85	543.85	543.86
263.4		544.2	544.2	544.21
263.4	**Sangli K. T. Weir**			
263.3		543.79	543.8	543.79
263.2		543.86	543.87	543.86
260.0		543.67	543.67	543.67
259.0	**Warna Confluence**	543.67	543.67	543.67
257.3		543.5	543.5	543.5
252.9		543.05	543.05	543.05
248.9		542.91	542.92	542.91
247.0	**Miraj Arjunwad**			
245.6		542.25	542.26	542.25
241.4		542.08	542.09	542.08
238.1		541.87	541.88	541.87
234.9		541.27	541.28	541.27
234.7		541.23	541.24	541.23
234.7	**Mhaisal KT Weir**			
234.7		541.21	541.22	541.21
234.4		541.17	541.18	541.17
232.5		540.94	540.95	540.94
230.4		540.56	540.56	540.56
227.2		540.14	540.14	540.14
224.2		539.62	539.62	539.62
221.3		539.36	539.35	539.36
221.3		539.34	539.34	539.34
220.5	**Panchanga Confluence**	539.34	539.34	539.34
218.8		539.14	539.14	539.14
215.6		538.83	538.82	538.83

Chainage (km)	Location	W.S. Elevation in (m)		
		Without Almatti Dam	Almatti FRL 519.6	Hippargi BC
212.8		538.4	538.39	538.4
211.3		538.18	538.18	538.18
211.0	**Rajapur KT weir**			
210.5		538.16	538.16	538.16
209.9		538.03	538.03	538.03
209.7		538.01	538.01	538.02
205.9		537.79	537.79	537.8
198.3		536.72	536.72	536.72
188.3		**535.09**	**535.11**	535.1
170.7		533.02	533.03	533.05
165.9		532.62	532.62	532.66
155.1		531.46	531.47	531.53
150.1		530.87	530.86	530.96
141.5		529.77	529.77	529.95
135.1		528.72	528.72	529.04
119.9		526.52	526.53	527.33
119.0	**Hippargi Bridge**			
119.0		**526.53**	**526.54**	526.46
110.5		526.25	526.27	526.18
103.0		524.95	525	524.9
92.2		523.68	523.79	523.7
81.0		522.54	522.74	522.65
72.5		521.22	521.58	521.49
62.7		519.37	520.08	520
53.0		517.82	518.96	518.9
40.2		515.66	517.94	517.91
29.3		513.64	517.51	517.49
21.5		511.04	517.35	517.34
16.2		509.22	517.3	517.29
5.0		505.26	517.26	517.26
1.0		499.94	517.26	517.25
0.0	Almatti	496.86	517.1	517.1

Note: 1) It is seen that the backwater profile of Almatti dam, merges with water profile in natural condition of the river Krishna at km 119 from Almatti dam i.e. Just downstream of Hippargi reservoir, in the territory of the State of Karnataka. Similarly, effect of Hippargi extends 70 km upstream of Hippargi i.e. 22 km downstream of the State boundary.

2) Simulation study, with Almatti FRL at 519.6 m, indicated water surface elevation upstream of Hippargi barrage, as 526.53m. However, level actually observed at this location is 527.33 m.

Scenario IV : Unsteady state analysis 2019 flood event by controlling the discharge of river Warna (75%, 50%,25%), Almatti reservoir at 519.6 m.

Table 5.11 below shows the details of above scenario.

Table 5.11

Chainage (km)	Location	W.S. Elevation in (m)			
		Warna Discharge			
		100 % (9 August discharge)	75%	50%	25%
367.4		561.93	561.93	561.93	561.93
366.0		561.76	561.76	561.76	561.76
366.0	**Karad Bridge**				
365.9		561.75	561.75	561.75	561.75
365.8		561.74	561.74	561.74	561.74
363.2		561.46	561.46	561.46	561.46
360.8		561.15	561.15	561.15	561.15
359.8		560.92	560.92	560.92	560.92
358.0		560.62	560.62	560.62	560.62
357.4		560.52	560.52	560.52	560.52
353.6		560.12	560.12	560.12	560.12
348.8		559.19	559.19	559.19	559.19
345.3		558.71	558.71	558.71	558.7
344.3		558.57	558.57	558.57	558.57
337.3		557.53	557.53	557.53	557.52
333.2		557	557	557	557
331.0		556.81	556.81	556.81	556.81
327.1		555.92	555.92	555.92	555.92
324.0		555.76	555.76	555.75	555.75
320.7		555.43	555.43	555.42	555.42
319.1		555.37	555.37	555.37	555.37
316.6		555.21	555.21	555.2	555.2
314.2		555.12	555.12	555.12	555.12
313.7		555.11	555.1	555.1	555.1
312.8		555.07	555.07	555.07	555.07
311.6		555.02	555.01	555.01	555.01
310.0		554.98	554.98	554.98	554.98
307.3		554.68	554.68	554.68	554.68
306.2		554.48	554.48	554.48	554.48
303.1		554.08	554.07	554.07	554.07
300.9		553.94	553.94	553.94	553.94
297.7		553.72	553.72	553.71	553.71
293.4		553.27	553.27	553.26	553.26
291.9		553.15	553.14	553.14	553.14
288.4		552.96	552.95	552.95	552.94
286.7		552.85	552.85	552.84	552.84
284.8		552.69	552.69	552.68	552.68

Chainage (km)	Location	W.S. Elevation in (m)			
		Warna Discharge			
		100% (9 August discharge)	75%	50%	25%
282.1		551.26	551.24	551.23	551.21
279.7		549.02	548.99	548.95	548.92
276.8		547.53	547.5	547.47	547.44
274.6		545.8	545.74	545.68	545.62
271.6	**Yerala Confluence**	545.6	545.54	545.48	545.41
271.6		545.58	545.52	545.45	545.39
270.7		545.42	545.35	545.28	545.21
264.9		544.5	544.41	544.32	544.23
264.9	**Sangli Bye pass**				
264.9		544.43	544.34	544.25	544.15
264.6		544.49	544.4	544.3	544.19
263.9		544.34	544.24	544.15	544.05
263.9	**Irwin Bridge**				
263.9		544.28	544.19	544.09	543.99
263.5		543.85	543.75	543.65	543.54
263.4		544.2	544.11	544.01	543.91
263.4	**Sangli K. T. Weir**				
263.3		543.8	543.69	543.59	543.48
263.2		543.87	543.76	543.66	543.55
260.0		543.67	543.56	543.45	543.33
259.0	**Warna Confluence**	543.67	543.56	543.45	543.33
257.3		543.5	543.39	543.28	543.16
252.9		543.05	542.95	542.85	542.75
248.9		542.92	542.82	542.72	542.61
247.0	**Miraj Arjunwad**				
245.6		542.26	542.14	542.02	541.9
241.4		542.09	541.97	541.86	541.74
238.1		541.88	541.76	541.65	541.53
234.9		541.28	541.17	541.07	540.97
234.7		541.24	541.14	541.04	540.93
234.7	**Mhaisal KT weir**				
234.7		541.22	541.12	541.02	540.91
234.4		541.18	541.07	540.97	540.87
232.5		540.95	540.85	540.75	540.65
230.4		540.56	540.48	540.39	540.3
227.2		540.14	540.06	539.98	539.89
224.2		539.62	539.55	539.48	539.41
221.3		539.35	539.29	539.22	539.15
221.3		539.34	539.28	539.21	539.14

Chainage (km)	Location	W.S. Elevation in (m)			
		Warna Discharge			
		100 % (9 August discharge)	75%	50%	25%
220.5	**Panchganga Confluence**	539.34	539.28	539.21	539.14
218.8		539.14	539.08	539.01	538.94
215.6		538.82	538.76	538.7	538.63
212.8		538.39	538.33	538.27	538.2
211.3		538.18	538.12	538.06	538
211.0	**Rajapur KT Weir**				
210.5		538.16	538.1	538.03	537.97
209.9		538.03	537.97	537.91	537.85
209.7		538.01	537.96	537.89	537.83
205.9		537.79	537.74	537.68	537.61
198.3		536.72	536.66	536.61	536.55
188.3		**535.11**	**535.04**	**534.99**	**534.94**
170.7		533.03	532.97	532.92	532.87
165.9		532.62	532.56	532.51	532.46
155.1		531.47	531.41	531.36	531.32
150.1		530.86	530.82	530.77	530.72
141.5		529.77	529.72	529.67	529.62
135.1		528.72	528.67	528.62	528.57
119.9		526.53	526.5	526.45	526.41
119.0	**Hippargi Barrage**				
119.0		526.54	526.51	526.46	526.42
110.5		526.27	526.23	526.19	526.15
103.0		525	524.96	524.91	524.87
92.2		523.79	523.76	523.71	523.67
81.0		522.74	522.71	522.66	522.62
72.5		521.58	521.55	521.5	521.47
62.7		520.08	520.05	520.01	519.97
53.0		518.96	518.93	518.91	518.88
40.2		517.94	517.93	517.91	517.9
29.3		517.51	517.5	517.49	517.48
21.5		517.35	517.35	517.34	517.34
16.2		517.3	517.29	517.29	517.29
5.0		517.26	517.26	517.26	517.26
1.0		517.26	517.26	517.25	517.25
0.0	**Almatti**	517.1	517.1	517.1	517.1

Scenario V: Unsteady Analysis of 2019 flood event by controlling the discharge of Panchganga (25 %)

Table 5.12 below shows the details of above scenario.

Table 5.12

Chainage (km)	Location	W.S. Elevation in (m)	
		Panchganga Discharge	
		100 % (9 August discharge)	25%
367.4		561.93	561.93
366.0		561.76	561.76
366.0	**Karad Bridge**		
365.9		561.75	561.75
365.8		561.74	561.74
363.2		561.46	561.46
360.8		561.15	561.15
359.8		560.92	560.92
358.0		560.62	560.62
357.4		560.52	560.52
353.6		560.12	560.12
348.8		559.19	559.19
345.3		558.71	558.7
344.3		558.57	558.57
337.3		557.53	557.52
333.2		557	557
331.0		556.81	556.81
327.1		555.92	555.92
324.0		555.76	555.75
320.7		555.43	555.42
319.1		555.37	555.37
316.6		555.21	555.2
314.2		555.12	555.12
313.7		555.11	555.1
312.8		555.07	555.07
311.6		555.02	555.01
310.0		554.98	554.98
307.3		554.68	554.68
306.2		554.48	554.48
303.1		554.08	554.07
300.9		553.94	553.94
297.7		553.72	553.71
293.4		553.27	553.26
291.9		553.15	553.14
288.4		552.96	552.95
286.7		552.85	552.84
284.8		552.69	552.68
282.1		551.26	551.23
279.7		549.02	548.96
276.8		547.53	547.48

Chainage (km)	Location	W.S. Elevation in (m) Panchganga Discharge	
		100 % (9 August discharge)	25%
274.6		545.80	545.71
271.6	**Yerala Confluence**	545.60	545.5
271.6		545.58	545.48
270.7		545.42	545.31
264.9		544.5	544.37
264.9	**Sangli Bye pass**		
264.9		544.43	544.29
264.6		544.49	544.35
263.9		544.34	544.19
263.9	**Irwin**		
263.9		544.28	544.14
263.5		543.85	543.70
263.4		544.2	544.06
263.4	**Sangli K. T. Weir**		
263.3		543.8	543.64
263.2		543.87	543.71
260.0		543.67	543.51
259.0	**Warna Confluence**	543.67	543.51
257.3		543.5	543.33
252.9		543.05	542.86
248.9		542.92	542.70
247.0	**Miraj Arjunwad**		
245.6		542.26	541.90
241.4		542.09	541.71
238.1		541.88	541.46
234.9		541.28	540.79
234.7		541.24	540.75
234.7	**Mhaisal KT weir**		
234.7		541.22	540.72
234.4		541.18	540.67
232.5		540.95	540.4
230.4		540.56	539.94
227.2		540.14	539.39
224.2		539.62	538.67
221.3		539.35	538.25
221.3		539.34	538.23
220.5	**Panchganga Confluence**	539.34	538.23
218.8		539.14	538.03
215.6		538.82	537.72
212.8		538.39	537.3

Chainage (km)	Location	W.S. Elevation in (m) Panchganga Discharge 100 % (9 August discharge)	25%
211.3		538.18	537.14
211.0	**Rajapur KT weir**		
210.5		538.16	537.11
209.9		538.03	537.02
209.7		538.01	537.00
205.9		537.79	536.79
198.3		536.72	535.74
188.3		**535.11**	**534.22**
170.7		533.03	532.06
165.9		532.62	531.66
155.1		531.47	530.55
150.1		530.86	529.93
141.5		529.77	528.81
135.1		528.72	527.78
119.9		526.53	525.75
119.0	**Hippargi Barrage**		
119.0		526.54	525.77
110.5		526.27	525.50
103.0		525	524.17
92.2		523.79	522.98
81.0		522.74	521.97
72.5		521.58	520.87
62.7		520.08	519.49
53.0		518.96	518.56
40.2		517.94	517.73
29.3		517.51	517.4
21.5		517.35	517.29
16.2		517.3	517.25
5.0		517.26	517.22
1.0		517.26	517.22
0.0	**Almatti**	517.1	517.1

Scenario: VI: Unsteady Analysis of 2019 flood event by controlling the discharge of Karad (75 %, 50 %, 33 %) By controlling releases of Koyna

Table 5.13 below shows the details of above scenario.

Table 5.13

Chainage (km)	Location	W.S. Elevation in (m) Discharge at Karad			
		100 % (9 August discharge)	75%	50%	33%
367.4		561.93	560.45	558.64	557.05
366.0		561.76	560.28	558.46	556.86
366.0	**Karad Bridge**				
365.9		561.75	560.26	558.45	556.86
365.8		561.74	560.25	558.44	556.84
363.2		561.46	559.99	558.18	556.6
360.8		561.15	559.68	557.89	556.31
359.8		560.92	559.44	557.61	556.02
358.0		560.62	559.16	557.34	555.74
357.4		560.52	559.08	557.29	555.7
353.6		560.12	558.71	556.96	555.4
348.8		559.19	557.82	556.16	554.67
345.3		558.71	557.36	555.76	554.31
344.3		558.57	557.24	555.64	554.2
337.3		557.53	556.32	554.83	553.42
333.2		557	555.84	554.42	553.04
331.0		556.81	555.67	554.28	552.91
327.1		555.92	554.92	553.67	552.36
324.0		555.76	554.75	553.49	552.16
320.7		555.43	554.48	553.26	551.94
319.1		555.37	554.42	553.21	551.89
316.6		555.21	554.28	553.09	551.78
314.2		555.12	554.2	553.02	551.72
313.7		555.11	554.18	553.01	551.7
312.8		555.07	554.15	552.98	551.68
311.6		555.02	554.1	552.94	551.64
310.0		554.98	554.06	552.9	551.61
307.3		554.68	553.78	552.65	551.41
306.2		554.48	553.62	552.52	551.31
303.1		554.08	553.26	552.21	551.02
300.9		553.94	553.12	552.07	550.91
297.7		553.72	552.92	551.89	550.76
293.4		553.27	552.55	551.59	550.51
291.9		553.15	552.43	551.46	550.35
288.4		552.96	552.22	551.19	549.83
286.7		552.85	552.09	550.95	549.34
284.8		552.69	551.85	550.34	548.75

Chainage (km)	Location	W.S. Elevation in (m) Discharge at Karad			
		100 % (9 August discharge)	75%	50%	33%
282.1		551.26	549.71	547.87	546.77
279.7		549.02	547.53	545.58	544.38
276.8		547.53	546.28	544.42	543.16
274.6		545.8	544.67	543.08	541.92
271.6	**Yerala Confluence**	545.6	544.43	542.78	541.55
271.6		545.58	544.41	542.77	541.54
270.7		545.42	544.25	542.63	541.43
264.9		544.5	543.45	541.9	540.75
264.9	**Sangli Bypass**				
264.9		544.43	543.4	541.86	540.71
264.6		544.49	543.36	541.78	540.63
263.9		544.34	543.28	541.76	540.64
263.9	**Irwin Bridge**				
263.9		544.28	543.24	541.73	540.61
263.5		543.85	542.92	541.49	540.42
263.4		544.2	543.18	541.68	540.57
263.4	**Sangli K. T. Weir**				
263.3		543.8	542.8	541.37	540.3
263.2		543.87	542.87	541.43	540.35
260.0		543.67	542.68	541.23	540.15
259.0	**Warna Confluence**	543.67	542.68	541.23	540.15
257.3		543.5	542.51	541.06	540
252.9		543.05	542.12	540.7	539.68
248.9		542.92	541.97	540.51	539.48
247.0	**Miraj Arjunwad**				
245.6		542.26	541.27	540.05	539.07
241.4		542.09	541.1	539.87	538.88
238.1		541.88	540.9	539.68	538.7
234.9		541.28	540.39	539.28	538.35
234.7		541.24	540.36	539.26	538.34
234.7	**Mhaisal KT weir**				
234.7		541.22	540.34	539.25	538.33
234.4		541.18	540.3	539.2	538.29
232.5		540.95	540.1	539.06	538.17
230.4		540.56	539.78	538.79	537.93
227.2		540.14	539.41	538.48	537.65
224.2		539.62	538.97	538.14	537.4
221.3		539.35	538.73	537.94	537.25
221.3		539.34	538.72	537.94	537.25
220.5	**Panchganga Confluence**	539.34	538.72	537.94	537.25

Chainage (km)	Location	W.S. Elevation in (m) Discharge at Karad			
		100 % (9 August discharge)	75%	50%	33%
218.8		539.14	538.52	537.74	537.07
215.6		538.82	538.21	537.43	536.77
212.8		538.39	537.79	537	536.37
211.3		538.18	537.6	536.86	536.27
211.0	**Rajapur KT weir**				
210.5		538.16	537.57	536.84	536.25
209.9		538.03	537.46	536.75	536.17
209.7		538.01	537.44	536.74	536.16
205.9		537.79	537.23	536.52	535.95
198.3		536.72	536.18	535.49	534.95
188.3		**535.11**	**534.62**	**533.98**	**533.46**
170.7		533.03	532.51	531.82	531.27
165.9		532.62	532.11	531.42	530.87
155.1		531.47	530.99	530.33	529.78
150.1		530.86	530.39	529.69	529.11
141.5		529.77	529.28	528.59	528.01
135.1		528.72	528.24	527.58	527.08
119.9		526.53	526.15	525.6	525.15
119.0	**Hippargi Barrage**				
119.0		526.54	526.17	525.63	525.17
110.5		526.27	525.9	525.35	524.9
103.0		525	524.6	524.04	523.58
92.2		523.79	523.41	522.86	522.37
81.0		522.74	522.38	521.87	521.39
72.5		521.58	521.25	520.77	520.29
62.7		520.08	519.79	519.43	519.13
53.0		518.96	518.76	518.52	518.33
40.2		517.94	517.84	517.71	517.62
29.3		517.51	517.45	517.39	517.35
21.5		517.35	517.32	517.28	517.25
16.2		517.3	517.27	517.24	517.22
5.0		517.26	517.24	517.22	517.2
1.0		517.26	517.24	517.21	517.2
0.0	**Almatti**	517.1	517.1	517.1	517.1

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Chapter 6–Flood Mitigation Measures

Background:

The committee has meticulously **collected** all the required **data** from the various state and central government authorities regarding the flood event 2019 in Krishna sub-basin of Maharashtra. The necessary meteorological data, hydraulic and hydrological data, geographical and terrain data in the form of maps, have been collected. The detailed reports, based on their own authentic data have also been received from the various renowned central and state government institutions, non-governmental organisations, environmentalists and many experts working in related fields.

The site visits of all the committee members together were organised in the flood affected areas, particularly the Satara, Sangli and Kolhapur districts. Visit to the Almatti, and Hippargi dams in Karnataka state were also done. An aerial inspection of the flood affected area, by some of the members of the committee was also done going along the Krishna river, starting from Koyna dam in Maharashtra, part of Panchganga and upto Almatti dam in Karnataka. This was made possible using a small private plane, flying at the altitude as low as 1000 to 3000 ft. The date of flying was 9th August 2019, where the floods were at its peak, and its maximum inundation levels. Visit to IMD / IITM offices were given and detailed discussions were held with various officials / scientists and experienced retired director General Metrology further the visit to CWPRS was also given to update present aspects of River Engineering especially physical modelling.

Detailed discussions were held during the committee meetings held under the chairman of the committee, and also during the site visits with the Divisional Commissioner and concerned District Collectors along with revenue authorities, flood affected persons and elder citizens living in the areas.

After the detailed study and analysis by various experts, the various reasons responsible for this worst flood situation in the region were enumerated. All the natural and man-made issues have been considered. Those have been listed according to their most severity affecting for the cause. All these issues and findings have been discussed and formulated in the previous chapters in detail. Based on this, the Flood Mitigation Measures to be suggested were discussed. The measures suggested are sub-listed as **short term (urgent) and long-term measures** for every aspect. These are discussed below.

The various aspects considered to suggest the important measures are –

6.1 Advanced Flood Monitoring systems

6.2 Flood Forecasting

6.3 Restoration of Natural Land Drainage systems

6.4 Encroachments in Natural waterways

6.5 Flood Absorption measures

6.6 Disaster Management Units

6.1 Advanced Flood Monitoring Systems

The reservoirs having **gated spillways** within the upstream catchment areas of the flood affected plains are the great assets for the flood control, routing and absorption purposes. Unfortunately, there are great mis-concepts amongst the common citizens. Whenever the discharges are released over the dams by the dam operators, whatever small or large, the people believe (without knowing the situations and facts) that the floods have been created by these discharges. They commonly fail to understand that the releases through dams are made and are designed to release for the safety of the dam first. If the dam leaks or breaches after storing excessive floods than designed, the disaster that will occur will be much more than they feel by current releases. We have many examples of breach of dams, even ungated ones creating havoc and unprecedented damages – both fatal and property losses. Many such occurrences come even without sufficient warnings. To understand the use and importance of having dams (from flood mitigation point of view), one can imagine a no dam situation. This can also be proved from the historical records that large flooding had been occurred in the same area before these dams were constructed.

Thus, the **dams are biggest assets** available for us, the first tool to absorb initial hit of the floods (more absorption when its emptier during the heavy rains). Later, as per the situations and keeping safety of the dams as first priority, the dam operators release the controlled discharges over the spillways, thus reducing the potential hazard of the upcoming floods.

Moreover, it is also to understand that, most part of the **Maharashtra** is commonly known as **draught prone region**. Considering the rainfall pattern of the region with the help of historical data of previous 30 to 40 years, the dams are designed to store water during monsoon, for its use in multi-purpose sectors viz.

domestic, agricultural, industrial and many others. No dam is designed (except few in northern frequent flood affected states) in our country by keeping specific flood absorption capacity in the gross storages of dams. Our dams have been designed and constructed almost 20 to more than 100 years before, based on the then rainfall pattern observed previous to that. Due to global warming effects and climate changes, the regular natural cycle of our rainfall pattern has adverse and sudden effects creating new phenomenon like heavy to very heavy precipitation within very short durations, flash floods and many environmental disturbances.

Keeping all this in view and considering the given conditions, we have few options left for flood mitigation measures. The proper and planned flood monitoring through available dams and improving the existing systems can be one of them.

Adoption of the advanced flood monitoring systems is the first and important measure. The short-term measures and/or that can be urgently adopted and also some long-term measures are listed below.

6.1.1 Short term:

a. Fortunately, we have installed a **Real Time Data Acquisition System** (RTDAS) in Krishna sub-basin connecting all major dams which enables us to collect digitally all meteorological and hydrological data at real time (as fast as updated for every 15 minutes) at single monitoring centre at Pune. The same system can also install and maintained for rest of the Maharashtra as well as throughout the entire Krishna basin (including all its tributaries) covering areas in Maharashtra, Karnataka and Andhra Pradesh. (KWDT II, Further Report under XV-A had already ordered this action however, implementation is pending)

b. A refined/advanced RTDAS with **strong network**, both at upstream catchments of the existing dams and downstream free catchments of the basin can be installed. This can help for knowing the actual rainfall and other parameters almost instantly, before the flood waters reach dam sites and further down for better understanding of the situation and planning for the flood monitoring measures in advance.

c. **Efficient operation and maintenance** of the spillway gates and other components of the dams, installed systems and instrumentations is very much important and necessary. Especially electronic instruments need to be replaced / updated as per standardised practices.

d. **The Reservoir Operation Schedules** (ROS) of every dam can be improved or updated considering the updated parameters. This is prerequisite for achieving effective and fairly accurate acquisition of data. Revised survey for updating area capacity curves, stage discharge curves should be carried out at period intervals. At present, the ROS for all dams are prepared for the monsoon period (1^{st} June to 15^{th}October) of every year. Instead, those can be extended for the entire year for better planning of the operations of the reservoirs. Koyna dam as it has now become multipurpose dam, the emphasis in the preparation of ROS now may need change from hydropower dam to multipurpose dam and accordingly review need to be taken at Government level for considering the dependability for upper & lower curves. During heavy floods, the normal ROS should be switched over to the emergency flood moderation scheduled.

e. While releasing the discharges through dams, an **Integrated approach** can be adopted. For example, the releases from the dams on Krishna main river can be done prior to the releases through tributaries, which may enable to maintain the continuous movement of the flood mass flowing through the rivers. This will also help in lowering down the high flood levels (inundation) at the confluences and also minimising the stagnation period of the flood mass, particularly near habitats.

f. **The synchronised operations of all the dams** in the entire basin at macro level can be planned and executed by developing a software, so that minimum possible flood flow can be maintained in the main course of the river.

g. Induction and/or availing services of the dam experts, **hydrologists** can be necessary at every flood monitoring unit, which can help in quick and correct assessment, analysis and giving required support information to the decision makers for better flood monitoring.

6.1.2 Long Term:

a. **A Basin Level Flood Monitoring Board** be created for efficient and smooth flood operations during the monsoon period. This may be formed as a statutory mechanism by the consent of all-party states. The top-level decision makers of every controlling flood monitoring Units, representatives and counterparts from all three states along with a CWC official be the members of this board. They can ensure the fast and smooth communication of data to be exchanged, keeping continuous awareness of the flood situation and alarming each other about any kind of worst situation that may happen.

b. The latest and advanced **instruments** be installed on all major dams in the basin, to continuously monitor the health status of the dams, particularly when dams are full during the monsoon.

6.2 Flood Forecasting

Flood forecasting plays a major role in planning and monitoring the floods efficiently to avoid situations becoming worst and also to minimise the flood damages. With the help of advance forecasting of meteorological parameters and also by acquiring real time hydrological data, an efficient decision support system can be created.

6.2.1 Short term:

A Real Time Decision Support System (RTDSS) has been created for Krishna sub-basin with the help of customised software. The same can be upgraded and the advanced features be introduced with the help of experts.

6.2.2 Long term:

There is big scope for **improvement** in the **meteorological forecasting** as is done today. The area specific and short duration forecast systems like NOWCAST can be developed by the Institutions like Indian Meteorological Department (IMD), IITM etc. at the earliest. Considering the global warming effects and climate change effects on the atmosphere, the accurate and earliest flood forecasting may help the flood monitoring infrastructure and systems to make more flood resilient.

6.3 Restoration of Natural Land Drainage Systems

Natural Land Drainage Systems play an important role during flood mitigation. Its perfect natural flood resilient infrastructure, more effective than the artificial dam controls for flood monitoring. Due to the developmental works of the mankind, neglecting the further consequences, we have tampered it to greatest extent. Further it has already jeopardised the entire ecological balance of the nature. The erratic precipitation, flash floods, draughts, global warming etc. all are ill-effects of the same. Now, the time has come to make hard work towards restoration of natural imbalance to the original one, at least partially.

The basic principle of the natural land drainage system is to **drain out the rains** and the runoff to the lowest point on the land, i.e. to the sea level. The drainage system of any big river basin, comprising of confluence network of small nallas, tributaries, small rivers etc. ultimately go on meeting the main river which is flowing towards the sea. On the way, part of waters gets conserved by percolation to underground recharging the aquifers. On the way, it also feeds the eco-systems.

Another but important principle of this system is, to **quickly dispose off** the storm water of the surface making lands dry. Thus, it enables the open lands to receive further rains without much inundation/flooding. Wherever this doesn't happen, the area gets immediately flooded stagnating the water for long time. Thus, to avoid flooding we must restore this entire drainage system to its natural/original state.

This entire natural land drainage system can be divided in three-layer system, for ease of maintenance and execution of restoration works.

i. **Primary level**: the storm water drainage system of any building or a plot connecting to the open road side gutters in any city or habitat.

ii. **Secondary level**: the storm water drainage system of the city, open or underground, connecting all road side gutters to nearby nallas and small tributaries, upto confluence of main river.

iii. **Tertiary level**: main river course

6.3.1 Short term:

a. **For primary level drainage system:** The natural ground profile has been disturbed for developmental works by levelling, excavating the plots. The slopes of the open spaces and open grounds can be planned and the drains can be excavated so that storm water gets quickly disposed off, out of the plots. Rain water harvesting, bore well recharging techniques are some of the other methods too.

b. **For secondary level drainage system:** the city storm water drainage plan can be prepared and executed by civic authorities and the system can be properly maintained each year before the monsoon. All storm water can, thus, be collected through deep road side gutters, connecting them with proper slopes to nearby nallas/tributaries. The roads surface can also have proper camber, to quickly dispose off the rain water to side gutters. The slopes of nallas and underground pipes, if any, can be well designed and maintained annually, keeping them free of any debris, siltation, choking, undue vegetation, encroachments in water way by illegal constructions etc.

6.3.2 Long term:

a. For tertiary level drainage system: the original water way of the main river must be kept clean, free of any kind of obstacles across the river, encroachments, siltation etc. It can also be ensured that, the water way of the main river course can be sufficient to carry the designed flood discharge of the dam located at immediate upstream of the city and also to carry the runoff from the free catchments contributing to the same river from the downstream of the dam.

The restoration of natural water way of the main river course may be done by following ways –

i. **widening** of the waterway to its original cross section or to carry safely the designed flood discharge at that point

ii. **Deepening** of the river course upto sufficient level so as to accommodate incoming floods from all nallas and tributaries.

iii. **Desilting** of the river beds can be done, which may have accumulated from years together, particularly near confluences and meanders. This can help in reducing the high floods levels in habitat areas.

iv. Both the **banks** of the rivers can be **raised**, coupled with pumping facility, if necessary as per local situation, by earthen bunds, retaining walls etc. so as to limit the floods within the cross section of the main river and avoid the large inundations over adjoining plains or low-lying areas.

v. **River training works** can be properly planned and executed as per requirements.

b. Concept of Sponge cities: The restoration of old city lake ponds, small water bodies within the city areas and linking them to the nearby passing nalla or pipe drainage for temporary storage of the floods. This system can effectively be used to temporarily absorb the city floods, like sponge, and releasing back to the streams when floods recedes.

c. **Flood lines mapping:** As per the standard procedure of the WRD, the flood lines can be finalised and approved by competent authorities for the entire lengths of the main river courses for the guidance of city developments. These lines can be marked / superimposed on the city development plans and be kept accessible in public domain by the civic authorities. These lines can also be marked on ground, on permanent structures like bridges, old temples etc. so as to make public aware of the actual flood's situation and its further hazard potential.

For every river course, three such lines can be finalised. The Blue line, Red line and the Yellow line.

i. **Blue line:** normally, it's the line showing inundation level for 25 years flood. The area between the main river course and the blue line is called "Prohibited zone", where all kinds of constructions, obstacles are strictly prohibited.

ii. **Red line:** normally, it's the line showing inundation level for 100 years flood. The area between blue line and the Red line is called as "Restricted zone", where part constructions are allowed with certain restrictions.

iii. **Yellow Line:** This new concept is being introduced. At many places, it has been observed that the actually observed highest flood level, may be due to any back water effect, in a particular city is much above the red line, finalised by WRD. These lines can also be marked on city development plans as well as on the ground to make public aware of the **possible risks** while developing or investing in that zone. The area between Red line and the Yellow line can be called as "Alert Zone". However, no any kind of restrictions can be imposed within this Alert zone.

Alert zone will be in existence when the yellow line is above the red line. However all three lines must be marked and updated from time to time.

6.4 Encroachments in natural waterways:

As discussed above, restoration of the natural drainage system is the effective and permanent flood mitigation measure of all. Since long years, many developmental works have been done alongside the rivers and tributaries when such water courses pass through or bye-passing the city. Plot levelling, flattening of small nallas, restricting the waterways of streams passing through habitats by constructing retaining walls, ghats, river side roads, bridges or culverts by restricting waterway passing through pipes, throwing debris, disposables, domestic effluents into the nallas etc. are the major causes for flooding in the city areas and its long time stagnation.

6.4.1 Short term:

i. Strict restrictions on the **new developments** in and around the water courses can be put on immediately. The strict implementation of the legal provisions can be done by imposing heavy penalties.

ii. Removal of **existing encroachments** (structural and non-structural elements) need to be done, may be phase wise to clear and restore the original water ways. At many cities, the main river course has been so encroached upon that even the designed flood discharge of the immediate upstream dam also cannot pass through safely. This puts on extra restriction on the dam operators for releasing the required spillway discharge, thus compromising the safety of the dam itself. This situation may increase the hazard potential many folds.

iii. Regular **operation and maintenance** of all the existing cross structures like weirs, check dams, road bridges, footways etc. can be done. The floating debris coming with the floods may get arrested at such locations causing damages to the structures, as well as blocking the water way which results in increasing the flood level in the vicinity areas due to affluxes.

6.4.2 Long term:

i. The restrictions on constructions of new **river crossing structures** like bridges, footways and also new across structures like weirs, small check dams can be put on to avoid formation of local affluxes during floods.

ii. Strict implementation of the **ban** on sand, soil and other **mining materials** can be done. The constructions of water conservation works like deepening of nallas, underground weirs (below river beds, to check the underground currents), regradation of nalla beds etc. are done by locals by very unscientific manner. This adversely affects the natural eco-system and also creates obstructions for smooth flow of flood waters because of non-uniformity or reverse bed slopes.

iii. Natural **eco systems** of the rivers can be restored and maintained with the help of Environmental experts and NGOs, as it destroys the flora and fauna. Thus, maintaining the natural regime channel of the river course can also create the riparian flows in the rivers.

iv. **Creating awareness** for the importance of maintaining the natural eco-system amongst the citizens need to be done by special campaigns.

v. Physical and/or **mathematical modelling** can be done to assess effects of existing encroachment in natural water ways and to suggest further remedial measures.

vi. Detailed **sedimentation studies** for reaches of river near the confluences.

6.5 Flood absorption measures:

Creating specific spaces for the sole purpose of temporary storage of the floods is the flood absorption. This measure helps in flood mitigation, for synchronising the floods coming simultaneously from various tributaries and through main rivers.

6.5.1 Short Term:

i. In the absence of separate flood absorption space in the dams of Maharashtra, improving flood absorption capacities in the existing dams by **revising the Reservoir Operation Schedules** and keeping the margins may be done, before the water levels in the dams reach to FRL. Utilising the margin of encroachment of the water level in the dams upto HFL wisely can also be done as cautious decision. Increasing FRL of existing dams by extending spillway gate heights using flaps can also be done, wherever possible. In addition, this can also be done on ongoing and future dams with due considerations for additional submergence of land, rehabilitations, Interstate aspects and modification in designs wherever necessary.

ii. **Straightening of meanders**: By observing the alignment of the main Krishna river, particularly from downstream of Sangli city upto state boundary and beyond, many meanders have been formed to the river course. Due to sharp turns and curves of the river, the flow in the river slows down reducing the velocities of the flow. Also, heavy siltation has occurred, particularly at confluences and also to the inner side of the meandering curves deflecting the main flow towards outer curve. Due to heavy siltation occurred from many years, the river beds have risen to few meters from its original levels, thus increasing the high flood levels in the flood plains. With all such typical geographical phenomenon formed, the large flood plains have been created, forcing flood water to rush through low lying lands making direct cross connections with shorter lengths by connecting two sharp curves of adjacent meanders.

The actual **slopes** of the **river beds** in these stretches, when measured, found to be very flat, making the flood mass almost stagnant for longer durations. Studying the river stretches having meanders and following the path as shown by the actual flood flowing itself, **Straightening** of the meanders can be an effective mitigation measure. Detailed survey and planning can be done before taking up of such works.

Detailed **model study** can also be done before designing the re-alignment of river channels while proposing straightening of meanders.

In many countries in the world, for example in **Tokyo (Japan)** the works of strengthening (short cutting) of meanders (direction) had been implemented for achieving better flood control, water transport, salvaging rich / fertile land etc. Other supplementary works like group embankment, regulating reservoirs, retarding basin were also executed for above purpose.

6.5.2 Long Term:

i. **Construction of new flood dams**: On the principles of Sponge cities above (6.3.2 d), new flood dams can be created in the upstream catchment of the existing dams, wherever possible and also in the downstream free catchment area alongside of the river stretches. The flood waters can be diverted temporarily into these flood dams, while the peak floods are there. Later on, when receding, this water can be let back into the river streams.

ii. **Diversion of excess floods into adjacent sub-basins:** To have a permanent kind of flood mitigation measure on large scale, and with the concurrence of the downstream states, the excess flood water can be diverted to the adjacent sub-basins where there is no flooding at the same time. With the study of Krishna basin drainage system, the flood waters can be easily diverted into adjacent Bhima sub-basin, filling the enroute reservoirs and utilising their flood absorption capacities. The geographical study of the Krishna basin suggests that, the excess flood waters in Krishna sub-basin right from Panchganga river can be diverted to Bhima sub-basin upto Ujjani dam by gravity (without Pumping) by constructing series of tunnels connecting the rivers in between. This water, diverted into the Bhima sub-basin can ultimately coming back into the Krishna sub-basin where the river Bhima meets river Krishna in Karnataka state.

Also in order to reduce intensity of floods within Krishna Basin, during floods / monsoon period, priority can be given to optimum utilisation of potential envisaged on major LIS with due consideration of KWDT provision.

6.6 Disaster Management Units:

The Disaster Management Units, headed by **revenue authorities** (Collector at District level and Divisional Commissioner at Regional level) play very important role as flood mitigation authority. Their work starts much before the actual flood monitoring measures by the dam operators. They can co-ordinate the services of all respective agencies to limit the flood hazard potential and to control/reduce the actual flood damages. These units can always be alert and act literally on war footing so as to minimise the fatal casualties as well as minimising the flood damages of the properties, developmental infrastructures, cattles, flora and fauna within the flood affected areas and natural eco-systems. Their work continues even after the flood recedes until the normalcy prevails.

Here are the flood mitigation measures that can be improved upon by the disaster Management Units.

6.6.1 Short Term:

i. The disaster management units can be active 24 X 7, and alert throughout the year. They can keep themselves acquainted with all preliminary warnings and flood forecasting done by the respective agencies, time to time. They can be in constant touch with the respective agencies, by establishing direct connections through various media, taking continuous stalk of the situation. They can be prepared with the **basic data**, maps of the geographical area, contact numbers, direct communication links for emergency rescue operations as well as all necessary equipments required for flood relief activities.

ii. **EAP**: The emergency Action Plans for every disaster that may occur can be kept ready and be updated time to time.

iii. **SOP**: Standard Operating Procedures can be laid down in case of any kind of emergency that may occur. This also needs to be updated time to time.

In addition to prevalent plan & procedure, following actions are necessary

- **Timelines for preparation of EAP & SOP –**
  EAP & SOP can be prepared before gorge filling of the dam. Competent authority can not give approval to the gorge filling if EAP and SOP is not prepared. Uploading list of key personnel, their phone numbers and address must be done every year before end May. EAP should be reviewed when there is significant change in dam operation, downstream development etc. or every five years whichever earlier.

- **Demarcation of Maximum Observation Flood Line –**
  As per Dam Safety Manual (Chapter 8), the inundation map needs to be divided into three zones i.e. Prohibitive, Restrictive and Caution Zone. But the flood in Sangli and Kolhapur Districts crossed red line i.e. restrictive zone for which there were no preparedness for evacuation. This has resulted in more human and financial loss. Hence another zone needs to be created if maximum observed flood line crosses the red line. This flood line should be called as Yellow line and the zone as Caution 1. This is depicted in the **Annexure 2D.**

- **Disaster Management Cell –**
  The District Disaster Management Cell is headed by Disaster Management officer of the rank of Deputy Collector under District Collector. These district level cells work by and large efficiently during the flood situation in this year. It is recommended that there can be weekly review meetings during monsoon season and dam owner can communicate and coordinate with all Disaster Management Officers of concern districts.

- **Responsibilities of various Departments -**
  Responsibilities of various Departments like Revenue, Home Zilla Parishad, Post, Telecommunication, IMD need to be elaborated in detail as per **Annexure 2C**

iv. **Early Warning Systems:** Whenever the situation worsens, it's the prime duty of the head of the Disaster Management Unit to make citizens aware of the situations so that they can be prepared to face the disaster and minimise the damages. For this, with the help of weather and flood forecasting given by concerned authorities, the early warning systems can be developed. The installation of sirens, playing Davandies, mobile messages, media broad castings, wireless systems, Ham radios etc. can be done and maintained. The warnings to be given time to time, so as to give sufficient time and guidance to the affected citizens to save prospective damages.

6.6.2 Long Term:

i. The **public awareness campaigns** can be conducted time to time to inform them about the hazard potential for various emergencies. The mock drills can also be taken to make citizens habituated for the situation and they can not take the emergency as a surprise. Thus, they can always be prepared for any kind of situation to co-operate with the authorities and help control the situations without any panic. The fleet of volunteers be kept ready, with due training to them.

ii. The inundation maps / **vulnerability map** for various flood discharges be kept ready to help the decision makers, for all the important cities, villages, river stretches, confluences etc.(See Maps **9, 10 and 11**)

iii. **Emergency evacuation plans** be kept ready in the form of Evacuation Table as in **Annexure 2E**